

Research Master Thesis

Named Entity Recognition and Classification and Concept Extraction in the Legal-Tax Domain

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Abstract

This thesis investigates approaches to Named Entity Recognition and Classification (NERC) and Concept Extraction (CE) in the legal-tax domain, an intersection between the legal and tax domains. The legal-tax domain under investigation is a specialized domain comprising tax case law documents, integrating characteristics of both the tax and legal domains. The legal-tax domain is therefore highly specialized and faces the typical issues that domain-specific applications do: scarcity of annotated training data. This data shortage can be attributed to the high complexity of legal documents, which require domain expertise and time to annotate. Other mainstream domains also face this issue in more specific use cases and have resorted to domain adaptation using related datasets, large-language-model inference, and synthetically generated labels to mitigate reliance on large amounts of human-annotated data, unlike more traditional classification approaches. The legal-tax domain, however, remains unexplored in the application of these methods. To determine the best approaches to NERC and CE in this domain, a methodology that requires varying amounts of training data was adopted. This ranged from no training data, in the form of gazetteer matching and zero-shot prompting with LLMs, to little training data via few-shot prompting, to complete reliance on training data through fine-tuned encoders. To also assess the effects of annotation quality and domain matching between fine-tuning and testing data, the transformer models were fine-tuned on three different datasets: existing related-domain legal data, LLM-labeled in-domain data, and a mix of the two. The main goal of this study is to expand research in the legal-tax domain for NERC and CE, while shedding light on the impact of in-domain pre-training and fine-tuning data compared with human-annotated and synthetically generated labels. As the performance of the proposed methods is evaluated on a test set provided by the International Bureau of Fiscal Documentation (IBFD), the findings are intended to offer possible solutions to improve their current approach to identifying and classifying domain-specific named entities and concepts in tax case law documents. The findings may also provide insight into NERC and CE for other domains that also suffer from a lack of gold-annotated training data.

Declaration of Authorship

I, Manya Walavalkar, declare that this thesis, titled *Named Entity Recognition and Classification and Concept Extraction in the Legal-Tax Domain* and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a Research Master degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Date: August 15th, 2026

Signed:



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Chapter 1

Introduction

This thesis investigates Named Entity Recognition and Classification (NERC) and Concept Extraction (CE) for tax case law documents, evaluated on in-domain test data from the International Bureau of Fiscal Documentation (IBFD). Tax case law sits at the intersection of the legal and tax domains, combining the structural characteristics of legal texts with tax-specific terminology. For this reason, this thesis refers to it as the legal-tax domain. The main issue at hand, typical of such a domain adaptation use case, is the lack of human-annotated in-domain training data. To address this, three approaches requiring no or varying amounts of training data are compared for their NERC and CE performance. The experiments aim to determine which approaches perform best for both tasks, so the company can develop an automated method for identifying and classifying named entities and concepts in tax case law documents. The automated labels can also guide efforts to reduce the human verification needed to manually label documents from scratch.

The legal domain has undergone significant digitization recently, resulting in vast amounts of unstructured text that are impractical to process manually (Karamitsos et al., 2025; Kalamkar et al., 2022). In such cases, as noted in Kalamkar et al. (2022), it is imperative to harness the computational power of Natural Language Processing (NLP) to reduce strain on the judicial system and reduce pendency in legal decision-making, which otherwise increases due to manual processing. Though readily available digital documents make access to law easier, their format is often raw and unstructured, making it hard to search meaningfully without closer manual inspection, which quickly becomes tedious.

The IBFD is a company that offers high-quality information and education on international tax to its clients through access to its powerful Tax Research Platform (TRP) (n.d., 2026). Their aim is to disseminate the understanding of cross-border taxation, thereby providing tax practitioners with knowledge they can rely on to work faster and more effectively. As such, it is necessary for the TRP search queries to find specific, relevant documents. In certain cases, a simple keyword search does not suffice, as it may miss semantic relationships merely because they do not appear in a document's title or its body's text.

Their solution to this is to enrich documents with structured metadata that improves retrieval for free-text search queries. Tax case law texts are dense with information, including extensive jargon and documentation of legal actions between parties. As such, extracting and including these terms in a document's metadata can synthesize the covered content and inform users beyond the document's title. The two tasks that

facilitate this information extraction from unstructured data are NERC, the identification and classification of text spans into predefined categories (e.g: 'Article 43 of the TFEU [PROVISION] states that..') (Li et al., 2020), and CE, the identification and classification of domain terms or concepts into semantic categories (e.g: 'arm's length principle' [TAX_CONCEPT], 'transfer pricing' [TAX_CONCEPT]) (Qiu et al., 2019), which captures the semantic relationships that keyword string match searches would miss. Due to the fundamental similarities, the main task is framed as NERC. However, to account for differences in the characteristics of named entities and concepts, the two are evaluated separately.

NERC and CE are fundamental to this operation, as extracting these named entities and concepts enables the construction of a relational database and helps fill the gaps in the company's existing tax ontology, using Resource Description Framework triples, thereby demonstrating arbitrary relationships in the domain (Grainger et al., 2025; Goyal et al., 2018). Furthermore, these terms can later serve as part of the underlying data structure that enables traversal and semantic knowledge graph creation (Grainger et al., 2025).

As an initial approach to large-scale named entity tagging, the company adopted a hybrid approach. An automated system was created as a tool to pre-annotate documents, reducing the labor of manual annotation. First, general-domain NERC library tools and gazetteer matching were used for pre-annotation, followed by human annotators adding labels to any remaining uncovered items. Throughout the process, the human annotators validate these automated annotations by correcting missed or unassigned labels. Though general-purpose NLP tools (for example, SpaCy) are very helpful for automated entity detection, their utility is limited in domain-specific applications (Karamitsos et al., 2025) due to specialized vocabulary, unfamiliar document structures, and unconventional label taxonomies. Legal and tax domain documents like tax case law are a particularly challenging case: they tend to be long, syntactically complex, jargon-rich and vary in structure based on the originating jurisdiction (Karamitsos et al., 2025; Chalkidis et al., 2020; Hechtner et al., 2025). They include legal terminology, contain ambiguous terms that may be defined differently across legal institutions, and often include citations which carry substantial information that should not be ignored (Hechtner et al., 2025; Jain et al., 2021).

Adapting NLP tools to this domain requires instilling domain-specific knowledge needed to make informed predictions, which often takes the form of either custom rule-based pipelines added to pre-existing general-domain models or through models trained on in-domain text (Kalamkar et al., 2022). However, rule-based extensions require manual effort to construct and maintain patterns to match the task (Li et al., 2020; Yang et al., 2020) and training data is not always available for specific domains (Liu et al., 2021). Another obstacle to adapting models across domains is domain shift, in which model performance drops because the domain of its training or fine-tuning data differs from that of the task data it is applied to (Zhou et al., 2022). This issue highlights why in-domain or related-domain training is valuable, but that its impact can ultimately be limited by how closely the training data matches the target data. This means that in-domain or target-domain annotated data used to train models on target labels and data types remains the most reliable way to achieve task-specific performance in legal-tax domain NERC application.

Unfortunately, annotated legal data is scarce (Zin et al., 2024; Ariai et al., 2025). This can be attributed to the complexity of annotating such data, which must be

carried out by legal experts, making achieving annotator agreement a slow and laborious process, not to mention the resulting financial costs (Ariai et al., 2025). Domain experts are necessary for labeling to ensure that the annotated data is contextually accurate and aligns with current legal standards and interpretations. It is crucial to emphasize, as noted in Koenecke et al. (2025), that labeling legal documents in certain scenarios is high-risk, as incorrect labeling can misguide human interpretation of the true context and lead to an unwanted legal situation. As the TRP is intended to support tax practitioners, the thesis therefore aims to identify the best approach to NERC and CE in the legal-tax domain. The output is intended to provide the IBFD with an automated method for reliably extracting named entities and concepts from tax case law documents.

These constraints motivate the exploration of approaches that reduce or circumvent the dependency on large quantities of gold labels. On one hand, rule-based methods such as gazetteers are a transparent, annotation-free baseline that relies on handcrafted rules, exhaustive lexicons, and domain heuristics (Li et al., 2020). On the other hand, LLMs require much less to work with, performing inference with zero-shot and few-shot prompting (Wang et al., 2025), and even generating annotations for data that can then be used to fine-tune transformer models on downstream tasks (Bogdanov et al., 2024; Yazdani et al., 2026). In the middle, transformers offer a trade-off: they require a significant amount of unlabeled data for pre-training to capture language patterns, or some labeled data for fine-tuning to learn a specific task, compared to more traditional machine learning models that rely entirely on labeled data for training. Domain-specific pre-trained transformers are also available, reducing the need to provide models with large amounts of data. The successful performance of these methods has been documented in other specific domains such as biomedical, education, and geoscience (Šuvalov et al., 2025; Yang et al., 2020; Norouzi et al., 2025; Lv et al., 2022; Perera et al., 2020), but remains relatively underexplored in the independent legal and tax domains. NERC also remains considerably better researched than CE in the independent legal and tax domains.

This thesis, therefore, investigates which of these approaches is best for performing NERC and CE on tax case law documents in the legal-tax domain. Using the NERC test set provided by the IBFD, the experiment will evaluate gazetteers, LLMs, and fine-tuned encoders to gain insight into a limited-training-data context. These results will help shape an informed suggestion for the company on how to better identify and classify named entities and concepts in their tax case-law corpus.

The results of these experiments indicate that the LLM-labeled in-domain dataset for fine-tuning helps encoders achieve the highest NERC and CE performance, whereas using only existing human-annotated legal-domain data yields considerably less transferability, highlighting the importance of domain matching. GLiNER performed the best consistently, suggesting that its entity-to-span matching approach for NERC and CE is more effective than the traditional identification and classification used by other BERT-based models. Legal-BERT followed closely, suggesting that its legal domain-specific pre-training gave it an advantage, enabling it to substantially outperform RoBERTa across all fine-tuning datasets and to perform competitively with GLiNER under certain conditions. Although LLMs demonstrated comparable zero-shot performance and further improved in the few-shot condition, they were ultimately outperformed by encoders fine-tuned on the data they labeled, highlighting knowledge distillation. Lastly, the gazetteers alone addressed concepts, but when combined with the

automated components (SpaCy and Regular Expression filtering) that addressed NEs, performance was only nearly comparable to LLM zero-shot, indicating that general-domain NERC tools are not suitable for domain-specific applications like the legal-tax domain that involves abstract surface-form concepts.

1.1 Research Questions

The main research question addressed in this thesis is:

> **"What is the best approach to identifying and classifying named entities and concepts in the legal-tax domain?"**

Here, *task* refers to the identification and classification of named entities and concepts in the legal-tax domain. To answer the query, the following sub-research questions are proposed, grouped by their approach family:

- **Baselines**
  - SRQ1: How do the relevant baselines, consisting of the company baseline and out-of-the-box models, perform on the task?
- **Gazetteers**
  - SRQ2: How does a linguistic heuristic-based gazetteer perform on the task compared to the company's ontology-based gazetteer?
- **LLM prompting**
  - SRQ3: How do LLMs perform on the task in zero-shot and few-shot conditions?
- **Fine-tuning encoder models**
  - SRQ4: How does domain-matching affect fine-tuned encoder models' performance on the task across three training conditions: approximate legal domain NERC data, target domain tax case-law data with LLM-generated NERC labels, and a combination of the two?

1.2 Structure

The background required for this thesis is outlined in Chapter 2, which first defines the task of NERC and CE, the methods used to address it, the current state-of-the-art method, and how domain specificity affects model performance, before presenting research in the legal and tax domains. Following this chapter, a view of how the experimentation for this thesis was conducted is given in Chapter 3, starting with an overview of the data used: including the test set for the use case, followed by the existing approximate domain datasets, then the LLM procedure to generate synthetic labels for in-domain data, and finally combining both datasets for a mixed composition. The chapter also discusses the main methods, starting with the company baseline, heuristic-based gazetteer lookups, zero-shot and few-shot approaches, fine-tuned transformers, and finally the evaluation metrics used to measure performance. The results

are then presented in Chapter 4, first presenting the company's automated NERC baseline and its components, followed by gazetteers and out-of-the-box approaches, then the encoders by the fine-tuning dataset used. The chapter then offers a summary of the results and provides analytical insights into the error patterns the best-performing model consistently exhibited. Lastly, Chapter 5 closes with an overall discussion of the findings, offering recommendations for the company's use case, noting the limitations encountered and suggestions for future related studies, before a brief conclusion brings the paper to an end.

Chapter 2

Related Work

The aim of this chapter is to provide an overview of Named Entity Recognition and Classification and Concept Extraction as tasks, and address the typical approaches taken towards them. Beginning with Sections 2.1 and 2.2, the tasks of NERC and CE are defined. Subsequently prior work on the tasks are shown in Section 2.3, consisting of various machine learning approaches and then the current state-of-the-art. Then, the issue of domain adaptation is highlighted in Section 2.4. The legal and tax domains are introduced in Section 2.5, which defines the domains and then discusses NERC and CE in this domain-specific context.

2.1 Named Entity Recognition and Classification

A 'Named Entity' (NE) is a phrase that functions as a 'rigid designator', denoting a specific real-world referent, and is an atomic element belonging to a predefined semantic class, (Goyal et al., 2018). Named Entity Recognition and Classification, henceforth referred to as NERC, is a task in Natural Language Processing (NLP) that identifies and classifies such rigid designators within textual data, assigning them to different pre-defined categories such as organization, person, or location (Li et al., 2020). It is categorized as a sequence labeling task, as it involves mapping an input sequence to a corresponding labeled output sequence. This labeling is conducted using the Inside-Outside-Beginning scheme (IOB), which marks each token's position within an entity span.

Benchmark shared tasks in computational linguistics and artificial intelligence are collaborative competitions in which researchers evaluate their models on the same dataset using standardized metrics. It allows for objective comparisons of proposed model performances on specific tasks. An influential benchmark for this task was proposed by Sang and De Meulder (2003) in the form of the CoNLL 2003 shared task, which provided a language-independent NER dataset that is still widely used as a baseline for comparison in NERC. Though a recent study by Reiss et al. (2020) has contested label consistency in this dataset, CoNLL has left an undeniable impact as a standardized annotation scheme, more commonly known as IOB labeling. The most frequent way of using this labeling scheme is IOB2, in which the first token within an entity span is tagged with B-LABEL, and any tokens that follow it within the same span are tagged with I-LABEL. All other tokens that are not part of an entity are tagged as outside ('O'). An example sentence following this annotation scheme is seen in Table 2.1 below.

Token	NERC-label
Manya	B-PER
studies	O
HLT	B-MISC
at	O
the	O
Vrije	B-ORG
Universiteit	I-ORG
in	O
Amsterdam	B-LOC
.	O

Table 2.1: Example sentence labeled using IOB2 annotation

The conventional classes and most frequently used classes of entities in NERC systems are person (PER), location (LOC), organization (ORG), alongside the more loosely defined miscellaneous (MISC) class. Although generic entity classes are commonly used to benchmark systems, NEs can be domain-dependent, extending identification beyond the mentioned classes and proper nouns, since certain fields consider relevant nouns to be named entities as well (Zitouni, 2014). For example, the biomedical domain looks to extract PROTEIN, DRUG, and CHEMICAL entities (Perera et al., 2020), whereas the legal domain focuses on PROVISION, LAW, and COURT (Kalamkar et al., 2022), which expand past the traditional expectation that NEs are proper nouns.

NERC is a fundamental sub-task in information extraction, whose output enables various downstream applications such as information retrieval, question answering, and summarization (Goyal et al., 2018; Zitouni, 2014). Due to this utility, NERC is used across various domains like geoscience and biomedicine (Brandsen et al., 2021; Perera et al., 2020) but serves a particularly important role in the legal domain, as it compresses large documents into the relevant entities, which is useful for applications such as anonymization, citation extraction, and decision prediction (Karamitsos et al., 2025).

2.2 Concept Extraction

A 'concept' is defined as a collection of terms with the same semantic information (Qiu et al., 2019). Unlike NEs, which are specific mentions of real-world referents (Li et al., 2020; Goyal et al., 2018), concepts are abstract, serving as units of meaning expressed as words or phrases that can be linked to a controlled vocabulary, ontology, or task-specific semantic class (Fu et al., 2020). Concept mentions denote membership in a semantic category, so regardless of having varying forms, they can still share an overarching meaning. For example, the sentence: "*The battery life and shutter speed on the Canon EOS R6 is remarkable*," yields [battery life] and [shutter speed] both as mentions of a concept like Device Feature, whereas [Canon] is still a specific company mention that falls under the ORG named entity label.

Concept Extraction (CE), similar to NERC, is the process of identifying and extracting predefined concepts of interest from unstructured data (Fu et al., 2020; Li et al., 2020). Similarly, CE is also a sequence labeling task that classifies text sequences into predefined categories and, as a result, is often approached as NERC (Yang et al., 2020;

Fu et al., 2020). As such, the IOB labeling can also be applied to concepts (Yang et al., 2020). CE has also been investigated across many domains, most commonly in medical, clinical, and educational settings (Yang et al., 2020; Fu et al., 2020; Torii et al., 2011; Norouzi et al., 2025; Yang et al., 2025), as these fields often contain documents dense with specific jargon, but it remains relatively underexplored in the legal and tax domain.

An important distinction is that CE is more semantically and domain-oriented than NERC, as words are linked to concepts through meaning rather than surface forms; despite this, CE is often approached as NERC, which this thesis recognizes and adopts for its methodology.

2.3 Approaches to NERC and CE

As NERC and CE are both sequence labeling tasks, they have several main approaches in common (Yang et al., 2020). Earlier approaches were knowledge- and rule-based, relying on linguistic principles and lexical resources. Beyond these, unsupervised approaches that rely on algorithms that work without labeled training examples, and supervised approaches that traditionally relied on feature engineering were also employed. More recently, neural-network-based models learned semantic representations from raw text, reducing reliance on engineered features; and now, with the increasing generative capabilities of Large Language Models (LLMs), it is also feasible for such models to perform NERC and CE with little to no task-specific training.

The initial NERC and CE systems were primarily knowledge-based, relying on rules and gazetteers (Keraghel et al., 2024; Yang et al., 2020). These approaches allow systems to detect entities based on consistent patterns and surface forms (Keraghel et al., 2024). Gazetteers, specifically, were essential, as they serve as collections of key entities or terms that can be used for recognition (Keraghel et al., 2024). A disadvantage, though, is that these methods rely heavily on domain experts to identify patterns and manually design rules or lexicon that continuously capture NEs and concepts (Yang et al., 2020). While these may work robustly in well-defined domains and yield high precision, the handcrafted rules and lexicon come at the cost of perpetually incomplete dictionaries, expensive manual effort, low recall, and likely domain-specific applicability, limiting their generalization (Keraghel et al., 2024; Li et al., 2020).

Unsupervised machine learning approaches infer entities from lexical resources, patterns, and semantic distributions without labeled data, for example: clustering by contextual similarity (Li et al., 2020; Xie et al., 2023). For NERC and CE, feature-based traditional machine learning algorithms, such as Support Vector Machines, Conditional Random Fields (CRFs), and Logistic Regression, were also employed (Li et al., 2020; Xie et al., 2023). The features engineered for these tasks were usually linguistic features, such as word-level features (case, morphology, part-of-speech), list lookup features (gazetteers), and document and corpus features (local syntax, frequencies) (Li et al., 2020). Additionally, for CE, statistical and graph-based methods such as Term Frequency Inverse Document Frequency (TF-IDF) and Yet Another Keyword Extractor (YAKE) that rely on frequency and co-occurrence features were used (Xie et al., 2023; Norouzi et al., 2025).

The progression of deep learning led researchers to use neural networks, as these models could learn static feature representations from raw input, reducing reliance on engineered features (Li et al., 2020). For NERC and CE, this led to the successful

implementation of a Bidirectional Long Short-Term Memory (BiLSTM) neural network model with Conditional Random Fields architecture across domains (Li et al., 2020; Fu et al., 2020; Xie et al., 2023). CE was also conducted using models like EmbedRank, which ranked candidate concept phrases based on their embedding similarity to the input document (Xie et al., 2023). Further architectural changes enabled contextual modeling, allowing models to capture contextual representations of words.

2.3.1 Encoder-based Approaches for NERC and CE

Specifically, the discovery of transformers by Vaswani et al. (2017) introduced neural networks with self-attention layers, which allowed them to effectively capture and integrate contextual information globally. Bidirectional Encoder Representations from Transformer, known and introduced as BERT by Devlin et al. (2019) utilizes only the encoder component of the transformers architecture, and has the ability to capture bidirectional and dynamic representations of tokens, making it suitable for NERC as token entity types often depend on surrounding cues. Due to this advantage, BERT derivative models, such as DistilBERT and Robustly Optimized BERT Pre-training Approach (RoBERTa) are frequently employed in NERC and CE across varying domains, and have shown to outperform more traditional architectures such as the BiLSTM-CRF (Keraghel et al., 2024; Lothritz et al., 2020; Yang et al., 2020). BERT-based models can also be altered to have an additional linear classification layer, which is where the token is assigned a label. This layer is often a CRF as it enforces valid label transitions, which is why it is still used in combination with transformer models.

Despite this architectural advantage, BERT still requires general pre-training, which is why many studies either pre-train the model from scratch, continue pre-training from its initial general representations, or fine-tune it on a target domain to make it suitable for domain-specific NERC applications. Studies that use these methods have found that domain-matched pre-training and fine-tuning yield the largest increase in NERC performance, compared with models pre-trained on general domain corpora or fine-tuned on general domain datasets (Chalkidis et al., 2020; Yang et al., 2020; Brandsen et al., 2021; Au et al., 2022). Further pre-training of models with domain-specific data is known as domain-adaptive pre-training, which is explained further in Section 2.4 and has also been shown to significantly improve encoder performance in both domain-specific applications of NERC and CE. Some examples include RoBERTa-MIMIC (Yang et al., 2020), FinBERT Araci (2019) and LegalBERT-FP (Chalkidis et al., 2020), while a model such as SciBERT (Beltagy et al., 2019) demonstrates the same benefit through the more expensive and therefore less frequently employed from-scratch domain-specific pre-training instead.

Contrary to the traditional BERT-based models and their domain-specific variants often employed in NERC studies, Zaratiana et al. (2024) presents Generalist and Lightweight model for Named Entity Recognition (GLiNER) as a state-of-the-art (SOTA) model for NERC, which is a transformer-based (DeBERTa) architecture that frames NERC as an embedding similarity matching task between a candidate span and an entity label. Unlike traditional BERT methods, which classify token-by-token, and LLM prompting, which generates predictions token-by-token, GLiNER compares the vector of every possible span combination in a sequence to the vector of the target entity label. This matching mechanism is also well-suited to concepts, which are defined semantically rather than by surface form. Its matching mechanism and extensive continued pre-training on the Pile-NER, an enormous LLM-annotated dataset

containing entities from various domains enables it to display zero-shot performances that outperform ChatGPT and other instruction-tuned LLMs on NERC (Zaratiana et al., 2024).

In a similar light, NuNER, as introduced by Bogdanov et al. (2024), is a compact language representation model specialized for NERC. The authors used GPT-3.5 to annotate a corpus for entities that were used to train an encoder model via contrastive learning, aligning text embeddings with concept embeddings in the same vector space, similar to GLiNER, except it keeps input text and concept encoders independent, making it concept-agnostic. NuNER, as a token classifier, is optimized to capture arbitrary, long, and multi-word entities. Both models indicate that a BERT-derivative architecture can beat LLMs on NERC inference, despite being much smaller. They also indicate that the high-level reasoning of LLMs can be distilled into smaller models through synthetic labels, suggesting a possible solution for domains with low amounts of labeled training data available.

Simpler hybrid NERC systems are also possible, harnessing rule-based techniques to identify obvious entities, complemented by a machine learning model to detect more complex cases with less overt patterns. This process can also be reversed, as noted by Mehta (2025), who found that using a gazetteer as a post-processing step correcting misclassifications made by encoder models, harmonizes the powerful predictions of models with the gazetteer's guaranteed lookup for more difficult or abstract labels.

The thesis intends to use RoBERTa, LegalBERT, and GLiNER as encoder approaches to NERC and CE in the legal and tax domain, representing the impact of general domain pre-training, domain-specific pre-training, and span-entity embedding matching on performance.

2.3.2 LLMs for NERC and CE

Generative LLMs have progressed significantly since their start, with contemporary models having 100-10,000 times more parameters than BERT (Bogdanov et al., 2024).These autoregressive transformer models trained for next-word prediction can be prompted with natural language instructions to perform tasks (Zaratiana et al., 2024). With these constantly growing parameters come increasing capabilities, as their performance on zero-shot and few-shot learning improves (Keloth et al., 2024). This has led to studies employing models to perform zero-shot inference on tasks that otherwise require training data for models to learn from before classification, including NERC and CE (Hu et al., 2024; Keloth et al., 2024; Yang et al., 2025; Norouzi et al., 2025).

While LLMs offer impressive zero-shot capabilities, which are valuable in low-resource domains, studies have also shown that they struggle with fine-grained entity recognition unless supported by 'tailored prompts and post-processing' (Karamitsos et al., 2025). Similarly, when extracting complex, high-density entities from German pathology and radiology reports, Arzideh et al. (2025) found that despite instruction tuning, LLMs performed considerably worse than traditional BERT-based encoders with in-domain pre-training or fine-tuning. The authors found LLMs to be highly precise but with very low recall, indicating they are very conservative in domain-specific NERC as they are not familiar with the sub-language of medical texts and are unable to handle and distinguish between a large range of diverse entities. Similarly, Hu et al. (2024) tested general-purpose LLMs for their few-shot and zero-shot NERC capabilities in the medical domain using various in-domain benchmark datasets, and found that as the dataset entities increased in complexity and required more semantically

specific separation, the models struggled to assign correct entity boundaries and types. The authors also found that providing detailed task-specific instructions such as annotation guidelines and examples helped improve NERC performance. Keloth et al. (2024) also found that when performing NERC, LLMs do not always follow the given prompt strictly, resulting in missed or incorrect predictions. Haq et al. (2025) conducted a comparative analysis of LLMs as data annotators by examining their performance in NERC. The study found that zero-shot performance for generic, well-defined entities was higher than for more complex and diverse entities. Additionally, approaches such as providing examples in the prompt and retrieval-augmented generation supplemented examples substantially improved zero-shot performance, though ultimately data complexity and nuanced entities remain the biggest challenges that keep LLMs from approaching human-level performance in data annotation.

Matkin et al. (2024) also compared various fine-tuned BERT-based models to LLM zero-shot inference for skill extraction from Russian job vacancies, and found that the LLMs performed considerably worse, despite computing metrics by comparing the predicted entity's embedding to that of the gold entity for a fairer comparison. The same high precision-low recall effect was also observed, indicating conservative behavior from LLMs.

In contrast, studies that employed LLMs display competitive CE performance, though often against non-neural baselines rather than fine-tuned encoders. In particular, Norouzi et al. (2025) observed that their approach, ConExion, which prompts LLMs to extract domain-relevant concepts on two benchmark datasets of scientific abstracts (using the LLaMA-3-70B model), achieved higher F1 performance than other CE methods (including EmbedRank, MultPax, TF-IDF, and others). The authors attempted both zero-shot and few-shot prompting with varying phrasing, information amounts, and terminology. They discovered that the choice of 'search term' was important, and the model performed better when prompted for 'key phrases' than 'concepts' or 'entities'. They also applied exact lexical-match filtering, using a ground-truth concept list, to the LLM's output to reduce noise and hallucinations.

Another study that conducted CE was by Yang et al. (2025), which evaluated GPT-3.5, GPT-4o-mini, and GPT-4o for concept generation, concept extraction, and prerequisite relation identification. Their approach to CE prompted the model with varying amounts of 'information injections', starting with minimal context, increasing to include more information such as an example, the concept explanation, the educational course name, and its description. GPT-3.5 performed the best according to automated metrics, as it directly replicated keywords from the text. The other models, instead, produced summarized or reformulated concepts that were considered more valuable according to human evaluation. The findings also revealed that performance generally improved with increased contextual input compared with minimal prompts. Larger LLMs successfully leveraged additional context to improve their performance, whereas smaller models struggled, indicating that the model's ability to prioritize relevant content is as impactful as prompt length and density.

For NERC, some authors argue that prompt engineering can further boost LLM performance to levels that compete with traditional approaches. Wang et al. (2025) proposes GPT-NER, an innovative approach to NERC for LLMs: rather than the usual span detection task, which is difficult for LLMs (Yazdani et al., 2026), the model is instructed to reproduce the input sequence with special characters marking potential entity spans. Thus, the task is framed as sequence generation and is more aligned

with the capabilities of the autoregressive model. In the few-shot setting, the models were provided with a varying number of examples that are randomly retrieved or extracted based on the embedding similarity of the target sentence or entity being labeled. To address hallucinations, the models were prompted a second time to confirm their predictions. The GPT-NER prompting structure was applied to GPT-3 and evaluated on five widely adopted NERC datasets, with few-shot results comparable to fully supervised baselines under data-scarce conditions. Another methodological fix was proposed by Huang et al. (2025), who proposed GuideNER, a framework that replaces the input-label mapping function provided by few-shot examples with explicit annotation guidelines. The authors found that providing an LLM with incorrectly labeled text-entity pairs does not hinder it from performing NERC correctly, suggesting that the models already have a general grasp of classes and benefit more from better annotation guidelines than from examples. The framework thus provides the LLM with a summary and an annotated training dataset from which it generates rules, which it self-verifies and retains based on frequency per entity. The framework also uses shorter prompts after finding that an increased context length can hinder the model's ability to follow instructions, worsening performance, and that too many rules can introduce noise that distracts the model. When paired with examples to guide the output structure, the GuideNER prompt outperformed other LLMs across various benchmark NERC datasets, including CoNLL2003, ACE04, ACE05, and GENIA, even beating GPT-NER (Huang et al., 2025).

The thesis intends to use GPT-NER's prompt and output structure to investigate LLM zero-shot and few-shot inference for NERC and CE performance in legal and tax domains.

Synthetic Generation of Labels

In cases of data scarcity, recent studies highlight LLMs' ability to generate quality training data. Although using this data raises valid concerns about authenticity and potential harm, performance can improve in certain cases.

Dao et al. (2025) utilized LLMs (such as LLaMa-3 and GPT-4o) with biomedical domain-specific entities and their definitions, and instructed them to generate synthetic sentences with entities already tagged. This data was then used to further pretrain a BERT-based model before being fine-tuned on gold-standard data. The results showed that synthetic data pre-training significantly improved model performance and robustness compared with traditional augmentation methods such as synonym replacement, back-translation, and cross-domain adaptation. However, once the synthetic data reached a certain size, it introduced noise and overfitting, and as a result, it was outperformed by models trained on smaller, human-annotated data, highlighting that substitution for gold data may only work to a certain extent in low-resource domains.

Additionally, to circumvent privacy constraints on medical data, Šuvalov et al. (2025) used a locally trained GPT-2 model to generate synthetic medical text, which was then annotated for entities using the more powerful GPT-3.5-Turbo and GPT-4 models. Fine-tuning a smaller downstream XLM-RoBERTa model on this data yielded an F1-score of 0.69 for a domain-specific entity that it had encountered only in synthetic data, highlighting the use of this method in data-scarce domains. Zin et al. (2024) similarly used GPT-3 to generate synthetic contract law documents with annotated entities to combat legal annotated data scarcity and found it to outperform human-annotated legal data when used to fine-tune a BERT model, contrary to the findings

by Dao et al. (2025). The authors also found that GPT-3 produced more diverse entities, covering a wider scope that contributed to the downstream generalizability of the NERC model.

These fully synthetic approaches have been applied prevalently to the biomedical domain, but also show promise in the legal domain. Such success in generating data also raises the question of how well LLMs assign labels to real data, potentially enabling fully automated annotation or providing guidance to support human annotation.

Such ideas are proposed in Wang et al. (2024), which discusses leveraging the strengths of humans and LLMs such as GPT-3.5 to ensure annotation accuracy and reliability. The authors propose a pipeline with collaborative verification. First, an LLM is used to annotate unlabeled data, instructed to provide a label and a natural-language explanation for its decision. Then, a smaller verifier model assesses and ranks the quality of the LLM's output, returning only the lowest-scoring labels to human annotators. The authors intentionally made this choice to ensure that human labor went toward potentially incorrect annotations, avoiding wasted effort on confirmed correct labels. The results demonstrated that the verifier could improve overall dataset accuracy by at least 10% while requiring humans to re-annotate only 15% of the data, suggesting that this framework is worthwhile for the labor-intensive labeling characteristic of the legal domain. Similarly, Naraki et al. (2024) proposed a hybrid annotation approach combining manual annotation with LLM predictions to improve NERC datasets. The authors found that human annotations, supplemented by LLMs to address missing labels, helped reduce noise.

These studies demonstrate that LLM-generated data and labels can help mitigate issues to some extent. While they may be useful in scenarios with little to no data, their utility still does not match that of human-annotated data. However, this does not mean they cannot help alleviate the labor involved in labeling by human annotators.

Inspired by these studies, the thesis also intends to use LLMs to synthetically generate NERC and concept labels for unlabeled target-domain tax case law data to investigate their viability for zero- and low-annotation-resource cases. All in all, the thesis aims to utilize the approaches to NERC and CE as mentioned above, intending to demonstrate how varying amounts of labeled training examples impact task performance, seen by gazetteer lookups, out-of-the-box approaches, zero-shot and few-shot approaches, and fine-tuning across data regimes.

2.4 Domain Adaptation and NERC and CE in Low-Resource Domains

Applying NLP tasks to specific domains raises two issues: less available training data (Zhang et al., 2021) and reduced generalizability due to domain shift, defined as the distributional shift between the source and target domains (Zhou et al., 2022). Together, these lead to data scarcity: unless the target domain is mainstream and likely rich in labeled training data, the available training data is rarely a perfect match to the target domain and often does not include all the target labels (Liu et al., 2021; Zhang et al., 2021). Here, domain matching is the ideal solution to domain shift, and refers to the extent to which the pre-training, fine-tuning, and target domains align in terms of shared vocabulary, sentence structure, and label distribution. Closer alignment typically entails more effective task knowledge transfer to the target domain, as seen in the study by (Au et al., 2022). As a result, many methods cannot reuse datasets and

instead try to exploit unlabeled target corpora or small target samples instead of full re-annotation (Liu et al., 2021).

A way to approach this issue is with domain adaptation, a subfield of transfer learning in which a model that is trained on data from one domain (source domain) is adapted to perform well on data from another domain (target domain) (Farahani et al., 2021). The goal is usually to improve the performance on a desired task of a target model with insufficient or lack of annotated data by leveraging the knowledge from another related domain that has sufficient labeled data (Farahani et al., 2021). However, when employing this method for NERC, domain matching between the source and target domains remains an important factor. Kulkarni et al. (2016) note that NERC performance degrades across domains due to language conventions, differences in word usage and NE types being represented differently across domains. This issue also applies to CE, as concepts are more domain-bound than NEs. Additionally, domain-specific fine-tuned BERT models perform worse when applied to out-of-domain tasks as they may not capture the new domain's context and vocabulary as effectively, as noted by Keloth et al. (2024), a pattern also observed in the legal domain by Au et al. (2022), and discussed further in Section 2.5. Furthermore, the disconnect between the source and target domain in NERC can also be attributed to differences in word distributions and sentence patterns between the source and target domain (Zhang et al., 2021). So although domain-adaptation can take place in low-resource target domains by using general-domain source data, the performance is likely poor due to the lack of transferability, highlighting the importance of domain matching. In this way, in data-scarce domains, it may be necessary to utilize and combine smaller related-domain task-data if there are any overlaps in objectives. Keloth et al. (2024), however, argue that this results in inconsistent annotations with different entity types and annotation guidelines that likely affect the quality of the dataset and, in turn, the model's performance.

Low-resource domain adaptation also encompasses many other methods: zero-resource transfer (Liu et al., 2020), unsupervised adaptation (Jia et al., 2019), and data augmentation (Chen et al., 2024). A more prominent approach is domain-adaptive pre-training (DAPT), which continues language model pre-training on target-domain text (Chalkidis et al., 2020) before fine-tuning it for a specific task. In this procedure, models use a domain-specific corpus to continue masked language modeling, which shifts the model's general pre-training knowledge toward domain-specific terms and, in turn, yields better representations of the target domain. Low-resource domains with ample unlabeled target data can possibly employ this if their labeled data is held out for testing, or use existing domain-adapted models if their target domain is sufficiently researched or similar to one that is.

Liu et al. (2021) also investigated the effect of labeled fine-tuning data domains on cross-domain NERC performance, independent of DAPT. The authors used BERT-base to compare three approaches to separate general source-domain and target-domain labeled data: by fine-tuning on the target domain only, by sequentially fine-tuning on the source-domain first and then on the target domain, and by jointly fine-tuning on both the source and target domains. They found that sequentially fine-tuning consistently outperformed the other conditions, especially in lower-resource scenarios when target samples were limited. This performance degradation was attributed to the model's inability to focus on the target domain when fine-tuned on both domains simultaneously, whereas sequential fine-tuning allowed the model to first access general NERC task knowledge from the source domain before it specialized to the target domain. Fine-

tuning directly on the target data resulted in poor performance because the authors had a relatively small labeled sample, which was likely insufficient for extensive knowledge to transfer.

Although DAPT and fine-tuning a model can positively impact the performance for NERC, they are ultimately limited by domain matching between the target and source domain datasets. The former approach assumes the availability of unlabeled data, whereas the latter requires some amount of task-specific labeled data, both of which may prove to be challenges for low-resource domains such as the legal and tax domain. Furthermore, CE has not been extensively explored in the legal and tax domains. As a result, this thesis aims to perform NERC and CE in the low-resource condition described, to explore methods in how missing target-domain signals can be supplemented in the legal and tax domains.

2.5 The Legal and Tax domains

Specialized domain-task applications pose problems due to reduced availability of target-domain data (Zhang et al., 2021) and reduced generalizability due to domain shift (Zhou et al., 2022). As the data used for this thesis belong to both the legal and tax domains, they are discussed in general terms first, before being discussed in the NERC and CE contexts.

The legal and tax domains exhibit specific linguistic properties that make NERC and CE difficult. Both domains encompass texts that often contain specialized jargon and multilingual variations, which are attributed to document origin- and jurisdiction-dependent semantics, and can be difficult for generic language models without domain-specific fine-tuning to pick up (Karamitsos et al., 2025; Ariai et al., 2025; Hechtner et al., 2025). Additionally, the text also contains complex syntactical structures such as 'embedded clauses, conjunctions, and prepositional phrases' that compose long sentences that are difficult for models to parse and interpret properly without the right fine-tuning (Hechtner et al., 2025). Furthermore, sentences belonging to these domains often derive meaning from broader contexts, either from preceding sentences or related documents, which means that entities or concepts that rely on inter-textual reference for meaning may be difficult for generic models to capture (Hechtner et al., 2025). Legal and tax documents are digitized through full-text extraction, which often also extracts embedded tables and column layouts that can disrupt the text order and impair sequential task performance if done improperly (Hechtner et al., 2025).

Despite these challenges, Legal NERC has a substantial research history, while CE in the legal and tax domain remains sparse. Earlier studies used grammar-rule-based approaches alone or in combination with statistical approaches, such as CRFs, to leverage the rigid structure and patterns found in legal documents and to extract concepts, NEs, and their relations in Dutch and Luxembourgian legal documents (Schäfer, 2012; Andrew, 2018).

As NERC approaches shifted towards data driven models, the legal domain's lack of pre-existing labeled datasets meant that the authors across multiple studies had to create their own dataset (Leitner et al., 2019; Au et al., 2022; Kalamkar et al., 2022). This highlights a deeper issue than data scarcity alone, which is that domain-specific properties mentioned also require adapted modeling approaches. Leitner et al. (2019) addresses the scarcity of German legal data by creating and manually annotating a corpus of court decisions with 19 fine-grained semantic classes, some of which were found

to be important for legal reasoning but rare in general corpora, illustrating that generic NERC data may have some overlap, but fails to provide entity coverage in the legal domain. Au et al. (2022) also investigated the impact of general-domain fine-tuning on legal NERC by fine-tuning BERT on a manually annotated legal NERC dataset or the CoNLL-2003 NERC dataset before testing the model across domains. The authors found that across different architectures, all models performed better under domain-matching conditions, reinforcing that domain-matching fine-tuning data has a greater influence on performance than model architecture. Lastly, Kalamkar et al. (2022) created a broad-domain Indian judicial NERC dataset that also included tax-related judgments. They found that the RoBERTa model, combined with a transition-based parser, significantly outperformed the default RoBERTa model, which was also fine-tuned on the same dataset. They also implemented post-processing rules to capture document-level context to link legal references back to their full precedents, reflecting the inter-textual reference challenge discussed above.

These adaptations stress how domain-specific properties must be handled, as legal NERC, case outcome predictions (Ariai et al., 2025), and tax risk assessment all depend on well-annotated data, and without good-quality annotations, the models fine-tuned for such tasks may fail to generalize and produce misleading outputs used for real-world uses (Hechtner et al., 2025; Koenecke et al., 2025). Separately, limited labeled datasets can also increase the risk of overfitting and bias, as models will only reflect patterns they are exposed to, leading to unreliable or skewed results (Hechtner et al., 2025). Addressing this combination of scarce annotated data motivates domain-specific pre-training as a strategy, and also possibly using synthetic labels as mentioned earlier in Section 2.3.2.

One way to address this is through domain-specific pre-training, as general-purpose NERC systems struggle with 'nuanced legal expressions, multi-word entities, and rare domain terms' that are prevalent and key in legal texts (Karamitsos et al., 2025). To demonstrate this, Chalkidis et al. (2020) compiled a legal domain-specific dataset that they used to pre-train a BERT model from scratch, known as Legal-BERT-SC, and to continue pre-training (DAPT) a generic BERT model, known as Legal-BERT-FP. Both variants substantially outperformed base BERT on CONTRACTS-NER, a NERC dataset of $\sim$2000 legal contracts, confirming that generic pre-training is insufficient to handle the specialized language of legal text. (Singh et al., 2026) conducted a further comparative analysis of legal domain-specific models and found that Legal-BERT-SC remained better performing than other legal models and generalist models on legal contract classification. The authors also found that other legal models pre-trained on larger, more varied legal corpora likely performed worse on the target domain, since including data from other legal domains weakened the signals they had for handling the target domain appropriately.

More recently, Hussain and Thomas (2024) also applied various LLMs to perform zero-shot NERC on the aforementioned dataset from Kalamkar et al. (2022), providing a modern point of comparison to the frequently employed encoder approach described earlier.

Tax is a relatively understudied domain compared to the legal domain. As of recently, to the best of our knowledge, there has been little work published for NERC in the tax domain. Steinigen et al. (2023) investigated German tax CE by combining manually annotated datasets with domain-specific annotation schemes and then used them to train and evaluate transformer models for entity and relation extraction in

tax-legal texts. There is also no empirically established SOTA model for tax NERC; the only published model for tax-domain applications is TaxBERT (Hechtner et al., 2025), which applies the DAPT approach introduced earlier. It is a domain-adapted DistilRoBERTa model with continued pre-training on a large tax-text corpus, fine-tuned for tax-text classification and tax-risk sentiment assessment.

To this end, this thesis draws on the datasets in Au et al. (2022) and Kalamkar et al. (2022) as an approximate-domain dataset for the target legal-tax domain. Furthermore, Legal-BERT-SC is employed to demonstrate the impact of in-domain pre-training, alongside other approaches requiring varying quantities of training data, such as out-of-the-box model NERC models, gazetteers, and LLM zero-shot and few-shot. This variety is selected to provide the IBFD with an informed basis for deciding how to improve its current approach to identifying and classifying NEs and concepts. Additionally, while this chapter treated the legal and tax domain separately to reflect their differing research coverage, the remainder of this thesis addresses their intersection as the legal-tax domain.

Chapter 3

Methodology

This chapter describes the methodological decisions made to investigate the main research question: *"What is the best approach to identifying and classifying named entities and concepts in the legal-tax domain?"* Section 3.1 first discusses the specific use-case and its target domain that the thesis is based on. Section 3.2 provides an explanation of NERC and CE in the context of the target domain. Section 3.3 discusses the datasets used in this study, starting with the company test set, followed by approximate legal domain datasets, LLM-annotated data, and the process of combining both for a mixed dataset. Section 3.4 outlines the methods used: starting with the company baselines, followed by gazetteers, out-of-the-box approaches such as LLM zero-shot and few-shot prompting, and selected encoder models to fine-tune. Section 3.5 covers the evaluation metrics that are used to measure performance.

All the code used for this thesis can be found on this GitHub Repository[1].

3.1 Use-case

The use-case that motivates this thesis is presented by the IBFD. The IBFD intends to extract and categorize NEs and concepts from each document in its tax case law corpus and use them to enhance search query results. The NEs and concepts are expected to provide an overview of the content discussed in a document that would otherwise require manual examination or string-matching calculations, since their case law documents have short, laconic titles. As a result, the company aims to adopt an approach that reliably annotates its documents for NEs and concepts, preferably fully automated, as legal documents are laborious to annotate manually and require domain expertise. The current baseline is described in detail in Section 3.4.1 and is used to automatically produce NEs and concept annotations, which are then human-validated. Figure 3.1 displays what such annotations look like.

[1] https://github.com/manyawalavalkar/tax-case-law-NERC-CE

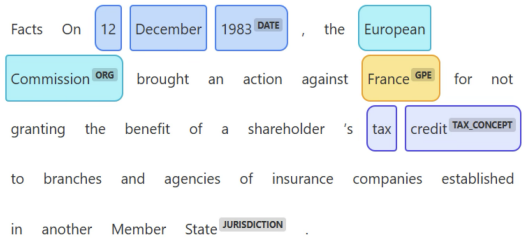


Figure 3.1: Automated and validated annotations applied to a tax case law document by the IBFD's annotators

3.1.1 Defining the domain

The target data for this use case is tax case law, which is situated within the broader legal domain. The documents to conduct NERC and CE on are case law, which, as defined by an IBFD tax specialist, is a '*category of legal domain documents constituting a body of law made up of the interpretation and application of the law as established by judges and courts*'. Case law can encompass topics such as administrative and public law and, as in this use case, tax law. Tax case law generally consists of judicial decisions interpreting tax statutes, regulations, tax-related court cases, and treaties. The content of these documents centers on tax, but their linguistic structure and features are also characteristic of the legal domain. Therefore, the target domain is considered to be the legal-tax domain.

3.2 Task definitions in the legal-tax domain

As such, the domain-specific application of NERC and CE entails that, rather than standard entities, specialized terms from industry jargon are targeted for identification. In this particular use-case, the extraction problem in the legal-tax domain are approached as two complementary tasks: NERC and CE, because the textual phenomena to be captured differ in behavior and demand respective modeling assumptions.

Table 3.1 below demonstrates the use-case test set's label distribution. It contains generic named entities such as `PERSON`, `ORG`, `GPE`, and `DATE`; domain-specific target named entities such as `COURT`, `LAW`, and `PROVISION`; and domain-specific target concepts such as `JURISDICTION`, `TAX_TYPE`, and `TAX_CONCEPT`. The most visible distinction is that the named entities each have explicit referential characteristics that categorize them, whereas the concepts do not, relying instead on the overarching category to which they belong to for classification.

The concepts are also not uniform in surface form. Some concepts can have many diverse surface forms, as different concept phrases can still belong to the same overarching category, for example, 'permanent establishment' and 'exemption' are still tax terminology despite their differences in explicit definition. On the other hand, some concepts are more morphologically regular: `TAX_TYPE` instances contain the word 'tax' in many instances, `JURISDICTION` also similarly follows a pattern of consistency and includes the word 'state' in its instances.

As a result in this thesis, NERC is defined as the task of identifying and classifying spans that refer to discrete, real-world entities such as `PERSON`, `ORG`, `GPE`, `DATE`, `COURT`,

Category	Label	Description	Examples
Generic NEs	PERSON	People, judges, advocates	"Advocate General Kokott", "Mr. Bosal"
	ORG	Companies, institutions (non-court)	"Bosal Holding BV", "European Commission"
	GPE	Countries, cities, states	"Netherlands", "Belgium"
	DATE	Dates and periods	"18 September 2003", "tax year 2019"
Target NEs	COURT	Courts and tribunals	"Court of Justice", "referring court"
	LAW	Named laws, treaties and directives	"TFEU", "Parent-Subsidiary Directive"
	PROVISION	Specific articles and sections	"Article-49 TFEU", "Section 3(2)"
Target Concepts	JURISDICTION	Abstract jurisdictional references	"Member State", "residence state"
	TAX_TYPE	Specific types of taxes	"corporation tax", "income tax"
	TAX_CONCEPT	Tax terminology (umbrella terms)	"permanent establishment", "exemption"

Table 3.1: List of generic, target named entities and concepts in the use-case test set

`LAW`, and `PROVISION`. CE is defined as the task of identifying and classifying spans that denote domain-specific concepts such as `JURISDICTION`, `TAX_TYPE`, and `TAX_CONCEPT`.

The central motivation for a two-task design is based on the distinction between named entities and concepts as classes. The key similarity between the two that has led CE to be treated as NERC in past work is methodological: both tasks are addressed with the same span-identification and sequence-tagging architectures (Yang et al., 2020).

However, past this similarity, the two diverge in their mode of reference. Named entities are referential and extensional, referring to a particular instance or object in the world, whose identity is fixed by that reference (e.g: `PERSON`: 'John Smith", `COURT`: 'European Court of Justice") (Goyal et al., 2018). Concepts are intensional, which means their content can be defined as the 'combination of the distinctive features of which a superordinate concept comprises', often determined by its place in a domain-specific context (Temmerman, 2000). This means that its identity is based more on its semantic grouping rather than its referent. For example, `TAX_CONCEPT` differs from `TAX_TYPE` as its key disambiguating characteristic is that it covers general tax terminology (i.e., 'Basic rate', 'Thin capitalization'), whereas the latter covers mentions of general tax types (i.e., 'corporation tax', 'withholding tax'). Although `TAX_TYPE` may resemble an NE due to covering specific mentions of taxes, they are not tightly restricted to a real-world

referent and instead are general references.

The two are different kinds of linguistic reference, which is why a method that works for one may not transfer completely to the other. Although both can be addressed with semantic modeling, CE requires greater semantic depth to extract nominal phrases and map their attributes to a superordinate concept, whereas NERC relies more on surface-level features to identify the prototypical category to which it belongs (Temmerman, 2000; Yang et al., 2020; Keraghel et al., 2024).

Although the company posed the problem as NERC and both tasks are presented as sequence labeling, collapsing them into a single NERC schema overlooks the linguistic differences between NEs and concepts, which can mask the challenges models face when predicting. In this thesis, because the aim is to find the best-performing approach across all target labels, we treat CE as an NERC schema. However, we evaluate concept label and named entity label predictions separately to account for their differences and the distinct challenges they pose for the model.

3.3 Datasets

This section first describes the test set, then explains how we created training and validation splits from available approximate legal domain data and LLM-annotated target domain data, and how we combined them into mixed training and validation splits.

Domain-specific applications, as in this thesis, typically face a shortage of labeled training data, especially when the domain is specialized. In such cases, domain adaptation is used, where models are trained on a source domain similar to the target domain and transfer knowledge to perform well in an unfamiliar domain close to what they have learned (Farahani et al., 2021).

Existing datasets can help address this scarcity of gold-annotated data for NERC and CE in the legal-tax domain; however, they often come from broader domains, raising the question of how well can the domains transfer. The available approximate domain datasets found and selected for this task are the IndianLegalNER by Kalamkar et al. (2022), which comprises Indian general case-law documents (including tax case-law), and E-NER by Au et al. (2022), which contains general legal company filings from the US Securities and Exchange Commission's EDGAR dataset. The datasets are from an approximate domain, are human-annotated, and share some labels in common with the test set. They were selected to represent the typical solution adopted in the past when dealing with target domain training data scarcity, and to demonstrate to what extent approximate domain data can be helpful for a domain-specific application.

Synthetically generated datasets and labels are a more contemporary solution to training data shortages, with many authors proposing prompts to harness the full generative potential of LLMs. Therefore, a second approach to producing target domain training data is to exploit the company's corpus of unlabeled tax case law documents and use LLMs as annotators. This makes it possible to use target domain training data, with the trade-off being lower annotation quality. To examine the capabilities of LLMs as annotators, both zero-shot and few-shot prompting are used to address scenarios where little to no gold-annotated data is available.

Although Keloth et al. (2024) states that combining NERC datasets can negatively impact performance, we carried out this step to demonstrate data scarcity in the legal-tax domain, which requires multiple datasets to provide sufficient coverage of

target NEs and concepts. Furthermore, it presents an opportunity to combine legal domain datasets that approximate the target domain, rather than completely unrelated or generic domain datasets, which are expected to be more detrimental to model performance due to their decreased generalizability in a specific domain.

Both the approximate domain and target domain datasets are used to fine-tune encoder models and aim to show how domain matching improves performance. Therefore, using both together as a mixed dataset could be insightful, as it would help cases where data scarcity can be mitigated through synthetic label augmentation alongside existing human-annotated data.

To fine-tune encoder models with these datasets and ensure fair evaluation, we either retained data splits from the original source, where available (for IndianLegalNER), or created a 90:10 fine-tuning validation split where needed (for E-NER, LAD, and Mixed data).

3.3.1 Test set

The dataset used for evaluation in this thesis is provided by the IBFD. The IBFD specializes in cross-border taxation documentation and thus has a wealth of tax case law that it intends to extract entities and concepts from to enhance its search queries. Table 3.2 below demonstrates the distribution of labels and concept instances in the test dataset.

Table 3.2: Token-level NE and concept label distribution in the IBFD test set, split by annotation quality (left), and corpus-level descriptive statistics (right). *Silver*: pipeline-annotated and verified by two annotators; *Gold*: fully manual single-annotator subsets

Label	**Silver**	**Gold**	**Final**	**%**
B-COURT	44	62	106	6.28
B-DATE	55	77	132	7.82
B-GPE	189	110	299	17.71
B-JURISDICTION	125	28	153	9.06
B-LAW	29	36	65	3.85
B-ORG	119	102	221	13.09
B-PERSON	110	76	186	11.02
B-PROVISION	65	22	87	5.15
B-TAX_CONCEPT	210	143	353	20.91
B-TAX_TYPE	44	42	86	5.09
TOTAL NEs	990	698	1,688	100

Statistic	**Value**
Total sentences	656
Avg token length (chars)	4.71
Avg sentence length (tokens)	34.65
Avg NE length (tokens)	2.17

The test dataset comprised tax case law documents from varying collections. This array of documents covered English tax-related case law, ranging from international taxation to region-specific taxation, including decisions from the European Court of Justice and the US Internal Revenue Service. For the complete breakdown of collections used to create the test set, see Appendix A. These documents, as typical of the legal and tax domains, contain lengthy, syntactically complex sentences that use specialized jargon. Additionally, since these documents are made from scans, they may also convert figures such as tables and dividers into text, adding noise into the data. The current test set is found to have these as well.

Compared with benchmark NERC datasets such as CoNLL2003 and OntoNotes-5.0,

which contain 3,684 and 8,262 test sentences, respectively, the test set is considered small, consisting of only 656 sentences (Chen et al., 2019). Furthermore, two target NE labels (`LAW`, `PROVISION`) and one concept label (`TAX_TYPE`) have fewer than 100 instances, which compromises their evaluation. The generic NE labels dominate the distribution more than the target labels, seen mainly by `GPE` and `ORG`. The expanded version of this table can be seen in Appendix A.2, which also shows that all the labels are multi-token entities, some more than others, `DATE`, `PROVISION`, `ORG`, and `TAX_CONCEPT`, as compared to `GPE`, `LAW`, `COURT`, and `TAX_TYPE`, due to naturally occurring differences in surface forms.

The data was annotated using the company's current automated pipeline and then manually verified by human annotators, who added any missing labels or concepts. We were provided the test set data in subsets by the company, which we had to minimally clean so as not to disrupt the documents' content or original sentence boundaries. The first subset was pre-annotated using the automated pipeline and then manually verified by three annotators (A, B, C), serving as a silver set. Due to inconsistencies in the pre-annotated labels and the labor constraints typical of annotating legal-tax data, the remaining three individual subsets were manually labeled by a single annotator: two by B and one by C. These three subsets are considered gold. The gold and silver subsets were then combined to create the test set. Annotation reliability varies because the silver subset had three annotators, whereas the gold subset had two.

The Inter-Annotator Agreement (IAA) for annotators B and C is shown in Table 3.3 below. The IAA was calculated on the silver subset, which contains 5 sentences totaling 992 tokens; therefore, the reported values should be interpreted with caution, as they may not fully represent agreement across the entire dataset. As NERC annotations depend on identifying entity spans and assigning entity types, the agreement was assessed separately for these two components.

Overall, we measured agreement using the NERVALUATE strict matching criterion, which requires both entity boundaries and types to match; it measures agreement on complete entity annotation. Under this condition, the annotators achieved a precision of 0.69, recall of 0.63, and F1-score of 0.66. To isolate the agreement in entity-type assignment, Cohen's κ was calculated for the entities whose spans matched exactly between the annotators. The annotators shared 74 mutually identified entity spans, of which only four received different entity type labels, resulting in a Cohen's κ of 0.916, indicating very high agreement on entity types conditional on the annotators identifying the same span. Within this subset, it can be implied that disagreement was more strongly associated with span identification than with label disagreement.

Table 3.3: Inter-annotator agreement on the silver subset: Annotator B vs. Annotator C (5 sentences, 992 tokens)

Component	**Measure**	**Score**
Overall NER agreement	NERVALUATE strict F1	0.66
Entity-type agreement	Cohen's κ (shared spans)	0.916

3.3.2 Existing Approximate Legal Domain Data

In this section, the mentioned existing approximate domain legal datasets are introduced, discussing their origin, contents, purpose for creation, and use in the method-

ological framework of this thesis.

InLegal-NER

Kalamkar et al. (2022) proposed the original `InLegal-NER` dataset[2], containing annotations, metadata, IDs per text, and the raw text of roughly 11,000 rows for training and 1,070 for validation. The text for this dataset was retrieved from an Indian search engine for Indian law documents with search queries made for 8 distinct case types (Tax, Criminal, Civil, Motor Vehicles, Land & Property, Industrial & Labour, Constitution, and Financial) and 29 varying courts. These documents were selected after careful planning to control case type, ensuring English was the only language used.

The dataset consists of Indian court judgments and their preambles, annotated with 14 legal named entities [`COURT`, `PETITIONER`, `JUDGE`, `LAWYER`, `DATE`, `ORG`, `GPE`, `STATUTE`, `PROVISION`, `PRECEDENT`, `CASE_NUMBER`, `WITNESS`, `OTHER_PERSON`]. The annotation process was cyclical, with pre-annotation of the data in each cycle to reduce annotation effort. First, the data was processed using SpaCy's NER pipeline, with custom labels added because the legal entities exhibited consistent patterns. After these automated labels were reviewed and corrected, they were used to train a modified RoBERTa model, which then provides predictions on the following dataset, which is then manually corrected before being added back to the model's training pool. This process was repeated four times, iteratively adding back to the training set to obtain the final dataset, which was then split into training, development, and test sets (Kalamkar et al., 2022).

We selected this dataset for this use case because its domain (general case law or legal) approximates the target domain (tax case law or legal-tax), making it closer to the target domain distribution than general-domain data. Furthermore, it was selected as legal domain NERC data in English with labels that overlap with the target NEs, and concepts are not prevalent. The text in this dataset comprises case law, which partially alleviates the issue of domain mismatch with the test set, which contains tax case law. This dataset contains diverse forms of case law documents, differing in structure according to state-specific jurisdictional requirements, which would also help the model learn entity variations. Although the dataset does not include tax NE labels, the included tax case law documents are still helpful, as the model can learn about NEs and text structures in that context.

We decided to use a modified version of the `InLegalNER` dataset (Kalamkar et al., 2022). The modified `Indian_Legal_NER_dataset` (Mukund, 2024) was accessed via Hugging Face[3]. This dataset is a cleaned replica of the original `InLegalNER` dataset, removing irrelevant metadata and retaining only sentences extracted from Indian case law and preamble texts, along with its dictionaries containing character-level entity spans and label information. The data was converted to CoNLL by using SpaCy to tokenize the text and aligning the character-level entity spans with the resulting tokens for IOB2 labeling. The text's original case was retained, as they convey important signals for the models.

A limitation to keep in mind is that the dataset offers broad case-law domain coverage (criminal, civil, tax, labor, etc.), which means the entity mentions per label may be broader than the typical legal and tax NEs and concept labels in the test set, and can possibly indicate challenges to pick up target domain specific representations. However,

[2]https://huggingface.co/datasets/opennyaiorg/InLegalNER
[3]https://huggingface.co/datasets/AjayMukundS/Indian_Legal_NER_Dataset

this applies to the generic NE labels as well, as greater diversity among them is better for the model (DATE: '19.8.1988" vs '28 March 1959", or ORG: 'Hong Kong Bank" vs 'Saurabh Medical Hall").

Since the dataset was created to address the broader legal domain entities, its labels had to be mapped to fit those present in the test set. Table 3.4 demonstrates the pre-mapping entity composition presented in the Indian_Legal_NER_dataset dataset.

Table 3.4: Label mapping scheme for the Indian_Legal_NER_dataset dataset. Counts are original (pre-mapping) B- token frequencies

Original	Mapped	Train	Val
Court	Court	2,367	282
Date	Date	1,885	216
Gpe	Gpe	1,398	178
Org	Org	1,441	155
Provision	Provision	2,384	257
Statute	Law	1,804	221
Judge	Person	2,325	154
Lawyer		3,505	506
Witness		881	58
Other_Person		2,653	263
Case_Number	O	1,040	118
Precedent		1,351	173
Respondent		3,862	266
Petitioner		3,068	181

From the listed entities, certain labels were retained as they were common with the test set, or mapped due to label-definition overlap. In the first case, the target NE labels COURT and PROVISION were retained exactly. The NE label STATUTE was mapped to LAW since the two share similar label definitions. Since the Indian_Legal_NER_dataset also has multiple granular labels that can be combined into one, JUDGE, LAWYER, WITNESS, and OTHER_PERSON were mapped to PERSON. While there were other entities like PETITIONER and RESPONDENT, they contained both PERSON and ORG instances and for convenience were mapped to 'O'. It is kept in consideration that marking potential PERSON and ORG instances as 'O' is likely to cause confusion for the model. Similarly, labels irrelevant to this use-case were also changed to 'O', like PRECEDENT and CASE_NUMBER. The remaining default labels, such as DATE, ORG, and GPE, were retained as they were. With these updated labels, the dataset consisted of 10,995 sentences of text taken from varying documents. A similar process was applied to the validation split made available by the authors, resulting in 1074 sentences. Table 3.5 below demonstrates the final mapped label distributions in the training and validation set.

Table 3.5: `Indian_Legal_NER_dataset`: token label distributions (left) and dataset statistics (right)

Label	Train	%	Val	%
B-COURT	2,367	11.47	282	12.31
B-DATE	1,885	9.13	216	9.43
B-GPE	1,398	6.77	178	7.77
B-LAW	1,804	8.74	221	9.65
B-ORG	1,441	6.98	155	6.77
B-PERSON	9,364	45.36	981	42.84
B-PROVISION	2,384	11.55	257	11.22
TOTAL NEs	20,643	100	2,290	100

Statistic	Train	Val
Total sentences	10,995	1074
Avg token length (chars)	4.10	4.20
Avg sent length (tokens)	55.27	57.37
Avg NE length (tokens)	2.6954	2.6039

E-NER

The `E-NER` dataset was created to demonstrate the benefits of approximate-domain fine-tuning, as using general English-domain data degrades NER performance on legal text (Au et al., 2022). It was accessed through the author's GitHub Repository[4]. This dataset consisted of legal company filings taken from the US Securities and Exchange Commission's Electronic Data Gathering, Analysis, and Retrieval (EDGAR) dataset. Pre-processing 52 raw documents left 11,644 sentences for annotation. These were ultimately manually annotated by one of the paper's authors due to significant discrepancies found in the labels produced by crowd workers. This was attributed to the difficulty of annotating legal data for non-experts (Au et al., 2022). Unfortunately, because the dataset was labeled by a single person, it is not possible to calculate the IAA or gauge annotation quality. The dataset is available as a single large set, with no splits.

The dataset offered two sets of labels, seen in Table 3.6, one following the CoNLL 2003 shared task entities, which were `LOCATION`, `PERSON`, `ORGANIZATION`, and `MISCELLANEOUS`, and one that retained `LOCATION` and `PERSON` as is, but fine-grained the `ORGANIZATION` labels into `BUSINESS`, `GOVERNMENT`, and `COURT`, and `MISCELLANEOUS` into `LEGISLATION/ACT` and `MISCELLANEOUS`.

Following the authors' motivation, we selected this dataset to investigate the benefits of approximate domain training. Compared to the `Indian_Legal_NER_dataset`, which covers tax case law in its general case law corpus, this dataset is broader but still expected to provide models with coverage of the generic legal domain entities. Furthermore, it offers two target NE labels (`COURT` and `LEGISLATION/ACT`) in common with the test set, as well as legally oriented generic NE instances that can also be informative for broader model coverage. In Table 3.6 below, the original and mapped labels are displayed alongside their respective counts.

Although the E-NER dataset followed a CoNLL-style token-label structure, its original file used comma-separated formatting and followed the IOB1 structure. The IOB1 annotations were converted to IOB2 to ensure consistency with the annotation scheme used in the other dataset. The E-NER dataset's text was minimally reformatted and cleaned for conversion and was mostly maintained as is as it did not undergo tokenization or case normalization. Fine-grained labels were mapped to the target label set where possible. From the original dataset, the default labels `LOCATION` and `PERSON` were

[4]https://github.com/terenceau1/E-NER-Dataset

Table 3.6: E-NER label sets (left) and the mapping scheme applied to reach the target labels (right). Counts are post-mapping and post-conversion (IOB1 to IOB2) B- token frequencies

CoNLL 2003 Set	Fine-grained Set
Location	Location
Person	Person
Organization	Business
	Government
	Court
Miscellaneous	Legislation/Act
	Miscellaneous

Original	Mapped	Count
Court	Court	30
Person	Person	635
Location	Gpe	1,027
Legislation/Act	Law	1,171
Business	Org	3,779
Government		580
Miscellaneous	O	1,599

retained, and LOCATION was mapped to GPE. The labels BUSINESS and GOVERNMENT were mapped to ORG, and COURT was retained as is. Finally, LEGISLATION/ACT was mapped to LAW, and MISCELLANEOUS was mapped to 'O'. The resulting dataset contained a total of 11,644 sentences. Since the original dataset was only available as a single large dataset, it was decided to create a validation split. As COURT was the minority label in this dataset, the documents were split based on whether they contained the entity, and the resulting documents were proportionally distributed when creating the validation set, yielding a 90:10 split. The resulting document sets were randomly shuffled while preserving document boundaries and written to CoNLL. This produced a training set of 10,260 sentences and a validation set of 1384 sentences. It should be noted that creating a validation split from the overall training data carries a risk of data leakage, even if the texts were split on a document level. The resulting label counts can be seen in Table 3.7 below.

Table 3.7: E-NER dataset: token label distributions (left) and dataset statistics (right)

Label	Train	%	Val	%
B-COURT	15	0.25	15	1.32
B-GPE	951	15.64	76	6.67
B-LAW	1,033	16.99	138	12.11
B-ORG	3,514	57.78	845	74.12
B-PERSON	569	9.36	66	5.79
TOTAL NEs	6,082	100	1,140	100

Statistic	Train	Val
Total docs	46	6
Total sentences	10,260	1,384
Avg token length (chars)	4.70	4.70
Avg sent length (tokens)	35.08	31.54
Avg NE length (tokens)	2.46	3.36

3.3.3 Combining approximate legal domain data

To examine how human-annotated approximate legal domain data affects a model's performance on NERC and CE through fine-tuning, we combined the selected datasets into an approximate-domain training set and validation set.

The training and validation splits from the Indian_Legal_NER_dataset and E-NER datasets were combined. The E-NER document boundaries were preserved, while the Indian_Legal_NER_dataset was treated as individual sentences as it did not have any document boundaries. Before combining each split, both datasets were randomly shuf-

fled, using a fixed seed to ensure reproducible ordering. This resulted in a single approximate domain training set and validation set with data from both selected datasets.

As both these datasets had similarly sized splits, no additional changes were made to further balance the dataset. This created a combined dataset containing approximate legal-domain NERC data, referred to as the `Approximate Legal Domain Data (ALDDD)`; Table 3.8 below shows the differences in statistics across their splits.

Table 3.8: `Approximate Legal Domain Data (ALDDD)`: token label distributions (left) and dataset statistics (right)

Label	**Train**	**%**	**Val**	**%**
B-COURT	2,382	8.91	297	8.66
B-DATE	1,885	7.05	216	6.30
B-GPE	2,349	8.79	254	7.41
B-LAW	2,837	10.62	359	10.47
B-ORG	4,955	18.54	1,000	29.15
B-PERSON	9,933	37.17	1,047	30.52
B-PROVISION	2,384	8.92	257	7.49
TOTAL NEs	26,725	100	3,430	100

Statistic	**Train**	**Val**
Total sentences	21,255	2,458
Avg token length (chars)	4.32	4.41
Avg sent length (tokens)	45.52	42.82
Avg NE length (tokens)	2.64	2.86

3.3.4 LLM-annotated data

Data augmentation and synthetic labeling used in data-scarce domains can yield positive outcomes as reported by Šuvalov et al. (2025), Dao et al. (2025), and Yazdani et al. (2026), prompting the implementation of this approach for the use-case. Using LLM-annotated data offers a valuable trade-off: the data can belong to the exact target domain, ensuring a domain match, at the cost of lower labeling quality.

To prevent overlap with the test set, a file containing metadata for all test set documents was used to ensure no overlap occurred when retrieving unlabeled tax case law documents. The test set metadata file included the Document IDs, collection name, and the countries mentioned for each document. All subsequent queries to retrieve data were made via the Elasticsearch Index. To keep the synthetic dataset as close as possible to the test set, the metadata file was first used to extract the names of the test set's document collections, yielding 11 distinct legal document collections. These collections covered tax-related case-law documents, spanning global, European, and American jurisdictions. Following this, each document retrieved per collection underwent a matching criterion: it would proceed only if it did not have a matching document ID to an existing document in the test set, and if its associated country set matched a country set for one of the test set documents. The retrieved documents were separated by sentences and tokenized with SpaCy to separate punctuation conjoined to text, while case was retained, maintaining the decision taken for the `ALDD`.

Initially, the retrieval procedure aimed to reproduce the original composition of the collections used in the test set when creating the training and validation sets; this approach significantly narrowed the document pool and, when combined with the matching criteria, yielded a much smaller dataset than expected. Thus, the retrieval target was readjusted to obtain a dataset of comparable size to the combined approximate legal domain dataset, while loosely maintaining the test set collection proportions. Document boundaries were retained throughout the process to later facilitate document-level shuffling.

Subsequently, to label the documents per collection, GPT-5.4-mini was called via the Azure OpenAI endpoint (OpenAI, 2026). Using the best-performing output format and prompt from the few-shot condition, each collection's sentences were submitted as individual queries to the model, with up to five requests processed concurrently to accommodate the API rate limit. Temperature, the parameter that controls the randomness in the model's predictions, was set to the default 0 to ensure more predictable and consistent outputs (Hu et al., 2024). We prompted the model to act as a NER classifier and assign tags from the target label set (NEs and concepts); we then converted these predictions into IOB2-formatted annotations and wrote them to CoNLL.

Finally, each synthetically labeled collection was split at document boundaries, and its documents were randomly shuffled using a fixed seed before being split into training and validation sets using a 90:10 split of 21,364 and 2267 sentences, respectively. These final splits were written to CoNLL files, and their label counts can be seen in Table 3.9 below. This dataset is referred to as the LLM-Annotated Data (LLMAD).

Table 3.9: `LLM-annotated dataset`: token label distributions (left) and dataset statistics (right)

Label	**Train**	**%**	**Val**	**%**
B-COURT	3,890	9.78	228	5.45
B-DATE	3,649	9.18	481	11.51
B-GPE	4,628	11.63	437	10.45
B-JURISDICTION	2,559	6.44	166	3.97
B-LAW	4,720	11.87	532	12.73
B-ORG	4,677	11.76	646	15.45
B-PERSON	3,564	8.96	311	7.44
B-PROVISION	3,940	9.91	428	10.24
B-TAX_CONCEPT	6,052	15.22	692	16.56
B-TAX_TYPE	2,084	5.24	259	6.20
TOTAL NEs	39,763	100.0	4,180	100.0

Statistic	**Train**	**Val**
Total sentences	21,364	2,267
Avg token length (chars)	4.37	4.34
Avg sent length (tokens)	25.22	21.50
Avg NE length (tokens)	2.51	2.62

3.3.5 Combining approximate domain legal data and LLM-annotated target domain data into a Mixed dataset

To examine how `ALDD` and `LLMAD` interact when used together to fine-tune a model, the datasets were combined using their respective training and validation splits.

The training splits of `E-NER` and `Indian_Legal_NER_dataset` and `LLMAD` were combined to form a Mixed training set, and their validation splits were combined to form a Mixed validation set. `Indian_Legal_NER_dataset` sentences were treated as individual units, and `E-NER` and `LLMAD` were kept as document-level units as they had document boundaries. This choice was made as the `ALD` had a fixed order of E-NER documents first, followed by `Indian_Legal_NER_dataset` sentences. The combined training and validation units were then shuffled using a fixed random seed and written to CoNLL format. This created a mixed dataset with both human-annotated approximate domain and LLM-annotated target domain data, which is referred to as `(MIX)` from hereforth. In Table 3.10, the training and validation set split label counts for the dataset are presented.

Table 3.10: `Mixed` dataset: token label distributions (left) and dataset statistics (right)

Label	Train	%	Val	%
B-COURT	6,272	9.43	525	6.90
B-DATE	5,534	13.92	697	9.16
B-GPE	6,977	17.55	691	9.08
B-JURISDICTION	2,559	6.44	166	2.18
B-LAW	7,557	19.01	891	11.71
B-ORG	9,632	24.22	1,646	21.63
B-PERSON	13,497	33.94	1,358	17.84
B-PROVISION	6,324	15.90	685	9.00
B-TAX_CONCEPT	6,052	15.22	692	9.09
B-TAX_TYPE	2,084	5.24	259	3.40
TOTAL NEs	66,488	100.0	7,610	100.0

Statistic	Train	Val
Total sentences	42,619	4,725
Avg token length (chars)	4.34	4.38
Avg sent length (tokens)	35.35	32.59
Avg NE length (tokens)	2.56	2.72

3.4 Methods

This section discusses the methods used for NERC and CE in the target legal-tax domain, organized by the range of required training data. To start, the company baseline is first established in 3.4.1. Subsequently, gazetteers are presented in Section 3.4.2, followed by LLM zero-shot and few-shot prompting in Section 3.4.3, and finally by fine-tuning encoder models in Section 3.4.4

3.4.1 Company baseline

To establish a baseline for comparing experimental performance, the company's current pre-annotation pipeline is evaluated on the test set.

The company currently uses an automated pipeline to pre-annotate data. To start, an HTML document is parsed for text, which is processed with SpaCy's `en_core_web_sm` model for default label NEs, targeting conventional entities such as `PERSON`, `ORG`, `GPE`, `DATE`, and `LAW`. For the `COURT` label, the `ORG` labels yielded from the SpaCy model underwent Regular Expression (RegEx) filtering, assigning `COURT` to `ORG` spans containing COURT-related words (i.e, *tribunal, high court, hoge raad, bundesfinanzhof*). Similarly, the `PROVISION` label was tagged by filtering `LAW` labels that matched RegEx-filtered provision patterns. In the end, the company's *Organizations & Treaties and Courts Glossary* served as a reference for manual `COURT` tagging; a comprehensive list of recognized organizations, treaties, and court names. Finally, an independent gazetteer lookup occurs, matching surface forms corresponding to `TAX_TYPE`, `TAX_CONCEPT`, or `JURISDICTION`. The gazetteer matches are combined with SpaCy predictions, and SpaCy predictions take precedence.

To closely examine how each component in the automated pipeline affects performance, the components are separated into SpaCy, SpaCy with RegEx filtering, and SpaCy with RegEx filtering and gazetteer lookup. The company's baseline files are adapted for each of these approaches.

3.4.2 Gazetteer Approach

Gazetteers are well-suited for most domain-specific NERC and CE applications, with finite terminology and low compute, since they perform simple string matching to find

target terms. They perform well with predictable surface forms, yielding high precision (Keraghel et al., 2024; Li et al., 2020). However, the concept labels in this use-case are not uniformly abstract: `TAX_TYPE` and `JURISDICTION` have relatively consistent surface forms, whereas `TAX_CONCEPT` is variable. A simple dictionary lookup is expected to perform better on orthographically consistent concepts than on abstract ones, resulting in high precision on known forms and lower recall for varying forms that are not recorded (Keraghel et al., 2024; Li et al., 2020).

This approach is selected specifically as it is part of the company's existing baseline for the target concept labels. Their gazetteer is built on a curated taxonomy, raising the question of how gazetteer construction can impact performance and coverage. To investigate this, a heuristics-based gazetteer was created for comparison with the company's taxonomy-based gazetteer. Both gazetteers are derived from the company's International Tax Glossary (ITG), a large XML file containing 3,471 diverse terms related to, but not limited to, the tax, legal, and financial domains.

The ITG is visited as a webpage by customers, where they can access each term's definition. The source file behind this display is organized as a nested taxonomy as a source file. These are displayed in Figure 3.2. The nested taxonomy declares which overarching general term each specific term belongs to. Each specific term in the glossary contains its name, definition, and 'see also' terms. Each specific term in the glossary is also manually assigned related terms, known as 'tax topics', each with a confidence score between 1 and 3 indicating the strength of the relationship between the term and the topic.

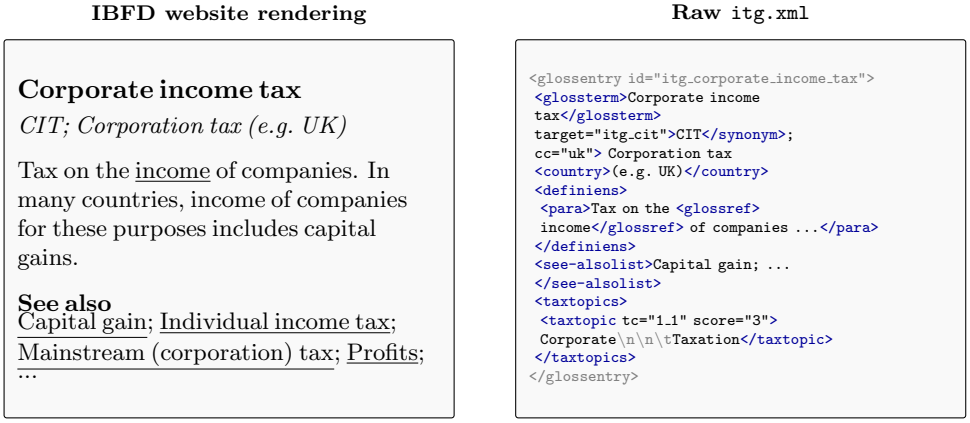


Figure 3.2: The 'Corporate income tax" entry as rendered on the IBFD website's ITG page and in its raw itg.xml glossary file.

The Company's Taxonomy-based Gazetteer

The Company Gazetteer utilizes the ITG file's taxonomy to classify terms into `TAX_TYPE` or `TAX_CONCEPT`. From the glossary file, terms with at least one attributed tax topic with a confidence score of 3 were selected to create a `TAX_TYPE` list, maximizing precision and minimizing the false positives in the downstream lookup. This list was also used to filter the glossary, retaining only the candidates for the more abstract `TAX_CONCEPT` label. The `JURISDICTION` entities were curated manually by experts. All the classified concept instances were then added to one file for collective lookup, resulting in a total of 2,102 `TAX_CONCEPT`, 869 `TAX_TYPE`, and 7 `JURISDICTION` lookup terms. The gazetteer also

contained other fine-grained TAX concept labels, whose surface forms were remapped to TAX_CONCEPT, further increasing coverage to 2,557 terms. All these term entries retained specific glossary details, most importantly, their surface form variations (derived from synonyms), which enabled broader coverage during lookups. An example entry from this gazetteer is presented in Figure 3.3.

company gazetteer entry

```
{
  [key]"term_id"; : ‘itg_corporate_income_tax",
  [key]"canonical_form"; : ‘Corporate income tax",
  [key]"surface_forms"; : ["Company income tax",
   ‘Corporate income tax"],
  [key]"entity_type"; : ‘TAX_TYPE",
  [key]"country_code"; : null,
  [key]"taxtopics"; : [],
  [key]"definition"; : ‘Tax on the income of
   companies. In many countries ...",
  [key]"see_also"; : ["itg_underlying_tax_credit",
   ‘itg_profits", ‘itg_capital_gain", ...],
}
```

Figure 3.3: The ‘Corporate income tax” entry in the company gazetteer, which includes the term_id, its canonical and surface forms, entity type, country code, assigned tax topics (null in this case), definition, see also terms and abbreviation.

The taxonomy, the topic assignments, and the full glossary were all produced by IBFD’s in-house tax specialists, grounding the gazetteer in domain-expert annotation.

Heuristics-based gazetteer

The heuristics-based gazetteer examines how well more traditional approaches, such as rule-based patterns and lexicons, perform when applied to challenging domain-specific NEs and concepts.

The steps taken to convert the glossary XML file into two TAX_CONCEPT and TAX_TYPE lists is visualized in Figure 3.4. To process the large company glossary XML file, as done with the company baseline, Claude (Anthropic, 2024) was first used to parse it into a more human-readable format, since the raw file was not suitable for direct extraction. The prompt used for doing so can be found in the Appendix B.1. The parsing produced a legible file listing all terms in the glossary.



Figure 3.4: Processing of a single glossary entry through the heuristic gazetteer pipeline, using 'Corporate income tax" as an example. The raw `itg.xml` entry is parsed into a structured record, then classified by the two-layer rule set. Here, the final token 'tax" matches a Layer 1 type headword, so the term is labeled `TAX_TYPE` and written to `tax_type_terms.txt`

To classify the glossary terms into their respective categories, heuristic rules were created for each label that focused on each glossary term's headword. Headwords dictated the main rule, as `TAX_TYPE` and `TAX_CONCEPT` have recurring patterns in their final words, which can be a useful signal for classification. These findings came from frequency statistics on the glossary terms and informed the heuristics. The resulting headword lists that comprise this gazetteer were manually curated based on these frequency patterns and their suitability for the concept they belonged to. A workflow of how the heuristics are applied is displayed in Figure 3.5.

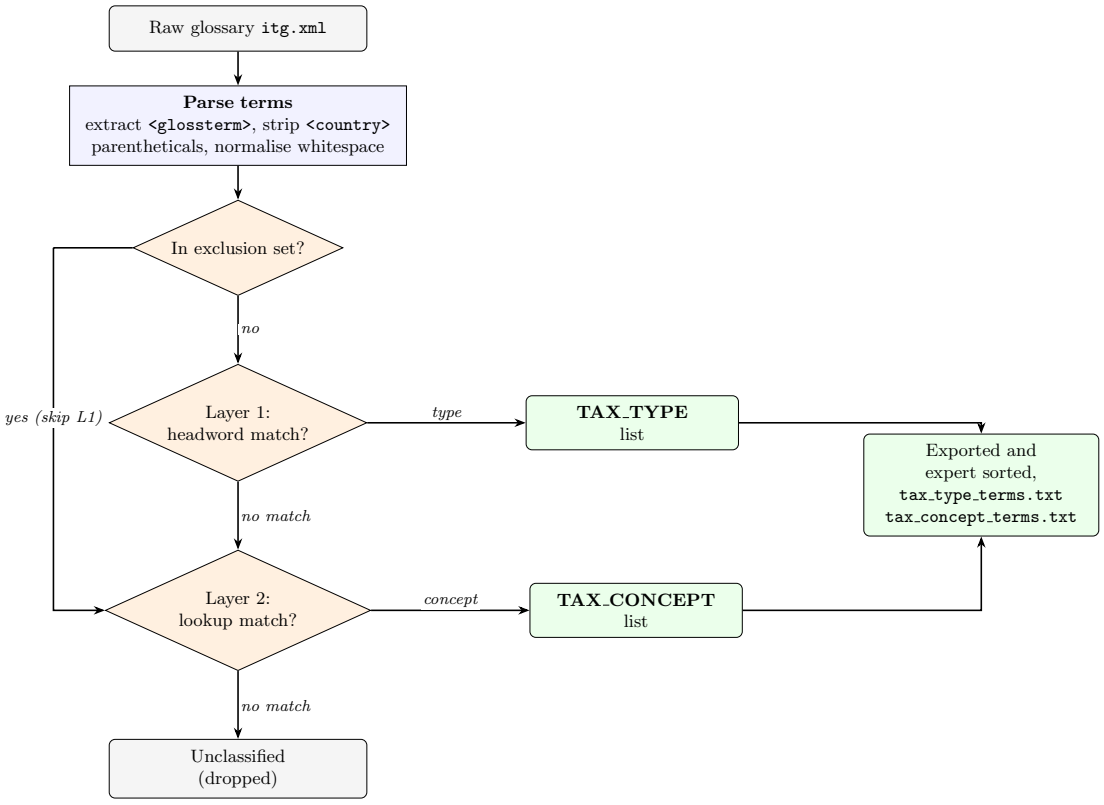


Figure 3.5: Visualization of how the two-layer heuristics-based classification occurred to create the `TAX_TYPE` and `TAX_CONCEPT` lists

Before any heuristics are applied, the terms are lower-cased, stripped, and split into words, with the last word being isolated. First, the exclusion criteria are applied, string-matching the entire term to ensure that ambiguous and non-applicable terms are excluded. These include instances such as the entire term consisting of generic words like 'tax' or 'levy', as well as others that were identified as unsuitable for automatic classification. Subsequently, the first heuristic layer is applied, matching the last word of non-excluded terms to lists of expected headwords curated separately for `TAX_TYPE` and `TAX_CONCEPT` headwords. The `TAX_TYPE` list contained recurring terms ending in 'tax', 'duty', 'levy', as well as other related terms such as 'surcharge', 'tariff', and 'rate', and the `TAX_CONCEPT` list included terms endings such as 'method', 'principle', 'rule', which match the broader morphological regularity observed for the two concepts.

A second look up layer was then applied to terms that did not receive a label through the first layer. This layer was created using a manually curated list of well-known `TAX_TYPE` and `TAX_CONCEPT` terms, including acronyms ('apa', 'gaar', 'vat', 'gst') and established terminology ('arm's length', 'pay as you earn'). This allowed the classification of terms that did not have a headword eligible for the first layer. Terms that did not match on either layer were unclassified and excluded from the gazetteer list.

Running these rules identified two candidate lists of 357 `TAX_TYPE` and 208 `TAX_CONCEPT` entities, which were further corrected and verified by a tax expert who made additions and removals where necessary. This led to a final domain-validated Heuristics-based Gazetteer, comprising 279 tax-type terms and 244 tax-concept terms.

For the `JURISDICTION` concept, the original `JURISDICTION` lookup list was manually crafted by experts and is part of the Company Gazetteer. Due to time and domain-expertise constraints, all these `JURISDICTION` entries were placed into their own file and applied separately alongside the Heuristics-based Gazetteer. A visualization of an entry is seen below in Figure 3.6

Jurisdiction-only gazetteer

```
[
 {
  "term_id"; : ‘jurisdiction_member_state",
  "canonical_form"; : ‘Member State",
  "surface_forms"; : ["Member State", ‘Member States",
    ‘EU Member State", ‘EU Member States"],
  "entity_type"; : ‘JURISDICTION",
  "country_code"; : null,
  "taxtopics"; : ["EU Tax Law"],
  "definition"; : ‘A country that is a member
    of the European Union."
 },
 {
  "term_id"; : ‘jurisdiction_residence_state",
  "canonical_form"; : ‘residence state",
  "surface_forms"; : ["residence state",
    ‘state of residence", ‘resident state", ...],
  "entity_type"; : ‘JURISDICTION",
  "taxtopics"; : ["Tax Treaties", ‘Double Taxation"],
  "definition"; : ‘The state in which a taxpayer
    is considered resident for tax purposes."
 },
 { ... source state, contracting state, host state,
   home state, third state ... }
]
```

Figure 3.6: The manually curated `JURISDICTION-only` lookup gazetteer, applied separately from the Heuristics-based Gazetteer. Each entry shares the same schema as the Company Gazetteer

To compare performances between the gazetteers, the Company Gazetteer and the Heuristics-based Gazetteer were evaluated independently on the test set. Evaluation for this approach takes place by matching gazetteer surface forms or terms against the test sentences using case-insensitive whole-word and whole-phrase string matching. Longer expressions are prioritized, and once a span has been matched, overlapping matches within the same span are discarded. The resulting matches were converted to IOB2 labels in the CoNLL format.

Since the Heuristics-based Gazetteer does not contain `JURISDICTION` entries, the `JURISDICTION`-only gazetteer (Figure 3.6) is run in combination with the Heuristics-based Gazetteer for the gazetteer experiments. To see the effect of the Heuristics-based Gazetteer on the company baseline performance, it replaced the Company Gazetteer, while retaining the SpaCy and RegEx components .

3.4.3 Zero-shot and Few-shot approaches

As established by previous work discussed in Chapter 2, the improved capabilities of LLM's to perform tasks without requiring specific training data make them a worthwhile investigation for a low-resource NER domain like legal-tax. To investigate LLM efficacy and the effect of annotated examples, zero-shot prompting and few-shot prompting are conducted to observe how general domain LLMs perform when provided with specific target-domain example predictions.

Prompt Engineering and Output Structures

Prompt engineering is one of the ways to manipulate LLM performance, using varying natural language instructions and examples in different formats to help the models make improved predictions on specific tasks Hu et al. (2024). Wang et al. (2025) designed GPT-NER, a state-of-the-art prompt approach, which has often been used as a benchmark in LLM NERC labeling studies (Naraki et al., 2024; Huang et al., 2025). Their work revealed that framing NERC as a generation task for LLMs mitigates the difficulties these models have with instructions that reflect more traditional interpretations of NERC, such as span detection and categorization. As a result, GPT-NER instructs the LLM to return the input string as is, with added markers to indicate the beginning and end of an entity, as seen in this example from the original study: INPUT: [`China says Taiwan spoils atmosphere for talks.`], OUTPUT: [`@@China## says @@Taiwan## spoils atmosphere for talks.`] Since this output format (alongside other factors) demonstrated LLM results comparable to supervised baselines across five of the most widely used NER baselines, the approach was employed in this use-case. The authors also created a data store that holds entity-, sentence-, and token-level embeddings for training and input sequences, thereby facilitating the retrieval of semantically similar examples for few-shot prompting. Due to complexity and time constraints, we use only their general prompting format and structure, with natural language modifications to match the task's domain specificity. When we employed the GPT-NER prompt, the model was instructed to return its predictions as the original input string with special-character markings and a sentence index.

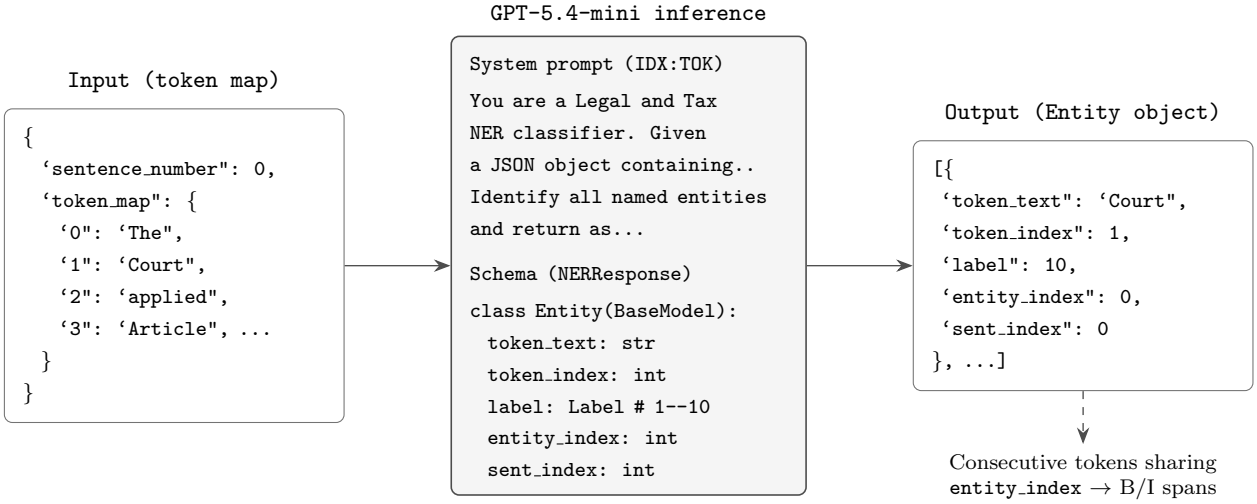


Figure 3.7: Input/output structure of the IDX:TOK LLM prompting pipeline. The model receives a sentence as a token map, is constrained by the IDX:TOK system prompt and the `NERResponse` Pydantic schema, and returns a flat list of per-token `Entity` objects. Each `label` is enumerated: `PERSON` = 1, `ORG` = 2,.... `COURT` = 10); consecutive tokens with a shared `entity_index` are reconstructed into BIO spans in post-processing

The authors also mentioned another output format that asked the LLM to return the predicted entity and its index in the sentence, but this method exhibited the same span-detection issue they were trying to prevent with GPT-NER (Wang et al., 2025). For

experimental purposes, and with guidance from a company data scientist, we formulated a similar approach called IDX:TOK. In this prompt structure, the model receives a dictionary that maps indices to each token in the input sentence. It is prompted to process the sentence for NERC and return a structured output as entity dictionaries, an embedded list of dictionaries for each detected entity. These include the token text, index, entity label, entity index, and sentence index, which together help separate sentences and group tokens into their respective entity spans for IOB2 tagging later. An example of this prompt's processing can be seen in Figure 3.7.

Although it would have been ideal to use GuideNER (Huang et al., 2025), given that it outperformed GPT-NER, its prompt generation requires a labeled training dataset for LLMs to first derive annotation guidelines from, which defeats the purpose of a zero-shot approach that is supposed to work without any training examples.

For both output formats and prompting conditions, the LLM was asked to return its predictions in a structured format, so that its predictions could be mapped directly back to the test set. The predictions were converted into IOB2 annotations and written to CoNLL files. Although recent studies on synthetic label generation for NERC using LLMs (Šuvalov et al., 2025; Wang et al., 2025) utilized GPT-3, GPT-3.5, and GPT-4-Turbo, these models are considered outdated compared to the current generation LLMs from OpenAI. This was further confirmed also by Yang et al. (2025), who observed that larger, more advanced models (GPT-4o) handle zero-shot CE scenarios in a more stable manner in comparison to older or smaller ones (GPT-3.5, GPT-4o-mini). After consulting with the company's data science team, it was therefore decided to use a current-generation model that is viable for deployment in a future pipeline. As a result, GPT-5.4-mini was used to perform all zero-shot and few-shot prompting.

Zero-shot prompting strategy for NERC and CE inference

After deciding on the output formats, the zero-shot approach was first implemented, differing only in the specific annotation instructions tailored to the desired output structure. To decide on the labeling guidelines to provide to the model, findings by Huang et al. (2025) served as a framework. The authors found that concise annotation guidelines were beneficial for in-context NERC, emphasizing that explicitly defining the list of entities through specific rules is more impactful than relying on large sets of examples, which can often result in longer prompts that crowd the model's context window without proportional gains. In light of these claims, two labeling guideline conditions were used.

For the first condition, the company's current simple manual annotation guidelines were provided to the model in their original form, with an inclusion and exclusion criterion for most labels. The guideline excerpt for `PROVISION` is provided as an example in Figure 3.8. The prompt also included boundary rules, providing the model with guidance for certain complex labels such as `TAX_TYPE` and `PROVISION`, as well as for overlapping spans, and output format rules, which provided an overview of the structured output it had to return, including the per-entity features mentioned earlier. This prompt can be found in Appendix C.1 and Appendix C.2.

In the second condition, the data scientists provided a more detailed explanation of the simple label definitions, intending to give the model additional information that might improve its predictions. An excerpt of this detailed version is shown in Figure 3.9. This expansion led to all labels now having inclusion and exclusion criteria, and to in-text mentions of viable entities being included, which were removed to keep the

```
PROVISION
Covers: specific articles, sections, paragraphs, or clauses within a law,
including the law name when it appears as part of the reference.
```

Figure 3.8: Zero-shot simple labeling guideline excerpt for the PROVISION label

prompt strictly zero-shot. Other than these extended label descriptions, the prompt remained the same. This prompt can be found in Appendix C.3 and Appendix C.4.

```
PROVISION
Covers: specific articles, sections, paragraphs, or sub-paragraphs within
a named instrument, including the host instrument when cited together.
Spans cover the full citation as a single entity --- pinpoint plus host
instrument plus subdivisions --- and coordinated pinpoints sharing one host
as one span.
Excludes: the host instrument cited alone without a pinpoint (use LAW),
case-paragraph references, and recital references unless project-scoped in.
```

Figure 3.9: Zero-shot detailed labeling guideline excerpt for the PROVISION label

Additionally, GLiNER1 (described in Section 3.4.4), is evaluated for zero-shot, as its span embedding matching mechanism allows inference without fine-tuning. This provides a state-of-the-art encoder baseline for zero-shot performance against which the LLM zero-shot performance can be compared, indicating which is better suited to out-of-the-box NERC and CE on the **tax case law domain**.

Few-shot prompting strategy for NERC and CE inference by LLM

Both the simple and detailed annotation guidelines were used for the few-shot prompt, as their zero-shot performances were variable. To examine the interaction between the level of detail in annotation guidelines and specific examples on model performance, few-shot prompting was conducted.

For this approach, both the original simple and detailed guidelines were retained from the zero-shot, except that detailed guidelines had their in-text examples added back. Handcrafted examples were added to these prompts to address tricky cases for the LLM. These included diverse cases, such as when a sentence has no entities, when dealing with difficult labels such as PROVISION that is very similar to LAW, or differentiating between TAX_TYPE and TAX_CONCEPT, or other seemingly misleading combinations of subsequent entities. We created the examples by exploring the company's internal corpus in the test-set-only collections and selecting suitable sentences for guidance. It should be noted that we are not legal or tax experts and simply followed the annotation guidelines to label the prompt examples, which can introduce noise or inconsistency into the demonstrations if incorrect. These prompts can be found in Appendix D.1, Appendix D.2, Appendix D.3, and Appendix D.4.

3.4.4 Fine-tuned encoders

Unlike the preceding approaches, fine-tuned encoder models learn task-specific representations from labeled data, building upon the linguistic and general domain knowledge already acquired during pre-training. As such, transformers show how performance depends on the pre-training and fine-tuning domains. To investigate the effect of these variables, we select models with general pre-training, such as RoBERTa, domain-specific pre-training, such as LegalBERT, and the state-of-the-art GLiNER model. To observe the influence of different fine-tuning data on NERC and CE performance in the legal-tax domain, datasets composed of ALDD, LLMAD, and a mix of both are used. This approach is expected to reveal whether models with approximate domain pre-training and models fine-tuned on target domain data improve the detection and classification of target-domain NEs and concepts.

All models used the respective validation split for each fine-tuning dataset (as described in Subsections 3.3.3, 3.3.4, and 3.3.5) to compute the evaluation loss. Evaluation loss provides insight into the model's performance on the validation set, allowing the comparison of model checkpoints across epochs to see when the loss stagnates or increases, thereby reducing the risk of selecting a model that may have started overfitting. The fine-tuning script identifies the model checkpoint with the lowest validation loss, which was then used for evaluation on the test set. A k-fold grid search was initially considered but proved computationally expensive. Therefore, fixed hyperparameter values were used for fine-tuning both RoBERTa and LegalBERT, with the learning rate set to 2×10^{-5}, and remaining parameters specified in Table 3.11. Finally, when predicting on the test set, both models' predictions for the first subword of each original token are considered the token-level prediction, and any subsequent subword predictions are ignored. The resulting token-level predictions were written as IOB2 annotations to CoNLL.

Table 3.11: Fine-tuning hyperparameters for Legal-BERT and RoBERTa across all fine-tuning datasets

Hyperparameter	**Legal-BERT**	**RoBERTa**
Base checkpoint	`legal-bert-base-uncased`	`roberta-base`
Learning rate	2×10^{-5}	2×10^{-5}
Weight decay	0.01	0.1
Train batch size	16	16
Eval batch size	16	16
Epochs	6	6
Random seed	67	67
Best-model criterion	Lowest eval loss	Lowest eval loss
Eval / save strategy	Per epoch	Per epoch

Finally, as will be explained further in Section 3.5, each lowest-loss model checkpoint was evaluated using the SeqEval macro average F1-score, across the different fine-tuning conditions. Note that, since the primary evaluation metric is macro F1, the checkpoint selected based on validation loss is not necessarily the macro F1-optimal one.

RoBERTa-base[5] and Legal-BERT-base-uncased[6], both BERT-based transformers,

[5]https://huggingface.co/FacebookAI/roberta-base
[6]https://huggingface.co/nlpaueb/legal-bert-base-uncased

were selected to isolate the impact of general versus in-domain pre-training (Liu et al., 2019; Chalkidis et al., 2020). GLiNER1-large_v2.1[7] is evaluated for its embedding similarity matching framing of NERC, which also suits CE (Zaratiana et al., 2024).

RoBERTa

Despite having the same architecture as BERT, RoBERTa performs better on downstream tasks due to its extensive pre-training (Wang et al., 2020; Bai et al., 2024). It is trained on large general-domain datasets such as BookCorpus, English Wikipedia, CC-News, OpenWebText, and Stories, using only dynamic masked language modeling, which allows it to create richer contextual embeddings for the words it encounters than a traditional BERT model (Liu et al., 2019). Because of its general-domain pre-training, the model is expected to perform worse on target NEs and concepts that require domain-specific knowledge (`COURT`, `TAX_CONCEPT`) than on labels with more general-domain association (`PERSON`, `DATE`, `GPE`), after fine-tuning. As a result, this model serves as a baseline for general-domain pre-training against which in-domain pre-training can be compared.

Legal-BERT

Legal-BERT is a family of BERT-based transformers, specifically trained on legal corpora, enabling it to capture the terminology and stylistic conventions particular to law-domain documents (Chalkidis et al., 2020). Chalkidis et al. (2020) released both Legal-BERT-SC, pre-trained from scratch and Legal-BERT-FP, a general variant that was further pre-trained on legal domain specific data created for their study. This is particularly useful to the use-case, as case law, like other legal text, has vocabulary and syntax that general web text barely contains. In comparison to a general-domain model's tokenizer, an in-domain model's tokenizer is less likely to fragment domain-specific, unfamiliar spans since the extensive pre-training would have exposed it to them. In particular, the Legal-BERT corpus comprises over 355,000 documents from European, British, and American contracts, legislation cases, and court cases. The pre-training using legal documents also exposes the model to seeing sentences with intricate syntax and reasoning patterns similarly seen in tax law documents, which is useful for this study. As there is no explicit mention of tax-law coverage in its pre-training corpus, the target NE labels: `COURT`, `LAW`, and `PROVISION`, which are legal-oriented, may likely be better represented than the tax-related concept labels. The Legal-BERT-SC has been shown to consistently outperform generalist and other legal domain-specific models on legal contract classification tasks (Singh et al., 2026). We select Legal-BERT-SC for a fair comparison against RoBERTa, as both are trained from scratch, and to show how in-domain pre-training can provide an advantage that further improves performance with target domain fine-tuning.

GLiNER

GLiNER, short for Generalist and Lightweight model for Named Entity Recognition, is a state-of-the-art DeBERTa-based architecture that treats NERC as matching entity-type embeddings to textual span representations in the same vector space, rather than the traditional token-by-token recognition and classification approach of BERT-based

[7] https://huggingface.co/urchade/gliner_large-v2

models like RoBERTa and LegalBERT, or auto-regressive generation approaches of LLMs (Zaratiana et al., 2024). The model encodes the input sequence and candidate entity labels together, computing representations for all candidate spans using token embeddings and entity-type representations from candidate labels, marked by a special token. The dot product between each of the candidate spans and entity label representations is computed and passed through a sigmoid function to obtain their similarity score. These similarity scores represent how confident the model is in matching the span to its label and can be set to a threshold for stricter or more lenient classification.

This means that GLiNER is likely better suited to handle the target concept labels (JURISDICTION, TAX_CONCEPT, TAX_TYPE), classifying by meaning rather than reference. Because the model uses embedding similarities to match spans to entities, it enables zero-shot classification and is therefore valuable for low-resource use cases. However, its domain generalizability to the legal-tax domain remains a question, given its extensive pre-training on LLM-annotated data across many diverse domains.

For inference, GLiNER takes a list of target labels in natural language and assigns labels to candidate spans based on whether the similarity score between a span's embedding and a label's embedding exceeds the set threshold. Predicted spans are processed in descending order of this similarity score, so that the higher-scoring spans are prioritized. For this study, the labels are mapped as seen in Table 3.12, and the similarity threshold was kept at 0.5. Labels that contained acronyms such as ORG and GPE had to be simplified because their acronym forms performed worse. GPE was found best represented by COUNTRY, although this may reduce recall on non-country GPE spans. Similarly, representing TAX_CONCEPT and TAX_TYPE as single labels showed better performance than treating them as individual words (i.e., 'TAX CONCEPT', 'TAX TYPE'). GLiNER predicts character-level spans instead of token-level tags, which means that each predicted span is aligned back to its tokens using character offsets. Thus, tokens whose character ranges were contained within a predicted span were assigned B- or I- tags, while tokens without predictions were assigned `'O'`. In these cases, if these tokens were gold entities, they were considered 'missed'. These converted predictions were then written to CoNLL.

Table 3.12: Mapping between the natural-language entity labels provided to GLiNER and the target label set used for evaluation

GLiNER Label	**Target Label**
Person	PERSON
Organization	ORG
Country	GPE
Law	LAW
Date	DATE
Provision	PROVISION
Court	COURT
Jurisdiction	JURISDICTION
Tax_Concept	TAX_CONCEPT
Tax_Type	TAX_TYPE

GLiNER can also be fine-tuned (Zaratiana et al., 2024) on domain-specific annotated data, resulting in its parameters to adapt to domain-specific text and target entity types. This allows us to inspect how domain-specific fine-tuning affects the per-

formance of a model with a non-traditional architecture. The hyperparameters used for fine-tuning are presented in Table 3.13 below.

Table 3.13: Fine-tuning hyperparameters for GLiNER-large-v2.1 used across all fine-tuning datasets, following the settings of (Zaratiana et al., 2024)

Hyperparameter	**GLiNER**
Base checkpoint	`gliner_large-v2.1`
Learning rate (backbone)	5×10^{-6}
Learning rate (other layers)	1×10^{-5}
Weight decay	0.01
Train batch size	8
Eval batch size	8
Epochs	6
Warmup ratio	0.1
Random seed	67
Best-model criterion	Lowest eval loss
Eval / save strategy	Per epoch

3.5 Evaluation Metrics

As the aim of the thesis is to find an approach that works for named entities and concepts, every approach is evaluated for its performance on the full NE label and concept set, even if it was designed not to capture them. Since performance in NERC requires both span identification and correct label assignment, multiple metrics are used to compute precision, recall, and F1-score to evaluate model predictions. Table 3.14 below provides an example sentence to illustrate how these metrics work.

Table 3.14: Token-level gold and predicted IOB2 tags for an example sentence

Token	**Gold**	**Predicted**
The	O	O
Court	B-COURT	B-COURT
of	I-COURT	I-COURT
Justice	I-COURT	I-COURT
of	I-COURT	O
the	I-COURT	O
European	I-COURT	O
Union	I-COURT	O
applied	O	O
Article	B-PROVISION	B-PROVISION
49	I-PROVISION	I-PROVISION
.	O	O

SeqEval

SeqEval (Nakayama, 2018) is a library for evaluating sequence labeling, assessing performance on tasks such as NERC, part-of-speech tagging, and semantic role labeling. It

is designed to handle common token labeling formats characteristic of CoNLL datasets. The framework offers macro and micro average precision, recall, F1-score, and accuracy in two modes: DEFAULT and STRICT. In the DEFAULT mode, the model is evaluated leniently, allowing incorrect span boundaries as long as they have the same label as the gold entity. In STRICT mode, the model is evaluated precisely, requiring exact alignment, and any invalid or unaligned boundaries are counted as incorrect. Both of these modes enable inspection of the model's ability to correctly identify spans and choose the right label to assign.

Table 3.15: SeqEval per-label and macro-average scores for the example in Table 3.14, under DEFAULT and STRICT modes

Mode	Label	P	R	F1
DEFAULT	COURT	0.00	0.00	0.00
	PROVISION	1.00	1.00	1.00
	Macro avg	0.50	0.50	0.50
STRICT	COURT	0.00	0.00	0.00
	PROVISION	1.00	1.00	1.00
	Macro avg	0.50	0.50	0.50

As seen in Table 3.15 where SeqEval evaluation is presented, the library by default excludes 'O' when calculating metrics. Due to the truncated predictions assigned to the COURT instance, both SeqEval conditions treat the prediction as insufficient, hence the 0.0 across both DEFAULT and STRICT conditions. If the spans aligned but the concept label differed, the DEFAULT version would consider it correct, but the STRICT version would not.

To ensure the evaluation requires both a correct NE or concept label and correct boundary assignment, SeqEval-STRICT is used to produce overall metrics. The macro-average precision, recall, and F1-score are used as the primary metrics, as this measure treats all classes equally, regardless of their frequency in the dataset (Keraghel et al., 2024), which was necessary for this use case as the generic NEs outnumber most of the target NEs and concepts.

NERVALUATE

While SeqEval is informative, it is limited as it only offers a token-to-token comparison, essentially visualizing how the model does per label. For a more in-depth look at span-level behavior, the NERVALUATE (Batista and Upson, 2026) library is also used to analyze the errors of the best-performing model, as it provides insight into the most challenging instances in the test set.

This module's output provides a more in-depth look at the spans predicted by the model. Specifically, it scores the predictions in two ways. Firstly, per label, the results are sorted by if they were `correct`: span and entity type matching gold instance, `incorrect`: boundary found but wrong entity, `partial`ly correct: only part of the entity was found or misaligned boundary, `missed`: unidentified span in the gold data or `spurious`: predicted an entity that was not there. Secondly, results are sorted by the matching scheme, as is defined in Table 3.16 below: `strict`, `exact`, `partial`, and

`type`. For both of these measurements, the framework also provides recall, precision, and F1-scores per label, from which macro and micro values can be computed.

Table 3.16: The four matching schemes used by NERVALUATE to evaluate span predictions.

Scheme	Matching rule
Strict	Both the entity boundary and the entity type must match the gold span exactly.
Exact	The entity boundary must match the gold span exactly, regardless of the predicted entity type.
Partial	The predicted span must overlap the gold span, regardless of entity type; an exact boundary is not required.
Type	The entity type must match, as long as there is some overlap with the gold span, regardless of boundary alignment.

From the Table 3.17 below, it can be seen how NERVALUATE diverges from SeqEval. In the Per Entity `strict` condition, which examines how predictions and gold labels compare in terms of correct span boundaries and labels, it yields the same results as SeqEval: 0.0 for COURT and 1.00 for PROVISION. However, the Matching scheme condition shows span matching in more detail: the `strict` matching has one entity predicted correctly (PROVISION) and one incorrectly (COURT); the `exact` matching has one entity that was correct in span boundaries (PROVISION); the `partial` matching had one predicted span correctly overlapping the gold span (PROVISION), and another partially overlapping the gold span (COURT); and lastly the `type` matching had two correct predictions that shared the same label with the gold span, regardless of their boundaries.

Table 3.17: NERVALUATE scores for the example in Table 3.14. *Per entity* shows results under the strict scheme for each label; *matching scheme* aggregates across both entities. `Cor`: correct, `Inc`: incorrect, `Par`: partial, `Mis`: missed, `Spu`: spurious. Recall equals precision throughout in this single-prediction-per-gold example.

		Outcome counts						
View	**Label / scheme**	**Cor**	**Inc**	**Par**	**Mis**	**Spu**	**P/R**	**F1**
Per entity (strict)	COURT	0	1	0	0	0	0.00	0.00
	PROVISION	1	0	0	0	0	1.00	1.00
Matching scheme	Strict	1	1	0	0	0	0.50	0.50
	Exact	1	1	0	0	0	0.50	0.50
	Partial	1	0	1	0	0	0.75	0.75
	Type	2	0	0	0	0	1.00	1.00

In this way, these metrics complement the token-level scores provided by SeqEval, demonstrating what overall performance looks like when not penalized by strict span boundaries or entity-type matching.

Chapter 4

Results

To answer the main RQ: *'What is the best approach to identifying and classifying named entities and concepts in the legal-tax domain?'*, the results for each approach attempted are presented, displayed in order of increasing training data required. Section 4.1 first discusses the company baseline, presenting each component. Section 4.2 then presents the gazetteers, differentiating between the Company gazetteer and the Heuristics-based gazetteer. Section 4.3 displays zero-shot and few-shot approaches, including GLiNER and LLM inference. Section 4.4 then shows the fine-tuned encoder models, separated by each fine-tuning data source. Finally, a summary of the results is provided in Section 4.5, followed by Section 4.6, which presents an error analysis of the best-performing model.

4.1 Baselines

This section answers SRQ1: *'How do the relevant baselines, consisting of the company baseline and out-of-the-box models, perform on the task?'* In particular, SpaCy, SpaCy+RegEx, and SpaCy+RegEx+the Company gazetteer are evaluated. Table 4.1 below presents their results.

The baselines build on one another, with each added component improving performance; the Company baseline, which encompasses all three components, achieved the highest F1-score of 0.37. The SpaCy baseline alone performs poorly, addressing only the generic NEs it was trained on, leaving other target NEs, such as `COURT` and `PROVISION`, and target concepts undetected. Adding the RegEx rule-filtering component recovers `COURT` (0.36 F1) and `PROVISION` (0.37), and the Company gazetteer recovers the target concepts to varying degrees, indicating that each baseline element is complementary in capability.

All three baselines struggle consistently with the `LAW` (0.02) and `ORG` (0.22) NEs the most, despite the labels being part of SpaCy's generic label capabilities, likely due to domain-specific surface forms such as statute names and multilingual organization names in legal text (e.g: 'Directive 2008/78', 'Skatteministeriet v T Danmark') which possibly diverge from the surface forms present in its generic domain English text training distribution. Furthermore, despite the gazetteer contribution, the `TAX_CONCEPT` (0.12) and `TAX_TYPE` (0.12) concepts labels are detected poorly overall.

`JURISDICTION`, in contrast, has the largest performance gain (0.00 to 0.71) overall. This is likely attributable to the fact that 124 out of its 153 total instances are differing forms of 'Member State'. These minor variations are well accounted for in the

Table 4.1: F1 per NE and concept label for the company's baseline pipeline: SpaCy alone, SpaCy with RegEx post-filtering, and the company baseline (SpaCy+RegEx with the Company gazetteer). Best F1 per label in bold. Precision and recall in Appendix E.1

	F1-score			
Label	SpaCy	SpaCy+RegEx	Company baseline	Sup.
Named entities				
COURT	0.00	**0.36**	**0.36**	106
DATE	**0.57**	**0.57**	**0.57**	132
GPE	**0.66**	**0.66**	**0.66**	299
LAW	0.01	**0.02**	**0.02**	65
ORG	0.20	**0.22**	**0.22**	221
PERSON	**0.53**	**0.53**	**0.53**	186
PROVISION	0.00	**0.37**	**0.37**	87
Concepts				
JURISDICTION	0.00	0.00	**0.71**	153
TAX_CONCEPT	0.00	0.00	**0.12**	353
TAX_TYPE	0.00	0.00	**0.12**	86
Macro avg	0.20	0.27	**0.37**	1688

`JURISDICTION` gazetteer (Figure 3.6), which contains varying forms for each term (e.g, for the term 'member state', the surface forms: 'Member State', 'Member States', 'EU Member State', and 'EU Member States' exist). These surface form lookups had larger coverage in comparison to the `TAX_CONCEPT` (e.g, for the term 'asset intensity adjustments', the available surface forms were: 'working capital adjustments', 'balance sheet adjustments') and `TAX_TYPE` (e.g, for the term 'branch profits tax', the available surface forms were: 'BPT', 'branch tax'), whose stored surface form variations varied in terms of synonyms, not accounting for simpler variations such as number. Furthermore, many of the available surface forms for the `TAX_CONCEPT` and `TAX_TYPE` concept labels were simply the terms themselves, which reduces their coverage. In sum, the difference in performance likely indicates that `JURISDICTION` may be easier to capture, as it exhibits more stable surface forms and better gazetteer coverage than `TAX_CONCEPT` (e.g., 'double taxation', 'exemption', 'distributed profits') and even `TAX_TYPE` (e.g., 'corporation tax', 'withholding tax', 'state tax'), despite it having some regularity such as ending in 'tax'.

As the results demonstrate, the company's current pipeline reaches a 0.37 F1-score, which can be attributed to its individual components, but is mainly driven by the generic NE labels and the `JURISDICTION` gazetteer, while performance across the target NE and concept labels still remains low, motivating the alternative approaches evaluated in the following sections.

4.2 Gazetteers

This section answers SRQ2: *'How does a linguistic Heuristics-based gazetteer perform on the task compared to the company's ontology-based gazetteer?'* by evaluating the Heuristics-based and Company gazetteers separately, each combined with the

JURISDICTION gazetteer. The Heuristics-based gazetteer is also replaced in the company baseline to demonstrate its impact in the current pipeline. Table 4.2 below displays these results.

Table 4.2: F1-score per NE and concept label for the Heuristics-based and Company gazetteers in isolation and combined with the SpaCy+RegEx pipeline. Best F1-score per label in bold. Precision and recall in Appendix E.1

	Gazetteer only		With SpaCy+RegEx		
Label	Heuristics	Company	Heuristics	Company	Sup.
Named entities					
COURT	0.00	0.00	**0.36**	**0.36**	106
DATE	0.00	0.00	**0.57**	**0.57**	132
GPE	0.00	0.00	**0.66**	**0.66**	299
LAW	0.00	0.00	**0.02**	**0.02**	65
ORG	0.00	0.00	**0.22**	**0.22**	221
PERSON	0.00	0.00	**0.53**	**0.53**	186
PROVISION	0.00	0.00	**0.37**	**0.37**	87
Concepts					
JURISDICTION	**0.74**	**0.74**	0.71	0.71	153
TAX_CONCEPT	**0.16**	0.11	0.13	0.12	353
TAX_TYPE	**0.48**	0.11	**0.48**	0.12	86
Macro avg	0.14	0.10	**0.40**	0.37	1688

The Heuristics-based Gazetteer outperformed the Company gazetteer when evaluated in isolation and when combined with the SpaCy+RegEx rules. In the isolated evaluation, both gazetteers performed identically for JURISDICTION (0.74 vs 0.74) as the same gazetteer entries are used for this concept. TAX_CONCEPT remained a difficult concept, with the Heuristics-based Gazetteer scoring just 0.16 compared to the poor performance by the Company gazetteer (0.11). The performance for TAX_TYPE is more interesting, as the Heuristics-based Gazetteer (0.48) outperformed the Company gazetteer (0.11), despite the latter's inclusion of various surface-form variations that the former lacks.

Since the gazetteers only contribute to the concepts, the NE performances are identical for both when combined with the SpaCy+RegEx rules that actually detect NEs. When combining the gazetteers with the SpaCy+RegEx rules, JURISDICTION also yielded the same performance decrease (0.74 to 0.71) for both gazetteers. This is likely due to SpaCy NE labels being assigned first, which assigns generic NE labels to spans and reduces the overall applicability of the JURISDICTION lookup. This is possibly due to JURISDICTION overlapping with ORG and GPE entities assigned by SpaCy (e.g., 'EUROPEAN UNION' is a JURISDICTION gold entity but may elicit ORG or GPE for SpaCy). The Heuristics-based Gazetteer yielded the same performance (0.48) for TAX_TYPE. TAX_CONCEPT showed changes in performance, from 0.16 to 0.13 for the Heuristics-based Gazetteer, which further examination revealed was the same issue with SpaCy assigning ORG labels to gold tax concepts (e.g, 'APA', which stands for Advance Pricing Agreement, is assigned ORG, 'Transfer Pricing' or 'Income Tax', instances were also assigned ORG, possibly due misleading case). The Company Gazetteer interestingly saw an increase for both TAX_TYPE and TAX_CONCEPT.

The results imply that, with a Heuristics-based Gazetteer, surface-form-consistent concepts such as `JURISDICTION` and even `TAX_TYPE` are more suitable for lookup than the abstract `TAX_CONCEPT`. The overall average F1-score for the better-performing Heuristics-based Gazetteer remains low (0.14) because it does not address any of the NEs. It can be deduced that the gazetteer approach is not suitable for NERC, though it shows promise for some of the concepts.

4.3 Zero-shot and Few-shot approaches

This section answers SRQ1: *'How do the relevant baselines, consisting of the company baseline and out-of-the-box models, perform on the task?'* and SRQ3: *'How do LLMs perform on the task in zero-shot and few-shot conditions?'* by presenting GLiNER and LLM zero-shot and LLM few-shot results.

4.3.1 Zero-shot Inference by LLM and GLiNER

Table 4.3 below displays the results of all zero-shot approaches.

Table 4.3: Zero-shot F1-score per NE and concept label: GLiNER and the LLM prompt configurations (GPT-NER and IDX:TOK formats; s=simple, d=detailed guidelines), with the company baseline for reference. Best F1-score per label in bold. Precision and recall in Appendix E.2

	F1-score						
Label	GLiNER	GPT-NER (s)	GPT-NER (d)	IDX:TOK (s)	IDX:TOK (d)	Company baseline	Sup.
Named entities							
COURT	0.36	0.44	**0.47**	0.39	0.45	0.36	106
DATE	**0.75**	0.54	0.55	**0.75**	0.67	0.57	132
GPE	0.64	0.62	0.64	0.62	**0.68**	0.66	299
LAW	0.21	**0.31**	0.30	0.27	0.26	0.02	65
ORG	0.27	0.36	**0.40**	0.35	0.39	0.22	221
PERSON	0.59	**0.75**	0.71	0.67	0.68	0.53	186
PROVISION	0.12	0.35	**0.41**	0.33	0.40	0.37	87
Concepts							
JURISDICTION	0.11	0.27	0.23	0.30	0.15	**0.71**	153
TAX_CONCEPT	**0.32**	0.12	0.05	0.23	0.13	0.12	353
TAX_TYPE	**0.45**	0.11	0.04	0.31	0.20	0.12	86
Macro avg	0.38	0.39	0.38	**0.42**	0.40	0.37	1688

Across the various zero-shot approaches attempted, the IDX:TOK structured output (index-token dictionary output; see Figure 3.7) with the simple annotation guidelines yielded the highest F1-score (0.42), outperforming the Heuristics-based Gazetteer and SpaCy+RegEx rules (0.40), as well as the company baseline (0.37). It is followed closely by its detailed counterpart (0.40). This performance, as exhibited by the other prompts and the model, is attributed to the consistently high-scoring generic NEs such as `DATE`, `GPE`, and `PERSON`, which are likely easier to predict due to their relatively stable surface forms compared to abstract concepts.

In contrast, for the `TAX_CONCEPT` and `TAX_TYPE` concepts, GLiNER zero-shot demonstrates the best performance, yielding a higher F1-score than the best-performing LLM zero-shot prompt for `TAX_CONCEPT` (0.32 vs 0.23) and `TAX_TYPE` (0.45 vs 0.31). Although the IDX:TOK(s) outperforms GLiNER for `JURISDICTION` (0.30 vs 0.11), both

these approaches are still outperformed by the JURISDICTION-only gazetteer part of the company baseline (0.71). This suggests that for the `TAX_CONCEPT` and `TAX_TYPE` concepts, GLiNER's span-embedding matching approach may capture them better than prompt-based extraction in the zero-shot setting.

From these results, the IDX:TOK outperformed the GPT-NER (special character marked entity output; see Section 3.4.3) in the zero-shot approach (0.42 vs 0.39) overall, although GPT-NER still demonstrated better performance on certain individual NE labels. The level of detail in the annotation guidelines also does not appear to have a substantial impact on performance in the zero-shot condition.

4.3.2 Few-shot Inference with LLM

Table 4.4: Few-shot F1-score per NE and concept label for the GPT-NER and IDX:TOK prompt formats (s=simple, d=detailed guidelines), with the company baseline for reference. Best F1-score per label in bold. Precision and recall in Appendix E.3

	F1-score					
Label	GPT-NER (s)	GPT-NER (d)	IDX:TOK (s)	IDX:TOK (d)	Company baseline	Sup.
Named entities						
COURT	0.46	0.48	0.39	**0.49**	0.36	106
DATE	0.63	0.57	0.72	**0.74**	0.57	132
GPE	0.67	0.68	0.65	**0.71**	0.66	299
LAW	0.29	**0.30**	0.21	0.20	0.02	65
ORG	0.41	**0.44**	0.32	0.43	0.22	221
PERSON	**0.80**	0.67	0.58	0.49	0.53	186
PROVISION	0.47	0.46	0.35	**0.48**	0.37	87
Concepts						
JURISDICTION	0.47	0.62	0.52	0.63	**0.71**	153
TAX_CONCEPT	0.18	0.13	**0.30**	0.29	0.12	353
TAX_TYPE	0.43	0.42	0.47	**0.48**	0.12	86
Macro avg	0.48	0.48	0.45	**0.50**	0.37	1688

All the few-shot approaches outperformed the company baseline (0.37). The IDX:TOK output structure with the detailed annotation guidelines performed best, achieving an F1-score of 0.50 and outperforming its simpler version, which scored 0.45. Its performance across the NEs was less divided compared to the zero-shot approaches, consistently yielding the highest F1-scores for `COURT`, `DATE`, `GPE`, and `PROVISION`. GPT-NER's detailed version outperformed it on the `LAW` (0.30 vs 0.20) and `ORG` (0.44 vs 0.43), and GPT-NER's simpler version outperformed it on the `PERSON` label, by a large gap in F1-score (0.80 vs 0.49), despite it having less detail to guide annotation.

As for the concepts, the IDX:TOK output structure with the detailed annotation guidelines achieved the highest F1-score for `JURISDICTION` (0.63), followed closely by the detailed GPT-NER prompt (0.62), although both remain outperformed by the JURISDICTION-only gazetteer part of the company baseline (0.71). Their simpler versions performing worse on the concept (0.52 and 0.47) respectively, indicates that the detailed annotation guidelines likely help the LLM extract the concept better. Interestingly for IDX:TOK, `TAX_CONCEPT` and `TAX_TYPE` showed negligible differences (0.01 F1-score point) in performances when the annotation guidelines were simple or detailed showing that more detail is not as impactful for these two concepts as it was for `JURISDICTION`.

The improvement from zero-shot to few-shot performance also appears to be related to the labels represented in the provided examples (Appendix D.4). `JURISDICTION` and `TAX_TYPE` each were in two of the six few-shot examples, and received the largest F1-score gains across the four prompt configurations (+0.32 and +0.29 respectively). In contrast, LAW was not represented in any of the examples and showed no improvement, declining slightly further in three of the four prompt configurations (−0.04 on average). Interestingly, PERSON was represented but still also saw a slight decline in performance. This indicates that the example coverage may not guarantee improvement, and that the impact may be limited.

Comparing the results of zero-shot and few-shot approaches indicates that adding task-specific examples improves the LLM's performance regardless of the prompt output structure, showcasing substantial F1-score gains from zero-shot to few-shot for the detailed guideline variants (0.40 → 0.50 for IDX:TOK, 0.38 → 0.48 for GPT-NER). Additionally, combining the detailed annotation guidelines with the few-shot examples is more beneficial than using simpler annotation guidelines with the examples (0.45 → 0.50 F1-score for IDX:TOK(s) vs 0.42 → 0.45 for IDX:TOK(d)).

Among all zero-shot and few-shot approaches, the few-shot IDX:TOK prompt structure with detailed annotation guidelines achieved the highest overall F1-score (0.50). Thus, it was selected to label the unlabeled target domain tax case-law documents in the company corpus.

4.4 Fine-tuned models

To investigate the impact of domain-specific pre-training and fine-tuning, RoBERTa, Legal-BERT, and GLiNER were fine-tuned on human-annotated approximate legal domain data `(ALDD)`, LLM-annotated data `(LLMAD)`, and a mix of both `(MIX)`. Their results are presented per fine-tuning data type.

4.4.1 Approximate Legal Domain Data

To answer SRQ4: *'How does domain-matching affect fine-tuned encoder models' performance on the task across three training conditions: approximate legal domain NERC data, target domain tax case-law data with LLM-generated NERC labels, and a combination of the two?'*, the results for the `ALDD` fine-tuned encoders are presented in Table 4.5 below.

Table 4.5: F1-score per NE and concept label for RoBERTa, Legal-BERT, and GLiNER fine-tuned on the (ALDD), with the company baseline for reference. Best F1-score per label in bold. Precision and recall can be found in Appendix E.4

	F1-score				
Label	RoBERTa	Legal-BERT	GLiNER	Company baseline	Sup.
Named entities					
COURT	0.11	0.47	**0.66**	0.36	106
DATE	0.12	0.29	**0.53**	0.57	132
GPE	0.20	0.59	**0.67**	0.66	299
LAW	0.00	0.09	**0.28**	0.02	65
ORG	0.11	0.26	**0.39**	0.22	221
PERSON	0.30	0.70	**0.74**	0.53	186
PROVISION	0.11	**0.15**	0.11	0.37	87
Concepts					
JURISDICTION	0.00	0.00	0.23	**0.71**	153
TAX_CONCEPT	0.00	0.00	**0.29**	0.12	353
TAX_TYPE	0.00	0.00	**0.52**	0.12	86
Macro avg	0.10	0.26	**0.44**	0.37	1688

When using ALDD, GLiNER was found to perform the best, with an F1-score of 0.44, compared to Legal-BERT (0.26) and RoBERTa (0.10), which were outperformed by the company baseline (0.37). GLiNER outperformed both other models on nearly all NEs, except PROVISION (0.11), for which Legal-BERT yielded the highest F1-score (0.15). RoBERTa performs the worst overall (0.10), failing entirely on LAW and barely predicting COURT, ORG, and PROVISION, all at 0.11. The legal pre-training of Legal-BERT provided some advantage, as it performed better than RoBERTa on all NE labels and even outperformed GLiNER on PROVISION (0.15 vs 0.11), with the most comparable differences in performance being for PERSON, GPE, and COURT. It is interesting that despite the conflicting signals inherited from the (Indian_Legal_NER_dataset) dataset caused by mapping PETITIONER, which contained PERSON and ORG entities to 'O', PERSON was still the best predicted label for all models: RoBERTa (0.30), Legal-BERT (0.70) and GLiNER (0.74), while the ORG label seemed to have a more negative impact.

As for concepts, only GLiNER was able to predict them, as it matches span embeddings to labels, allowing it to predict concept labels it was not fine-tuned on, unlike RoBERTa and Legal-BERT. Despite this, it still performed poorly for JURISDICTION (0.23) and TAX_CONCEPT (0.29). However, it predicted TAX_TYPE better with an F1-score of 0.52.

The results illustrate a tradeoff that comes with using approximate domain data to fine-tune models for NERC and CE: while the models do learn some domain-oriented representations of the target labels, these are not directly transferable because the ALDD only approximates the target domain, resulting in poor overall performance. The results also reveal an advantage of domain-specific pre-training, which is seen by the difference in performance between RoBERTa and Legal-BERT, with the latter displaying the ability to predict the target and generic NEs consistently better than the former. Legal-BERT's domain-specific pre-training likely grants it a broader understanding of the complex linguistic patterns, syntactic structures, and jargon prevalent in the legal

domain, which likely works more harmoniously with the ALDD data for fine-tuning than the general-domain pre-training does when applied to the target domain. However, this advantage is still limited as the fine-tuning data neither exposed the models to the target concepts nor matched the target domain data, which is why traditional models like RoBERTa and Legal-BERT cannot predict concepts and underperformed overall in comparison to GLiNER.

4.4.2 LLM-annotated Data

To answer SRQ4: *'How does domain-matching affect fine-tuned encoder models' performance on the task across three training conditions: approximate legal domain NERC data, target domain tax case-law data with LLM-generated NERC labels, and a combination of the two?'*, the results for the `LLM-annotated data` fine-tuned encoders are presented below in Table 4.6

Table 4.6: F1-score per NE and concept label for RoBERTa, Legal-BERT, and GLiNER fine-tuned on the `ALDD` and on the `LLMAD`, with the company baseline for reference. Best F1-score per label in bold. Precision and recall in Appendix E.5

	Approximate Legal Domain data			LLM-annotated				
Label	RoBERTa	Legal-BERT	GLiNER	RoBERTa	Legal-BERT	GLiNER	Company baseline	Sup.
Named entities								
COURT	0.11	0.47	0.66	0.47	0.54	**0.55**	0.36	106
DATE	0.12	0.29	0.53	0.72	**0.78**	0.75	0.57	132
GPE	0.20	0.59	0.67	0.59	0.68	**0.77**	0.66	299
LAW	0.00	0.09	**0.28**	0.19	**0.28**	**0.28**	0.02	65
ORG	0.11	0.26	0.39	0.25	**0.48**	0.43	0.22	221
PERSON	0.30	0.70	**0.74**	0.18	0.48	0.61	0.53	186
PROVISION	0.11	0.15	0.11	0.09	**0.47**	0.42	0.37	87
Concepts								
JURISDICTION	0.00	0.00	0.23	0.39	0.67	0.66	**0.71**	153
TAX_CONCEPT	0.00	0.00	0.29	0.08	0.25	**0.26**	0.12	353
TAX_TYPE	0.00	0.00	0.52	0.35	0.44	**0.55**	0.12	86
Macro avg	0.10	0.26	0.44	0.33	0.51	**0.53**	0.37	1688

Using `LLMAD` achieved higher performance across all encoders, outperforming both few-shot LLM inferences and encoders fine-tuned on `ALDD`, highlighting target domain knowledge transfer from few-shot to fine-tuning. GLiNER performed best (0.53), followed closely by Legal-BERT (0.51), both outperforming the company baseline (0.37). The best performances for NEs were split between GLiNER and Legal-BERT, with both models performing nearly identically for certain labels such as `COURT` (0.55 vs 0.54) and `LAW` (0.28 vs 0.28), and very close for others such as `DATE` (0.75 vs 0.78) and `ORG` (0.43 vs 0.48). RoBERTa, although still the worst performer (0.33), can now predict all labels, since the `LLMAD` unlike the `ALDD`, includes concept labels. Interestingly, though, the `PERSON` label saw a considerable drop in performance across all three models in comparison to their `ALDD` fine-tuned variants (Legal-BERT: 0.70 $\rightarrow$ 0.48), implying that the `ALDD` consisting of general case law and legal data contains a wider variety of `PERSON` role types, which provides the model with broader `PERSON` surface form coverage, whereas the LLM-annotated data references `PERSON` in much narrower legal contexts.

Even with exposure to the concepts in the fine-tuning data, TAX_CONCEPT remains difficult to predict across all three models, with the lowest F1-score observed by RoBERTa at 0.08, followed by Legal-BERT at 0.25 and GLiNER at 0.26. Since the LLM labels were generated by the IDX:TOK detailed variant, which only scored 0.29 on TAX_CONCEPT in few-shot, it is likely that poor upstream labeling by the LLM contributed to TAX_CONCEPT remaining difficult to predict across models. JURISDICTION sees an improvement across all three models compared to the ALDD fine-tuning variants, though none outperform the company baseline's performance for it (0.71). The best encoder performance for the concept is displayed by Legal-BERT (0.67), followed closely by GLiNER (0.66). TAX_TYPE also shows a similar pattern, with GLiNER (0.55) performing best, followed by Legal-BERT (0.44).

The results demonstrate that using LLM-annotated target domain data outperforms using ALDD on an overall level, with each model seeing substantial increase in performance: RoBERTa (0.10 → 0.33), Legal-BERT (0.26 → 0.51), GLiNER (0.44 → 0.53). These findings confirm the impact of aligning the domains of fine-tuning data with the testing data. It has a largely positive effect with a trade-off: the concepts were now detected by all the models, and the target NEs saw an improvement in performance, while some generic NEs dropped in performance, likely due to the domain-specific entity representations limiting broader entity coverage.

4.4.3 Mixed Data

To answer SRQ4: *'How does domain-matching affect fine-tuned encoder models' performance on the task across three training conditions: approximate legal domain NERC data, target domain tax case-law data with LLM-generated NERC labels, and a combination of the two?'*, the results for the mixed data fine-tuned encoders are presented below in Table 4.7.

Table 4.7: F1-score per NE and concept label for RoBERTa, Legal-BERT, and GLiNER fine-tuned on the approximate legal domain data, the LLM-annotated target domain data, and the two combined (MIX), with the company baseline for reference. Best F1-score per label across all conditions in bold. Precision and recall in Appendix E.6

	Approximate Legal Domain data			LLM-annotated			Mixed				
Label	RoB.	LegB.	GLi.	RoB.	LegB.	GLi.	RoB.	LegB.	GLi.	Comp. base.	Sup.
Named entities											
COURT	0.11	0.47	**0.66**	0.47	0.54	0.55	0.43	0.53	0.53	0.36	106
DATE	0.12	0.29	0.53	0.72	**0.78**	0.75	0.69	0.75	0.72	0.57	132
GPE	0.20	0.59	0.67	0.59	0.68	**0.77**	0.59	0.74	0.75	0.66	299
LAW	0.00	0.09	**0.28**	0.19	**0.28**	**0.28**	0.20	0.21	0.26	0.02	65
ORG	0.11	0.26	0.39	0.25	**0.48**	0.43	0.26	0.47	0.46	0.22	221
PERSON	0.30	0.70	**0.74**	0.18	0.48	0.61	0.30	0.38	0.45	0.53	186
PROVISION	0.11	0.15	0.11	0.09	**0.47**	0.42	0.11	**0.47**	0.38	0.37	87
Concepts											
JURISDICTION	0.00	0.00	0.23	0.39	0.67	0.66	0.46	0.67	0.62	**0.71**	153
TAX_CONCEPT	0.00	0.00	0.29	0.08	0.25	0.26	0.08	0.22	**0.32**	0.12	353
TAX_TYPE	0.00	0.00	0.52	0.35	0.44	**0.55**	0.26	0.48	0.51	0.12	86
Macro avg	0.10	0.26	0.44	0.33	0.51	**0.53**	0.34	0.49	0.50	0.37	1688

When using a MIX of approximate legal domain data and LLMAD target data, GLiNER

performed best with an F1-score of 0.50, followed closely by Legal-BERT (0.49); both outperformed the company baseline (0.37). RoBERTa remains the weakest performer (0.34). The mixed data improved RoBERTa by 0.01 F1-score point over the `LLMAD` fine-tuned variant (0.33), while GLiNER and Legal-BERT dropped from 0.53 to 0.50 and from 0.51 to 0.49, respectively. The best performance per NE is split between GLiNER and Legal-BERT, with near-identical performance for labels such as `COURT` (0.53 vs 0.53), `GPE` (0.75 vs 0.74), and `ORG` (0.46 vs 0.47).

For the concepts, a pattern similar to the `LLMAD`-fine-tuned variants was observed, with `JURISDICTION` predicted best (0.67) by Legal-BERT, followed by GLiNER (0.62). GLiNER then outperformed Legal-BERT on `TAX_CONCEPT` (0.32 vs 0.22) and `TAX_TYPE` (0.51 vs 0.48).

The results indicate that using a mix of human-annotated `ALDD` and `LLMAD` provides no benefit over using only `LLMAD`. It is likely that differing domains, labeling conventions, and (general-domain vs domain-specific) entity distributions between the human-annotated `ALDD` and the `LLMAD` do not complement each other but instead introduce noise.

Across all fine-tuning datasets, the LLM-annotated target domain data yielded the highest F1-scores overall, indicating that target-domain-specific data is most informative for encoder models. Although the models also learned generic legal domain NEs from `ALDD`, they were ultimately outperformed by the `LLMAD`, through exposure to the target domain NEs and concepts. Compared to the company baseline, the GLiNER fine-tuned on LLM-annotated data shows a 0.16-point increase in F1, indicating that an encoder model with span-embedding matching makes predictions that capture the nuances of NERC and CE better than a gazetteer or a general domain out-of-the-box NERC model with RegEx filtering rules does.

4.5 Summary

The results across all approaches demonstrated that no single method performs the best for every label. However, the highest overall performance was achieved by GLiNER fine-tuned on `LLMAD`, seen in Table 4.7. Table 4.8 below reports the best-performing approach for each NE and concept label, along with the performance of the mentioned GLiNER variant.

Table 4.8: Best-performing approach per NE and concept label across all experiments, with its precision, recall, and F1, along with the F1-score of the best all-round single model, GLiNER fine-tuned on LLM-annotated data. Bold marks the higher F1-score of the two. `ALDD` = Approximate Legal Domain Data, `LLMAD` = LLM-annotated data, ZS: zero-shot, FS: few-shot, GPT-NER: LLM output structure of special character marking for entities and concepts, IDX:TOK: LLM output structure of dictionary mapping for entity info per token

	Best-performing approach				GLiNER (LLMAD)
Label	Approach	P	R	F1	F1
Named entities					
COURT	GLiNER (ALDD)	0.64	0.69	**0.66**	0.55
DATE	Legal-BERT (LLMAD)	0.76	0.80	**0.78**	0.75
GPE	GLiNER (LLMAD)	0.71	0.84	**0.77**	**0.77**
LAW	GPT-NER ZS	0.22	0.51	**0.31**	0.28
ORG	Legal-BERT (LLMAD)	0.42	0.57	**0.48**	0.43
PERSON	GPT-NER FS (simple)	0.79	0.82	**0.80**	0.61
PROVISION	IDX:TOK FS (detailed)	0.37	0.68	**0.48**	0.42
Concepts					
JURISDICTION	Gazetteers only	0.65	0.86	**0.74**	0.66
TAX_CONCEPT	GLiNER ZS	0.32	0.33	**0.32**	0.26
TAX_TYPE	GLiNER (LLMAD)	0.50	0.63	**0.55**	**0.55**

For NEs, the best-performing approaches consist of fine-tuned encoder models and zero-shot, few-shot LLM inference. The highest performance for a label is seen by the GPT-NER simple annotation guidelines few-shot inference for `PERSON`, yielding an F1-score of 0.80 and outperforming the GLiNER-LLM variant (0.61). Subsequently, `DATE`, and `GPE` follow, as predicted by Legal-BERT (0.78) and GLiNER (0.77) when both fine-tuned on the `LLMAD`. GLiNER fine-tuned on `ALDD`, then performs best on `COURT`, at 0.66 F1-score. There is a larger drop in performance for the remaining NE labels: `ORG`, as predicted by Legal-BERT on the `LLMAD` (0.48), and `PROVISION`, as predicted by IDX:TOK detailed annotation guidelines prompt few-shot inference (0.48). Lastly, `LAW` was the highest predicted at 0.31 F1-score by GPT-NER zero-shot inference, indicating that no other approach learned it well either.

As for concepts, the best-performing approaches consist of a gazetteer and GLiNER. The highest performance seen is for `JURISDICTION`, both by the Company gazetteer (that already includes all `JURISDICTION` entries) and `JURISDICTION-only` gazetteer component of the Heuristics-based Gazetteer at 0.74 F1, outperforming the GLiNER fine-tuned on LLM-annotated data (0.66). `TAX_TYPE` is best predicted by GLiNER fine-tuned on the `LLMAD` variant (0.55), whereas `TAX_CONCEPT` is best predicted by GLiNER zero-shot inference at 0.32. Like `LAW`, it remained a difficult label for all approaches. These results indicate that the abstract `TAX_CONCEPT` and `TAX_TYPE` concepts are better handled by an approach that uses span-and-label embedding matching, whereas a gazetteer is best for the more stable surface forms of `JURISDICTION`.

The answer to the main RQ: *What is the best approach to identifying and classifying named entities and concepts in the legal-tax domain?* would be to use GLiNER fine-tuned on LLM-annotated data, as it achieved the highest overall performance. To maximize performance across all NEs and concepts, a suggestion is a hybrid method that

employs a GLiNER-LLM variant as a backbone and uses complementary approaches to maximize performance on target labels, such as a gazetteer for `JURISDICTION`. However, combining different systems like these raises the question of which system's predictions take precedence for overlapping or adjacent spans, an issue evident in the baseline results (Table 4.2), where the SpaCy+RegEX component assigned labels first, reducing the `JURISDICTION` and `TAX_CONCEPT` gazetteer lookup impact. This question is not addressed in this thesis and is instead left to future work.

4.6 Error Analysis for GLiNER-LLM

To investigate the prominent areas of difficulty in predicting on the test set, the best-performing model, GLiNER-LLM, is used to conduct an error analysis. The analysis is presented separately: on a Token-level in Section 4.6.1 to examine NE and concept label-type confusions after converting the model's span predictions into token-level labels; and on a Span-level in Section 4.6.2 to demonstrate entity recognition and boundary errors in the model's predicted spans. Both these approaches complement each other, by examining the different aspects of the model's errors and are addressed separately throughout.

For the token-level analysis, after model predictions were converted to IOB2 labels, the IOB labeling was stripped, and every misclassified token was grouped by its gold label-predicted label type pair. As a result, GLiNER is not evaluated as a token-classification model, but instead for how its span predictions map onto individual tokens. Up to the first 30 examples per gold-predicted label type pair were inspected. Pairs that occurred at least three times were inspected and ranked by frequency to highlight dominant patterns; however, interesting low-frequency patterns were also examined. The token-level confusion matrix is used to show label confusion, counting tokens rather than complete span instances.

For the span-level analysis, only predicted spans with the same label as an overlapping gold instance but with incorrect boundaries were grouped by label for inspection. From these, the first 10 span-level examples per label were inspected. The span-level per-label table from NERVALUATE is used to display correct, missed, and spurious entities or concepts, counting complete-entity-span instances rather than individual tokens.

4.6.1 Token-level Analysis based on Confusion Matrix

The Token-level analysis is conducted first by presenting a confusion matrix that displays the model's label-assignment capability and aims to demonstrate which labels are better predicted and which are sources of confusion. Then, frequently misclassified labels are investigated further for linguistic patterns that may pose difficulties or trigger confusion.

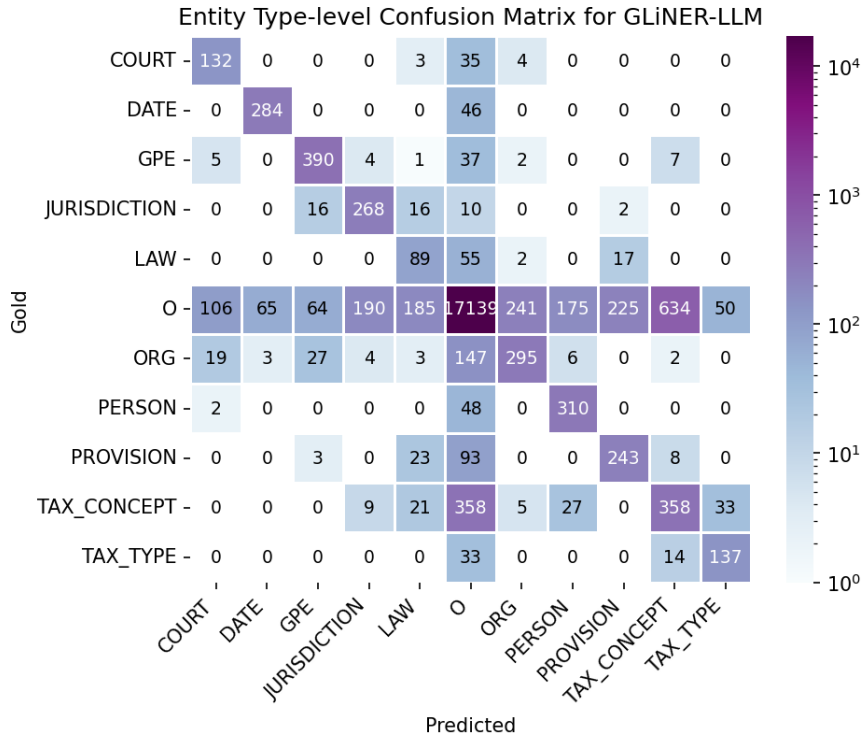


Figure 4.1: Entity-type-level Confusion Matrix for predictions made by the GLiNER-LLM variant on the test set

The confusion matrix in Figure 4.1 shows how the model performs at predicting the correct NE or concept label on a token-level. Specifically, it displays the direction of confusion: which gold labels are predicted as which labels, and their frequency is represented by the numerical values in the non-diagonal cells per gold label.

Across the token-level analysis, the following recurring mechanisms emerge as possible reasons for error: tokens with generic usage in the GLiNER pre-training corpus which can weaken its embedding proximity to the target label; non-English or multilingual tokens which are likely underrepresented in GLiNER's English-only pre-training; gaps in gold label validation which penalize GLiNER's plausible predictions; inconsistencies in LLM annotation as observed in the fine-tuning data set it labeled that were distilled into GLiNER's predictions; and lastly short acronyms which are context dependent for classification. These patterns are discussed in more detail per label where observed.

Generic NEs

'O': The `'O'` label represents tokens that were not part of GLiNER's predicted entity spans, and therefore mapped to `'O'`. This label is seen as confusing for the model in the token-level representation of its predictions. Although the `'O'` labels had 17139 tokens 'predicted correctly', it was also often predicted as other labels, most evidently seen by 634 tokens predicted as `TAX_CONCEPT`, 241 tokens as `ORG`, 225 tokens as `PROVISION`, 175 tokens as `PERSON`, 190 tokens as `JURISDICTION`, 185 tokens as `LAW`, 106 tokens as `COURT`, and 50 tokens as `TAX_TYPE`. Further examination of these errors revealed that GLiNER was mostly predicting valid tokens that were missed in the gold annotations. Examples

include: 'Deduction at source' and 'taxable income' predicted as TAX_CONCEPT, and 'Article 267 TFEU' predicted as PROVISION when their gold labels are 'O' despite other similar instances being tagged with labels. Similarly, tokens such as 'taxpayer' and 'commissioner' were frequently predicted as PERSON across several sentences, and tokens such as 'court' or 'Court' were predicted as COURT. This latter pattern is consistent with the LLM-annotated training data and the prompt used (Appendix D.4), which labeled both general and specific references to courts as COURT, suggesting that GLiNER is reflecting this pattern. This indicates a gold validation gap rather than a genuine model error, as several predictions correspond to plausible, relevant spans that are not labeled in the gold annotations.

DATE: The DATE label tokens achieved fairly straightforward predictions, with 284 tokens predicted correctly, and only 46 tokens missed. The most frequently misclassified tokens were '-', as this token is used for unconventional DATE surface forms, such as in the sentence snippet 'TNS:2007 - 01 - 23', which may not cross the similarity threshold for the DATE label that more conventional surface forms would, for example, '31 January 2013' or '11 Sept. 2008,' which it was able to predict correctly as DATE. However, interestingly, GLiNER also misses some of these forms too, like '1 March 2018' and '30 DECEMBER 1977', showing some inconsistencies in its predictions. Importantly, this highlights GLiNER's span-based architecture, in which candidate spans are scored against the embeddings of the provided entity labels. The results suggest that the highest-scoring DATE span may often be a prototypical entity that embeds more closely to the label, rather than one containing potentially skewing punctuation such as '-'.

PERSON: The PERSON label tokens also yielded straightforward predictions: 310 tokens were correctly predicted, with only 48 missed and 2 predicted as COURT instead. The frequently missed tokens were foreign last names and initials like 'Eckelkamp', 'J. Nymann - Lindegren', 'S.', 'T.' indicating that despite the LLM's ability to label multilingual PERSON entities as seen in the data it labeled, GLiNER's pre-training corpus is English-only (Zaratiana et al., 2024), which means that such spans are likely being less closely represented to the PERSON label embedding in comparison to more frequently represented English names and name shapes would.

GPE: The tokens for the natural language **COUNTRY** label mapped to GPE yielded mixed predictions: 390 tokens were predicted correctly, 37 were missed, 7 were predicted as TAX_CONCEPT, 5 as COURT, and 4 as JURISDICTION. The frequently missed tokens were 'the' when preceding spans such as 'United Kingdom' or 'Netherlands', followed by various acronyms like 'EEA', 'US', 'EU'. This may indicate that spans without 'the' are embedded closer to the COUNTRY label embedding than longer spans containing these components, as seen for DATE instances containing '-'. It was also intriguing that some acronyms were missed ('EEA') during prediction, even though they were labeled in the LLM training data, showcasing inconsistent behavior. Although this should be attributed to the LLM prompt (Appendix D.4) instructing exclusion of articles for this NE, the LLM-annotated data showed that 'the' was included in most GPE spans. **When misclassified as** TAX_CONCEPT tokens, the same pattern emerged: GLiNER paired 'US' with 'GAAP' that followed it, reflecting the training data pattern that tagged the only two instances of 'GAAPs'/'US GAAP' as TAX_CONCEPT, despite 'US' being a gold GPE. **When misclassified as** COURT tokens, the error appears to be a gold-validation gap,

as 'Provinciale 1 di Roma' is likely a COURT entity that was gold-tagged as GPE. The second error is for the 'brussels' in 'Tribunal de première instance of brussels', indicating that the instructions given to the LLM for labeling COURT are being correctly reflected by GLiNER (tribunals are included), but the model is penalized by a gold validation gap as only 'brussels' is tagged as GPE. Lastly, **when misclassified as** JURISDICTION, the same gold label validation errors are seen, the misclassified tokens being 'Member State' that were likely wrongly tagged as GPE.

ORG: The ORG label tokens also posed challenges for prediction, with 295 tokens predicted correctly, 147 tokens missed, 27 tokens predicted as GPE, 19 as COURT, 6 as PERSON, and 4 as JURISDICTION. Missed ORG tokens consisted of either determiners with nationality-modified organizations, such as 'the Netherlands Government', 'the Italian Government'. This may reflect the LLM's inconsistent application of the ORG annotation guidelines, as despite being instructed to exclude organizations with jurisdictional modifiers that localize them, the fine-tuning data was found to tag many instances such as 'the German Government' as ORG. This implies that GLiNER reflects the inconsistent labels inherited during fine-tuning. ORG tokens that were missed also composed multilingual company names, such as 'Denkavit II', 'Danmark', and 'Amurta'. Unknown or non-English tokens in candidate spans may also reduce their similarity to the ORG label embedding, making them less likely to be predicted as ORG entities, possibly due to GLiNER's English-only pre-training. There are also missed cases such as 'FII' and 'Thin Cap Group Litigation' that suggest an entity-coverage issue rather than a labeling error, as both are case names that fall outside LAW's guidelines for excluding case law and judgments and are not otherwise covered by other labels. **When misclassified as** GPE, actual GPE tokens embedded in an ORG span are being singled out, like in 'the Luxembourg Government', 'The Belgian State', and 'Swiss and Belgian tax authorities', reflecting the same inconsistency in following the exclusion criteria as seen earlier, suggesting that GLiNER may be reflecting the inconsistencies inherited from the data. Their gold ORG tag may be a validation gap. **When ORG is confused with COURT entities**, the issue seems to be acronyms that stand for actual courts and words more commonly associated with courts or tribunals. These appear in 9 of 19 misclassified tokens, such as 'ECJ', 'ITAT', which stand for the European Court of Justice and the Income Tax Appellate Tribunal, and in 'Dispute Resolution Panel' and 'Tax Adjustment Board'. These errors likely arise from the ORG label being strongly associated with various acronyms, which clash with the many surface forms of COURT that are also acronyms. **When misclassified as** PERSON, of 6 tokens, 3 were the acronym 'ITO', while the remaining 3 were non-English company names derived from surnames ('Du', 'Pont', 'Lucchini'), making them likely more associated with the PERSON embedding.

Target NEs

COURT: The COURT label tokens exhibit the most straightforward predictions compared to other NE target labels, with 132 tokens predicted correctly, 35 tokens missed, 4 tokens predicted as ORG, and 3 tokens as LAW. In these cases, the 35 tokens that were missed contained generic nouns or adjectives such as 'Chamber', 'Grand', 'The' and 'Court', that are often part of true COURT instances such as 'Grand Chamber' or 'the High Court', which is interesting as the LLM's annotation guidelines included name-constitutive modifiers as part of the span. This suggests that these tokens' span

embeddings, which are likely shaped by their predominantly generic usage in GLiNER's pre-training, do not embed close enough to the COURT label's embedding during fine-tuning. GLiNER demonstrates scattered performance, inconsistently predicting `COURT` for some of these modifiers or missing them otherwise. Furthermore, **when the `COURT` tokens were misclassified as ORG and LAW**, both of which show a similar pattern: errors were limited to acronym tokens. 'ITAT' and 'EC' were incorrectly predicted as `ORG` instead of `COURT`, possibly because acronyms are more likely associated with ORG. **When misclassified as `LAW`**, it seems more like a gold validation gap, with the 'EC' tokens tagged as gold `COURT` entities, when 'EC' refers to the European Commission and particularly to its treaty in its surrounding context (for example: 'same article of the EC treaty', 'the EC treaty provides...'), meaning that the GLiNER predictions are likely correct.

LAW: The `LAW` label tokens display a slightly similar pattern, with 89 tokens predicted correctly, 55 tokens missed, and 17 tokens predicted as `PROVISION`. The most frequently missed tokens were common nouns such as 'Directive', 'Tax', and 'Council', likely because these tokens have stronger generic-use associations in GLiNER's pre-training that prevent them from scoring a similarity high enough with the LAW label embedding. This was interesting as it reinforces the pattern seen for COURT. GLiNER also missed multilingual `LAW` tokens (e.g., 'Lov om indkomstbeskatning af aktieselskaber'), reinforcing that it likely embeds English entities best, which it was solely trained on. **When misclassified as `PROVISION`**, tokens were found to have a unique combination of special characters or common nouns, such as '2000/78' and 'Directive'. Gold validation gaps were also seen where, despite the true `PROVISION` instances not being tagged, GLiNER still predicted them correctly (for example: 'Article 267 TFEU'), sometimes predicting additional tokens for `PROVISION` instances that aligned with the LLM-prompt guidelines, like including 'second paragraph of' that preceded the gold 'Article 52'. This likely indicates that `PROVISION`, as labeled by the LLM following the given annotation guidelines, provides broad coverage, as reflected in GLiNER's performance.

PROVISION: The `PROVISION` label tokens had 243 tokens correctly predicted, 93 tokens missed, 23 tokens predicted as LAW, and 8 tokens as `TAX_CONCEPT`. Since `PROVISION` spans often contain short, low-content tokens such as prepositions, brackets and numbers such as 'of', '(', ')', '56', '63'., these tokens may be noisy and embed too weak to clear the similarity threshold for PROVISION, reinforcing the pattern seen for DATE and COURT, where non-prototypical or low semantic content span forms are not prioritized. Both `PROVISION` and `LAW` are also frequently confused with each other, which is expected, as `PROVISION` is essentially `LAW` but with a specific article or section according to the annotation guidelines (Appendix D.4), so a partial match would look like a `LAW` prediction when it is actually `PROVISION`. For example: 'Article 18 EC Treaty' and '..40 of the EEA Agreement' are gold `PROVISION` spans, but 'EC Treaty' and 'EA Agreement' were predicted as separate `LAW` entities. **When misclassified as `LAW`**, tokens were identified as acronyms or keywords such as 'TFEU', 'EC', or 'Treaty'. Since the examples provided to the LLM for both label definitions include overlap, like 'TFEU' vs 'Article 49 TFEU', it explains why there might be bidirectional confusion. Another source of confusion is also seen **when misclassified for `TAX_CONCEPT`**, occurring for the phrase 'Free movement of capital', suggesting that `PROVISION` spans

that include tokens related to tax-based terminology can possibly confuse GLiNER.

This tendency to select longer predicted spans for PROVISION is consistent with the test set token distribution (Appendix A.2), with 283 I- tokens versus only 87 B- tokens, indicating that its entities are substantially more multi-token than those of the other labels. This means that PROVISION spans are more likely to have predicted candidate spans that diverge from the gold span.

Target Concepts

TAX_TYPE: The TAX_TYPE label tokens show the least prediction difficulty compared to the other concept labels, with 137 tokens predicted correctly, 33 tokens missed, and 14 tokens predicted as TAX_CONCEPT. The misses seen were 'inheritance', 'gift tax', 'state tax', which interestingly appeared elsewhere in the test set and were predicted correctly otherwise. The errors occurred particularly when the tax-type-specific words are separated from 'tax' by conjunctions like 'inheritance *and* gift tax' and 'taxes *on* inheritance and gifts', causing the conjunction 'and' to be frequently excluded from the predicted spans. This indicates that the isolated function words within valid entity spans may make longer spans less likely to be selected, resulting in GLiNER selecting a shorter span that excludes certain obvious tokens.

TAX_CONCEPT: The TAX_CONCEPT label tokens, on the other hand, show a different pattern: 358 tokens are correctly predicted, and 358 are missed. Furthermore, it had 33 tokens predicted as TAX_TYPE, 27 tokens as PERSON, 21 tokens as LAW, 9 tokens as JURISDICTION, and 5 tokens as ORG. GLiNER is shown to predict TAX_CONCEPT instances that are tagged as 'O' in the test set (634 tokens), including instances such as 'Deduction at source', 'dividends', 'taxable income', which interestingly enough, do have gold TAX_CONCEPT tags otherwise across the test set, indicating gaps in the gold validation. Conversely, GLiNER also misses some tax-related common noun tokens, such as 'fraud', 'abuse', and 'avoidance', even though similar terms appear as part of correctly predicted gold TAX_CONCEPT tokens elsewhere in the test set, indicating inconsistency in handling borderline concept terms that appear as generic nouns. This is consistent with the recurring pattern that terms with a predominantly generic non-tax use may be more difficult for GLiNER to distinguish as TAX_CONCEPT instances. **When misclassified as** PERSON, the 'taxpayer' tokens accounted for 26 of the 27 misclassifications. The DeBERTa SentencePiece tokenizer (Appendix F.1) encodes 'taxpayer' and 'taxpayers' as single tokens, rather than segmenting them into 'tax' with a suffix, unlike correctly predicted instances like 'withholding tax' and 'tax credit', where 'tax' is a separate subword. This tokenization difference cannot alone establish the reason for the error. Looking into the fine-tuning data revealed that the LLM also labeled 'taxpayer' and 'taxpayers' as PERSON 171 times compared to TAX CONCEPT only 39 times, offering another stronger possible explanation for why these tokens were closely associated with the PERSON label. **When misclassified as** LAW, the tokens comprised legal-instrument nouns such as 'treaty' and 'APA' (Advance Pricing Agreement), reflecting a surface-form overlap between the two labels. **When misclassified as** JURISDICTION, the tokens contained 'residents', which is close to 'residence state', an example used for the LLM to label JURISDICTION instances, also seen in the fine-tuning data. **When misclassified as** ORG, GLiNER was predicting tokens that made general organizational references, such as 'tax authorities' and 'taxpayer company', which indicates that the presence of the 'tax' alone might be insufficient to distinguish

between TAX_CONCEPT and other entity types. However, this is also consistent with the LLM fine-tuning data, where 'tax authorities' instances are also labeled as ORG.

TAX_CONCEPT and TAX_TYPE also seem difficult for the model to distinguish, with 33 TAX_CONCEPT tokens predicted incorrectly as TAX_TYPE and 14 TAX_TYPE tokens predicted incorrectly as TAX_CONCEPT. This can be explained in part by the fact that although the label definitions provided in the prompt were framed as mutually exclusive, some expressions are inherently ambiguous between the two concepts. As a result, this ambiguity may also be learned by the encoder fine-tuned on such representations. Furthermore, TAX_TYPE instances followed a consistent pattern of containing the word 'tax', seen in 65 out of 86 TAX_TYPE instances, in examples like 'social solidarity tax', 'turnover based tax', and 'state tax'. This lexical pattern may contribute to the confusion between TAX_TYPE and TAX_CONCEPT, particularly for the TAX_CONCEPT spans that also contain 'tax'. This is seen in gold TAX_CONCEPT instances predicted as TAX_TYPE like 'withholding tax', 'shareholder's tax credit', and 'higher tax burden'. As the TAX_TYPE prompt annotation guideline actually counts 'withholding tax' as an example, it indicates that the gold labels might have validation gaps. Contrastingly, the gold TAX_TYPE tokens predicted as TAX_CONCEPT consisted of generic nouns, such as 'royalty', 'solidarity contribution'. 'GAAP' was another example, which is interesting since it was provided as an exclusion example when defining the TAX_CONCEPT label in the prompt annotation guidelines. Upon checking the LLM-fine-tuning data, it was found that the LLM tagged 'GAAPs'/'US GAAP' as TAX_CONCEPT, indicating that the LLM overrode the exclusion criteria (Appendix D.4) (which listed 'GAAP' as an exclusion for TAX_CONCEPT) when generating the training labels, which may potentially cause GLiNER to associate such an acronym with TAX_CONCEPT.

JURISDICTION: The JURISDICTION label tokens posed slightly less confusion for the model, with 268 tokens predicted correctly, 10 tokens missed, and 16 tokens each predicted as LAW and GPE. In the missed predictions for JURISDICTION, the frequently misclassified tokens are 'Member', 'EU', and 'the', indicating inconsistencies with GLiNER, as instances such as 'Member State' or 'Home State' were otherwise correctly predicted. **When misclassified as** GPE, instances such as 'Republic of', 'Kingdom of', and 'European Union' align with the LLM-annotated fine-tuning data, which correctly reflect the annotation guidelines exclusion criteria for JURISDICTION: label as GPE for named countries even in jurisdictional contexts, indicating that these predictions are likely correct but penalized due to validation gaps. All 16 of the tokens misclassified as LAW were 'European', 'Union' inside the predicted 'Charter of Fundamental Rights of the European Union', suggesting that the extended span may be more strongly associated with the LAW label than the shorter span expected by gold annotation.

4.6.2 Span-level Analysis

Since the preceding analysis investigated token-level classification independent of span boundaries, this Span-level analysis uses NERVALUATE's matching-type classification report. This analysis aims to test recognition, where a prediction should match both the gold span and label. The errors assign each prediction a category out of five: correct, incorrect, partial, missed, and spurious, and are evaluated by the extent to which the predicted spans match the gold spans.

The manually calculated macro-averaged[1] and micro-averaged strict-match F1-scores under NERVALUATE (0.53 and 0.52 respectively) match the corresponding SEQEVAL strict scores reported in Appendix E.5, providing a cross-check between the two evaluation tools. NERVALUATE is therefore used to break down scores into boundary and label errors. The matching scheme results are reported at the micro average to reflect aggregate performance across all labels, while the per-label results are reported separately and then subsequently averaged for the grouped analyses, to ensure that low-frequency labels are also represented equally.

Table 4.9: NERVALUATE matching schemes for GLiNER-LLM (micro-level, pooled over all labels). C=correct, I=incorrect, P_{art}=partial, M=missed, S=spurious. *strict* requires exact boundaries and the correct label; *ent_type* requires the correct label with any span overlap; *exact* requires exact boundaries, ignoring the label; *partial* credits partial span overlap, ignoring the label

Scheme	C	I	P_{art}	M	S	Prec.	Rec.	F1
`ent_type`	1209	174	0	305	1052	0.50	0.72	0.59
`exact`	1176	207	0	305	1052	0.48	0.70	0.57
`partial`	1176	0	207	305	1052	0.53	0.76	0.62
`strict`	1070	313	0	305	1052	0.44	0.63	0.52

Table 4.9 displays GLiNER's prediction performance in matching scheme types, isolating prediction errors by removing one requirement at a time. To start, the `strict` score, which requires both correct labels and span boundaries, achieves an F1-score of 0.52, with 1070 of 1688 entities and concepts correctly predicted. Removing the label requirement while keeping the correct boundaries raises this to 1176 correct instances, as indicated by the `exact` score (0.57), whereas removing the boundary requirement instead increases this to 1209 correct instances (0.59), as indicated by the `ent_type` score. This difference in performance suggests that GLiNER's errors are more frequently associated with span boundaries than with label errors, implying that the model generally assigns the correct label when the predicted span matches a gold entity, while boundary mismatches account for a larger proportion of the remaining errors, as indicated by the `partial` score. This may be because GLiNER selects candidate spans based on semantic similarity to the target NE or concept, which can lead to shorter or longer spans that are more strongly associated with the target label.

Label errors are therefore present but in the minority; removing the boundary requirement in `ent_type` increases the F1-score by 0.07, compared to an increase of 0.05 when removing the label requirement in `exact`. The `partial` scheme, which credits only partial overlap between the prediction and gold boundaries, reaches 0.62, suggesting that many boundary errors are partial overlaps rather than completely incorrect.

[1]NERVALUATE reports micro-averaged scores directly; macro-averaged scores were calculated by taking the unweighted mean of the per-label precision, recall, and F1-score values returned.

Table 4.10: Span-level (strict) per-label counts and F1-score for GLiNER-LLM. C=correct, I=incorrect, M=missed, S=spurious

Label	**C**	**I**	**M**	**S**	**F1**
Named entities					
COURT	78	7	21	95	0.55
DATE	108	8	16	41	0.75
GPE	252	17	30	88	0.77
LAW	32	6	27	123	0.28
ORG	108	28	85	142	0.43
PERSON	150	4	32	150	0.61
PROVISION	44	9	34	71	0.42
Concepts					
JURISDICTION	130	4	19	108	0.66
TAX_CONCEPT	114	47	192	362	0.26
TAX_TYPE	54	11	21	44	0.55

Table 4.10 shows a per-label count from the NERVALUATE `strict` evaluation. The counts in this table are not directly comparable to those in Table 4.9, as boundary errors can contribute to both the gold label's missed count and the predicted label's spurious count. Therefore, only the correct counts sum up accordingly across labels. The error counts in Table 4.10 reveal some trends in individual performance. The lowest-scoring labels `TAX_CONCEPT`, `LAW`, and `ORG` are dominated by missed or spurious predictions, each exhibiting a different source of difficulty. It can be seen that these lower-scoring labels also pose difficulty for GLiNER-LLM at the span level, as they did at the token-level: `TAX_CONCEPT` which contains abstract surface forms that vary in length, `LAW` which often contains common-noun tokens, and `ORG` where modifiers can affect span boundaries.

Table 4.11: Span-level (strict) macro-averaged F1-score by group for GLiNER-LLM, computed as the mean of member-label F1-scores. Generic NEs: DATE, GPE, ORG, PERSON. Target NEs + concepts: COURT, LAW, PROVISION, JURISDICTION, TAX_CONCEPT, TAX_TYPE

Group	**Macro F1**
Named entities	0.54
Concepts	0.49
Generic NEs	0.64
Target NEs + concepts	0.45

Table 4.11 reports grouped macro F1-scores, which assign equal weight to each label. It displays grouped performance by GLiNER using the NERVALUATE strict matching scheme, grouping the labels in two ways: NEs versus concepts, and generic NEs versus all target NEs and concepts. NEs (0.54) are predicted better than concepts (0.49), and generic NEs (0.64) substantially outperform the combined target NEs and concepts group (0.45). As the per-label scores in Table 4.10 showed, well-predicted generic NEs such as `GPE` (0.77), `DATE` (0.75), and `PERSON` (0.61) boost the NE average, likely due to their relatively stable span forms. Within the target NEs, `LAW` (0.28) and `PROVISION` (0.42) are particularly difficult compared to `COURT` (0.55). The low performance of

concepts can be attributed to TAX_CONCEPT (0.26), whose poor predictions diminish how much better JURISDICTION (0.66) and TAX_TYPE (0.55) are predicted. The gap between generic NEs (0.64) and the target NEs and concepts (0.45) suggests that GLiNER handles conventional NEs better but struggles with domain-specific target NE and concept labels.

The scheme comparison in Table 4.9 established that boundary errors dominate. To inspect these boundary errors directly, we collected all predictions that shared the same label with an overlapping gold span but lacked matching span boundaries, resulting in 167 boundary mismatches. These mismatches occurred predominantly for TAX_CONCEPT spans (61), followed by ORG (33), GPE (17), TAX_TYPE (14), and PROVISION (9). TAX_CONCEPT mismatches are nearly entirely truncations, where GLiNER selects a shorter candidate span that retains the most prototypical part of the label, dropping the modifiers or complements part of the span, such as 'double taxation unilateral relief', 'economic double taxation', 'the avoidance of double taxation' as only 'double taxation', and 'free movement of capital' as only 'free movement'. This pattern reflects the candidate-scoring mechanism first observed at the token level, suggesting that spans containing more prototypical tokens may be preferred over fuller spans with additional modifiers or complements.

PROVISION mismatches, in contrast, exhibit extended span predictions: gold PROVISION entities 'Article 242 QUATER' captured as 'Article 242 QUATER OF THE CODE GENERAL DES IMPOTS' and 'Article 52 OF THE EEC TREATY' as 'SECOND PARAGRAPH OF ARTICLE 52 OF THE EEC TREATY', indicating that shorter spans do not always embed closest to the label. These predictions reflect GLiNER's distillation of the LLM annotation guidelines, which asked for specific articles, sections, paragraphs, or sub-paragraphs for PROVISION spans.

GPE and TAX_TYPE mismatches also exhibit patterns seen on the token-level, dropping determiners or conjunctions: 'the Kingdom of Denmark' → 'Kingdom of Denmark' and 'inheritance and gift tax' → 'gift tax'. These examples suggest that GLiNER may favor different span lengths for each label: shorter for TAX_CONCEPT and longer for PROVISION.

Many of the boundary mismatches involved gold spans consisting of common nouns and prepositional modifiers. Such spans seem to be weaker cues for the label embedding, potentially causing GLiNER to settle on partially correct span candidates that embed more closely. Interestingly, casing did not appear as an obvious source of boundary errors in the examples examined, as the observed mismatches were mainly with function words, modifiers, and common nouns.

Taking the token-level and span-level observations together, the errors investigated cluster into the following categories: First, GLiNER's embedding-matching mechanism appears to contribute to its candidate spans being truncated or extended beyond the annotated spans to closely match the label embedding (TAX_CONCEPT truncation and PROVISION extensions). This may be related to the linguistic nature of these concept and NEs, which frequently contain common nouns, modifiers, and function words that likely provide weaker cues for distinguishing between shorter and longer candidate spans. Second, GLiNER also predicts valid concepts that are inconsistently labeled in the test set (TAX_CONCEPT spurious predictions), resulting in apparent over-predictions in the evaluation. These over-prediction errors cannot be attributed solely to GLiNER, as the model was fine-tuned on LLM-annotated data that demonstrated inconsistent labeling patterns. These predictions also reflect the gaps in gold-label validation that

penalize appropriate predictions. Third, adjacent labels exhibit genuine semantic overlap (e.g., `TAX_CONCEPT` versus `TAX_TYPE`, `LAW` versus `PROVISION`), which can be difficult for GLiNER, which uses embedding similarities for matching. Fourth, LLM prompting that fails to cover niche yet essential instructions, such as multilingual `ORG` or `PERSON` entities, combined with inconsistent guideline application, can introduce inconsistencies into the training labels and hinder the model's downstream performance. These findings suggest that improvements to the LLM-annotation process, such as prompt refinement and re-engineering to explicitly handle ambiguous surface forms and multilingual entities, can increase annotation quality, which may improve downstream performance. Additionally, the evaluation may become more representative of the model's performance if the gold-annotation validation gaps are mitigated.

Chapter 5

Discussion & Conclusion

In this chapter, the results from Chapter 4 are first reflected upon in Section 5.1 in relation to previous work mentioned in Chapter 2 to address the main RQ: *"What is the best approach to identifying and classifying named entities and concepts in the legal-tax domain?"*, highlighting the findings for each sub-research question as well. This is organized by approaches: baselines, gazetteers, zero-shot and few-shot, and fine-tuned encoders. Then, recommendations for the company on improving its automated NERC pipeline are provided in section 5.2, followed by the limitations addressed in section 5.3. Lastly, some future work suggestions are listed in section 5.4, and the chapter ends with a conclusion in section 5.5.

5.1 Main Observations

This section discusses the main observations and implications in the results produced for this thesis in the order of: baselines, gazetteers, zero-shot and few-shot, and fine-tuned encoders, also drawing on the error analysis insights found.

Baselines: Addressing SRQ1's baseline component, the baselines (Table 4.1) reveal that the current company baseline performs poorly on the domain-specific labels targeted in this study, achieving an F1 score of 0.37. Although the pipeline benefits from combining the generic SpaCy NERC model, RegEx rules, and gazetter matching, its performance is driven mainly by generic NEs and the concept of `JURISDICTION`. As such, it poorly addresses the target labels (`TAX_CONCEPT` and `TAX_TYPE`: 0.12 F1 score, `LAW`: 0.02 F1 score). This suggests that while the individual components are complementary, their combination does not sufficiently bridge the gap between general-domain NERC and the specialized legal-tax domain vocabulary, variation, and structures. While SpaCy provides useful coverage for the general NEs it is trained on, its performance does not extend well to the target labels, possibly due to domain-specific variation in entity forms, aligning with the claim made by Karamitsos et al. (2025). Furthermore, SpaCy predictions take precedence over gazetteer matches when they overlap, meaning that plausible matches can be suppressed by generic entity predictions. This means the gazetteer's effectiveness also depends partly on how its predictions interact with other components. As a result, the company baseline establishes that simply combining general-domain NERC tools with rule-based and gazetteer matching is insufficient to provide consistent coverage across the challenging legal-tax domain annotation scheme.

Gazetteer approaches: For SRQ2 addressing how gazetteers perform (Table 4.2), it is seen that the Heuristics-based Gazetteer outperformed the Company Gazetteer overall (by 0.04 F1 points), attributed to the better coverage of the target concept `TAX_TYPE` which saw a performance gain (0.48 vs 0.11). Although this advantage is retained when the gazetteers are integrated with the SpaCy+RegEx components, interactions between the gazetteer matches and the other components also affect performance, with drops in performance seen for `JURISDICTION` and `TAX_CONCEPT`. The overall findings indicate that lookups work better for concepts with surface-form regularity and recognizable surface forms, where lexical variation can be represented in the gazetteer, as seen by the highest performance for `JURISDICTION` across all approaches. The stronger `TAX_TYPE` performance of the Heuristics-based Gazetteer suggests that a lookup can be effective when it is made for concepts that exhibit recurring lexical patterns, whereas the more abstract `TAX_CONCEPT` is more constrained by the strictly lexical nature of gazetteer matching, as without consistently predictable surface forms, explicit string matching has limited generalization on unseen concept instances. In sum, this experiment's findings revealed that gazetteer matching alone cannot handle the complexities of concept extraction specifically for abstract concepts.

Out-of-the-box zero-shot approaches: For SRQ3's zero-shot component, out-of-the-box model (Table 4.3) zero-shot performances demonstrated by GLiNER (0.38) and GPT-5.4-mini (0.39-0.42) were better than the company baseline (0.37), bolstered by predicting the generic NEs well. These results broadly support the findings by Bogdanov et al. (2024), in which LLMs performed well in zero-shot NERC inference. Despite GLiNER performing only 0.01 F1 points better overall, it showed an enhanced capability to predict labels and concepts that were nearly undetected by the company baseline, such as `LAW` (0.21 vs 0.02 F1) and `TAX_CONCEPT` and `TAX_TYPE` (0.32 and 0.45 vs 0.12 and 0.12 F1). These results indicate that GLiNER's embedding-matching mechanism helps it address concepts more effectively than LLMs, whereas LLMs are able to handle NEs more effectively, perhaps due to their pre-existing understanding of the task and entities (Huang et al., 2025). This pattern also appears in recent studies showing that, in zero-shot conditions, LLMs perform better on well-structured and general entity types (Haq et al., 2025), while struggling with complex, ambiguous, or domain-specific entities (Haq et al., 2025; Hu et al., 2024). While the found results are not directly comparable to the CE studies by Norouzi et al. (2025) and Yang et al. (2025), it was interesting to see that GPT-5.4-mini, a newer and more advanced model, yielded CE performances in the same ranges as those reported by the authors when using older models such as GPT-3.5-Turbo, GPT-3.5 and GPT-4.o-mini, indicating that `TAX_CONCEPT`, `TAX_TYPE`, and `JURISDICTION` behaved as typical concepts and were equally difficult to extract, represented by their highest F1-scores similarly not exceeding 0.30, 0.23, 0.31 respectively. These findings indicate that legal-tax domain-specific NERC and CE remain a difficult task for LLMs to perform in the zero-shot condition.

Out-of-the-box few-shot approaches Additionally, for SRQ3's few-shot component (Table 4.4), it was found that adding a few-shot adjustment to the zero-shot prompts resulted in an overall improvement in performance (zero-shot performance ranges: 0.38-0.42 F1 $\rightarrow$ few-shot performance ranges: 0.45-0.50 F1) regardless of the prompt output structure. Including examples in the prompts improved performance for most NEs, as also observed by Hu et al. (2024) and Wang et al. (2025). Concept

performance also improved with few-shot examples, boosting JURISDICTION from 0.30 → 0.63 F1 and TAX_TYPE from 0.45 → 0.48 F1, but dropping slightly for TAX_CONCEPT from 0.32 to 0.30 F1. These findings aligned with studies by (Norouzi et al., 2025; Yang et al., 2025) that found few-shot prompting (one-shot specifically) to improve CE performance substantially compared to minimal instruction in the zero-shot condition.

Huang et al. (2025) argued that explicitly defining the entities through concise guidelines is more effective than relying on examples to convey input-to-label mappings or excessive rules that crowd the context window and hinder the model's ability to follow instructions. Our findings partially aligned with this, as in zero-shot settings, simpler guidelines outperformed detailed ones (0.42 F1 vs 0.40 F1), suggesting that instruction clarity is slightly impaired by prompt length. However, in the few-shot setting, detailed guidelines paired with examples achieved the highest performance, indicating that examples and guidelines are complementary, as Hu et al. (2024) also found. Additionally, Huang et al. (2025) suggested that LLMs already possess a general understanding of entity classes; however, our results indicate that, for certain abstract and semantically overlapping labels, examples were more effective than guidelines alone, as evidenced by the increase in performance for TAX_TYPE and JURISDICTION when prompts included examples. This implies the key role examples play in helping LLMs better identify concepts. Furthermore, based on the zero-shot and few-shot performance, the GPT-NER and IDX:TOK prompt output structures achieved comparable F1-scores overall: 0.38 and 0.39 vs 0.40 and 0.42 for zero-shot; 0.48 vs 0.45 and 0.50 for few-shot, respectively. A more consistent pattern was that GPT-NER tended to perform better on PERSON, while IDX:TOK is stronger on concepts, despite sharing the same annotation guideline instructions. These findings highlight that LLMs' NERC and CE inference improves substantially when prompts include sufficient instructions and examples to guide predictions. This is likely because the target label set contains challenging entities and concepts, allowing examples to compensate when guidelines may not suffice.

Approximate legal domain data for fine-tuning: Addressing SRQ4's first training data condition, of using existing legal data for fine-tuning encoders (Table 4.5) revealed that it provided exposure to legal-domain data, but not sufficiently to consistently improve performance over the company baseline (0.37 F1), as only GLiNER was able to exceed it (0.44). Legal-BERT's consistently stronger performance across all target NE labels compared to RoBERTa underscored the benefits of domain-specific pre-training. Simultaneously, the limits of domain matching are also visible since the existing legal domain knowledge is approximate to the legal-tax domain. As mentioned by Kulkarni et al. (2016); Zhang et al. (2021), NERC performance can degrade across domains due to differences in language conventions, word usage and distributions, sentence patterns, and entity representations. These properties are possibly why ELD, which encompasses general case law and legal company filings, does not reflect the characteristics of the tax case law data. This is particularly evident for the target NEs LAW and PROVISION, as, despite being represented in the source data, they remain difficult for RoBERTa and Legal-BERT. (Singh et al., 2026) also mentions that incorporating multiple genres of data during training can dilute the target-domain-specific signal, which is possibly what happened to the subset of tax case law included in the ELD. The more visible difficulties arise because the concepts JURISDICTION, TAX_CONCEPT, and TAX_TYPE are not covered in the ELD, and are consequently not

predicted by RoBERTa and Legal-BERT. This emphasizes the domain dependence of concepts compared to NEs, as NEs seem to recur across subdomains; however, concepts are more domain-constrained. The overall findings indicate that related domain-specific fine-tuning is informative but only in the areas where the source and target domains overlap.

LLM-annotated target domain data for fine-tuning: For SRQ4's second training data condition, using LLM-annotated target-domain data for fine-tuning (Table 4.6), resulted in each encoder achieving its best overall performance. All models saw improvements across every concept and nearly every NE label, likely attributable to domain-matched data, in alignment with Au et al. (2022), who saw legal NERC performance improve when fine-tuning and testing data belonged to the same domain. This is likely supported by the alignment in annotation guidelines that also provides the model with the needed coverage of target NEs and concepts. Even RoBERTa, the consistently under-performing encoder model, saw overall gains exceeding those of its existing human-annotated legal data fine-tuning variant, indicating that general-domain pre-training found LLM-annotated target domain data more informative to learn the task from than existing legal domain data. These variants performed better for `DATE`, `GPE`, `ORG`, and `PROVISION`, and `TAX_CONCEPT` and `TAX_TYPE`, reflecting the strengths of the LLM few-shot prompt, distilled into the encoder's downstream predictions. These results also align with findings by Yazdani et al. (2026) in the biomedical context, who found that pre-training GLiNER on LLM-annotated NERC data ultimately outperformed the LLM used to label its training data, highlighting the effective distillation of LLM knowledge into the encoder models trained on it.

The error analysis of GLiNER's predictions under this condition, which was its best performance overall (0.53), showed that boundary mismatches were more prevalent at the span-level than label confusion. This was attributed to GLiNER's architecture, which would overlook expected candidate spans to select the span most closely associated with the target label. The analysis also reveals GLiNER errors that originate from LLM labeling due to inconsistent application of prompt instructions, showing that certain annotation guidelines, such as inclusion and exclusion criteria, can be challenging for the LLM to apply, as also found by Keloth et al. (2024). Additionally, several overpredictions made by GLiNER in this fine-tuning condition pointed out gold-label validation gaps that dismissed valid predictions. GLiNER's span label embedding-matching architecture, which leverages plausible spans and distilled knowledge from the LLM-annotated target domain data, is likely a cause of these predictions.

Finally, Arzideh et al. (2025) and Matkin et al. (2024) found that traditional encoder models, particularly those pre-trained or fine-tuned on domain-specific data, outperformed LLM inference on NERC and skill extraction, a task similar to concept extraction. These findings were partially supported by our study: we found that Legal-BERT outperformed LLM inference when fine-tuned on target-domain data; however, RoBERTa still lagged behind, possibly due to its general domain pre-training. In sum, the results from this experiment indicate that target-domain fine-tuning improves the encoder performance of the domain-specific NERC and CE.

Mixed approximate domain and LLM-annotated target domain data for fine-tuning: Under SRQ4's third condition of using Mixed approximate domain and LLM-annotated target domain data for fine-tuning (Table 4.7), a drop from the high

performance for nearly all the NEs and concepts was exhibited in comparison to the LLMAD variants of models. Specifically, LegalBERT demonstrates this for the LAW NE label: its LLM-annotated data variant performed best for LAW (0.28), but when combined with approximate legal domain data in the mixed variant, its performance showed an overall drop (0.21). As the mixed data contains both varying domains and annotation sources, this drop may reflect conflicting labeling conventions (Keloth et al., 2024) between LLMs and human annotators, as well as broader differences (Liu et al., 2021) between legal (sub)domains, which may vary between tax case law and general legal documents. Liu et al. (2021) also states that jointly fine-tuning on both source and target domain data can hurt performance since models optimizing for two different domains struggle to concentrate on the target domain specifically. However, their study also found that this joint fine-tuning setting still outperformed fine-tuning directly on the target domain alone, which our findings do not align with, as the mixed condition underperformed the LLMAD fine-tuning data condition. These differences can be attributed to the authors' use of considerably smaller target-domain fine-tuning datasets, which may fail to provide models with sufficient signals to learn a domain-specific task. This may also be related to the nature of their source domain, which was a general domain dataset that may provide broader coverage of generic entities shared across domains, compared to the narrower related domain dataset we used as the source domain. The overall findings from this experiment indicate that combining datasets to boost target-domain label coverage may not benefit the model due to conflicts between domain characteristics and annotation conventions.

Domain adaptation: Domains play a key role in these findings. Legal-BERT's legal pre-training gives it a clear advantage over RoBERTa, as it outperforms the latter across all fine-tuning conditions. Being pre-trained on a broader, related domain allows Legal-BERT to learn contextual embeddings through an overall legal lens, making it better equipped to address the target subdomain due to similarities in complex sentence structures, jargon, and content. Interestingly, the legal pre-training also enables it to predict the legal-oriented concept of JURISDICTION better than other models when fine-tuned to detect it, although explicit tax concepts perform worse, as expected.

Additionally, domain-matching in fine-tuning has a large impact, as seen in the progression in performance from using approximate-domain data to mixed-domain data to target-domain data. The highest performance was achieved by all models when fine-tuned on the tax case law dataset labeled by the LLM, although this effect cannot be fully separated from the annotation sources (see Section 5.4 for a proposed method to examine this). The performance differences observed when using target-domain data for fine-tuning in comparison to approximate-domain data align with Kulkarni et al. (2016); Zhang et al. (2021); Au et al. (2022).

Furthermore, while both the approximate-domain data and mixed-domain data were composed by combining different domain datasets, the findings indicate that doing so to provide more target label coverage to the models is not always beneficial, as conflicting distributions and non-related domains can dilute the target training signal needed for optimal model performance (Keloth et al., 2024; Singh et al., 2026). These findings suggest that related-domain data can still be useful in providing some target-domain coverage, as seen in both the ELD conditions and Legal-BERT's pre-training, but ultimately, matching between the target domain and annotation scheme provides a stronger basis for NERC and CE.

Based on all the approaches and their results presented in this study, the main RQ is answered: the GLiNER model fine-tuned on LLM-annotated target domain data performed the best in NERC and CE, with a macro-average F1 score of 0.53, likely due to its span-embedding matching mechanism, which allows it to recognize and classify NEs and concepts better than other traditional approaches and architectures. The patterns demonstrated in the error analysis of this model indicate that the errors are present in both evaluation and training labels, rather than in model shortcomings, thereby motivating the limitations discussed in the upcoming sections.

5.2 Recommendations for the Company

The following proposed solutions aim to answer the use-case query: *how can the company's current baseline be improved?*

While it would be ideal to have a system that accurately labels NEs and entities in tax case law, the approaches tested in this thesis are not yet that well-performing, with the highest overall F1-score observed being 0.53. Since model outputs can still aid annotators by pre-annotating entities, it is in the company's best interest to prioritize approaches that achieve higher recall. Thus, the model is not expected to produce final entities and concepts, since its predictions would undergo manual review by the annotators; instead, it should present as many candidate entities as possible rather than fewer correct ones. In line with Wang et al. (2024), this would reduce overall human effort. A missed entity requires an annotator to notice the omission and add the label from scratch, which requires close reading of the dataset, whereas a spurious entity only requires the annotator to reject or accept the candidate, which takes comparatively less effort. However, this should be approached with caution, as labels with weak recall and precision will likely yield very few correct candidates in pre-annotation.

The main suggested approach would be to use the best-performing model (0.53 macro F1 score), GLiNER fine-tuned on LLM-annotated data, which had the highest total recall (0.65). Following, the `JURISDICTION` gazetteer can be applied as a post-processing step to correct any misclassifications, as proposed by Mehta (2025). To first gauge the utility of this pipeline, the lookup step can be applied to the GLiNER-LLM's predictions on the test set, and the results can then be evaluated against the gold labels.

In Table 5.1, the best-performing approach in terms of recall is presented alongside its respective label. Secondary suggestions are also proposed, since no single approach consistently predicted the highest score for each label, including GLiNER-LLM.

As seen in Table 5.1, the best-performing approaches differ. Another suggestion that takes each label into account is to use label-specific approaches: Legal-BERT-LLM for `COURT`, `ORG`; GLiNER-mixed for `DATE`; Legal-BERT-mixed for `GPE`; GPT-NER few-shot for `PERSON`, IDX:TOK few-shot for `PROVISION`, and the jurisdiction gazetteer for `JURISDICTION`. `LAW` and `TAX_CONCEPT` are excluded here due to low precision and discussed below. Given that the fine-tuned encoder models take precedence, showing the highest recall for six of ten NEs and concepts, using the three highest-F1 models (GLiNER-LLM, Legal-BERT-LLM, GLiNER-mixed) would be another alternative if the preference is to reduce reliance on LLM inference. This can be achieved by creating a majority-voting ensemble, which is briefly addressed in Section 5.4 but ultimately left to future studies.

`LAW` is considered an exception because no approach achieves usable precision. Even GLiNER, which was fine-tuned on `ELD` and achieved the highest recall for the label

Table 5.1: Recommended pre-annotation approach per entity class, selected by recall to prioritize entity coverage over precision. `ALDD` = Approximate Legal Domain Data. LLMAD = LLM-annotated data, ZS = zero-shot, FS = few-shot

Entity	Recommended approach	R	P	F1	Sup.
Named entities					
COURT	Legal-BERT (LLM)	0.79	0.41	0.54	106
DATE	GLiNER (mixed)	0.86	0.62	0.72	132
GPE	Legal-BERT (mixed)	0.85	0.66	0.74	299
LAW	GLiNER (`ELD`)	0.66	0.18	0.28	65
ORG	Legal-BERT (LLM)	0.57	0.42	0.48	221
PERSON	GPT-NER FS (simple)	0.82	0.79	0.80	186
PROVISION	IDX:TOK FS (detailed)	0.68	0.37	0.48	87
Concepts					
JURISDICTION	Gazetteers only	0.86	0.65	0.74	153
TAX_CONCEPT	GLiNER (mixed)	0.36	0.29	0.32	353
TAX_TYPE	GLiNER (LLM)	0.63	0.50	0.55	86

(0.66) only yielded 0.18 precision, making it likely that predictions made for this label will be higher in false positives than false negatives. This was also observed in the GLiNER-LLM span-level error analysis in Chapter 4, where spurious label predictions outnumbered missed entities (123 vs 27) for the label. This would undermine the purpose of helpful pre-annotation, as annotators would have to sort through excessive spurious candidates. Similarly, TAX_CONCEPT is also the weakest performing concept overall, with the lowest recall (0.36) and lowest precision (0.29); meaning that few true instances are identified, and a majority of what is identified is incorrect (362 spurious vs 192 missed in the GLiNER-LLM error analysis), raising the same issue as `LAW`.

5.2.1 LLM-annotation for target labels

Fine-tuning models with LLM-annotated target domain data yielded greater improvement than when fine-tuned with human-annotated approximate legal domain data. While these differences can be likely attributed to domain matching instead of the annotations, the synthetic labels prove to be a solution for when labeled data for target concepts are extremely scarce (like `TAX_TYPE`, `TAX_CONCEPT` which were not found in available existing datasets), as fine-tuned encoder model performances increased by substantial F1 points with this dataset compared to with the ELD (RoBERTA: 0.10 $\rightarrow$ 0.33, Legal-BERT: 0.26 $\rightarrow$ 0.51, GLiNER: 0.44 $\rightarrow$ 0.53). The trade-off is that LLM labels may be noisier than human-annotated labels, as the error analysis shows: GLiNER predicted spuriously, possibly misled by entities incorrectly labeled in the LLM-annotated training data. A similar phenomenon was observed in Wang et al. (2025)'s work, causing the authors to prompt the LLM post-inference to verify the entities they had labeled, as they often labeled irrelevant entities simply because they had knowledge of the entity label. If the company intends to run LLMs for NERC and CE inference, this validation step may help reaffirm the annotation guidelines to the model and reduce incorrect annotations.

5.2.2 Compute considerations

Using LLMs like GPT-5.4-mini for inference costs money, and since the intention is to extract NEs and concepts for all documents in the corpus, the company should estimate per-document labeling costs against its API pricing before committing to full-corpus inference. A cheaper alternative would be to use one of the fine-tuned GLiNER models for inference instead. Although, if financial constraints are fewer, given how positively LLM-few shot inference performed and how its labeling helped models through fine-tuning, it would be justified to use it to extract NEs and concepts from the unlabeled documents. An option in between would be to use the LLM to label more documents to increase the size of the LLM-annotated data available for fine-tuning GLiNER.

5.3 Limitations

This section addresses the main limitations faced during this thesis, revolving mainly around the test dataset, its gold label validation inconsistencies, issues with the few-shot prompting and fine-tuning datasets, and the lack of domain expertise in handling a domain-specific use case.

5.3.1 Test set size and class imbalance

As seen from the error analysis, the most pressing issue in the experiments was the test set, for a few reasons. First, 656 sentences containing only 1688 named entities constitute a small evaluation set compared to benchmark datasets such as CoNLL2003 and OntoNotes-5.0 (Chen et al., 2019), and may not represent the data or the model's actual performance. Additionally, generic NE (`DATE`, `GPE`, `ORG`, `PERSON`) label instances (838 total) are far better represented in comparison to the target NE (`COURT`, `LAW`, `PROVISION`) (258 total) and concept instances (`TAX_CONCEPT`, `TAX_TYPE`, `JURISDICTION`) (592 total). This can inflate the model's performance, since generic NE labels are prototypical entities with more consistent surface-form patterns that are likely easier to predict than domain-specific target NEs or concept variations. A larger, more evenly distributed test set would facilitate a fairer evaluation across all NE and concept types.

5.3.2 Gold annotation quality

In terms of gold annotations, there are also inconsistencies: annotation guidelines were not followed consistently. It is recommended to corroborate the labeling with other domain experts, especially when only two annotators annotate separate documents. The inter-annotator agreement may benefit from routinely re-solidifying their shared understanding of what each label constitutes. The IAA was calculated on an initial overlapping batch, but due to time and labor constraints in creating the remaining data batches for this experiment, documents were assigned across annotators without further overlap, and agreement could no longer be monitored on the rest of the dataset. The batch was small, with only 5 sentences totaling 992 tokens, so the IAA values should be interpreted with caution. Additionally, midway through the experiments, it was found that there was a mutual misunderstanding and inconsistencies among the independently annotated subsets regarding the domain-specific labels (`LAW` vs `PROVISION`, `GPE`) that had to be corrected before they could be used in the test set for evaluation. This indicates an important limitation of the IAA measure: the Cohen's κ of 0.916 indicated

high consistency in entity-type assignment among shared spans, but does not establish the validity of the annotation decisions themselves.

5.3.3 Few-shot prompt

Another issue was some gaps in the prompt engineered for LLM labeling of target domain data. The IDX:TOK structured prompt (see Figure 3.7), the few-shot variant with detailed annotation guidelines, was selected to label the training data, although in the zero-shot, its simpler-definition counterpart outperformed it. The naive assumption that more detail might help the model make better predictions was reflected by committing the few-shot detailed guideline prompt to label data before also testing the few-shot simpler guideline prompt. In hindsight, fine-tuning GLiNER with data labeled using the simpler guideline prompt would provide a direct comparison.

5.3.4 Fine-tuning dataset errors

The `Indian_Legal_NER_dataset` (Mukund, 2024) used was sourced from a re-hosted, reformatted version of the original `InLegalNER` corpus created by Kalamkar et al. (2022). A comparison with the original source revealed 99.78% exact correspondence in text and entity annotations; the remaining 0.22% had their entity annotations omitted in the reformatted version, likely due to a minor pre-processing error. Given the negligible difference, this is not expected to substantially affect the reported results, though it would be advisable for future replications to confirm the correct dataset before proceeding to the remaining methodology.

Upon close inspection of the LLM-annotated training data, it was found that the dataset had accidentally included one correctly labeled Dutch document, despite the language differences. Though this fault in the dataset was found too late, it occurred because it was safely assumed that the document collections were only in English. It is recommended to maximize the metadata filtering and use language as an exclusion criterion in the database to ensure this does not repeat.

5.3.5 Lack of domain-expertise

Although domain expertise was used to verify the custom gazetteers created for the tax labels, it would also have been beneficial when creating output examples for the LLM few-shot prompt. Instead, the author, who is not a tax or legal specialist, explored the corpus to find appropriate examples and labeled them using the annotation guidelines to the best of their ability. It would be suggested that domain expertise be sought for creating few-shot examples as well.

5.4 Directions for Future Work

This study has many areas of methodological improvement that could be taken on in future work.

It might be worthwhile to investigate the performance of TaxBERT (Hechtner et al., 2025) on the dataset regimes presented in this study, to see whether tax-domain-specific further pre-training is more informative about the tax concepts when paired with legal data fine-tuning. Additionally, since GLiNER performed best across all fine-tuning conditions, its successor, GLiNER2, may also show similar success (Zaratiana et al.,

2025). The new model can be applied to multiple tasks and can now also accept natural-language descriptions for entity types, creating its own form of zero-shot and few-shot classification that may be insightful to compare GPT-5.4-mini's capabilities to (Zaratiana et al., 2025). Finally, it would also be suggested to try NuNER, given its text-and-concept encoder architecture, extensive LLM-annotated pre-training, and reported performance that competes with GLiNER (Bogdanov et al., 2024).

While it was attempted earlier and discarded due to time constraints, another method worth exploring is embedding-based clustering or similarity. Given that GLiNER's embedding similarity mechanism performed well for concepts, this suggests that embeddings exhibit a degree of separation between `TAX_CONCEPT` and `TAX_TYPE` that warrants further investigation.

GPT-5.4-mini does demonstrate inconsistencies in labeling multilingual entities, predicting some but missing others. Since the model is multilingual, it is advisable to explicitly instruct it to consider all non-English entities, as the target domain often involves documents from various multinational jurisdictions. To further harness the utility of GPT-5.4-mini, it would also be recommended to experiment with the reasoning level hyperparameter that prompts the model to 'think' or use varying amounts of computational budget before making predictions. This results in a tradeoff between labeling quality, processing latency, and computational budget that should be navigated carefully if compute costs are a limiting factor. Furthermore, as Huang et al. (2025) suggests, instructing LLMs with fewer rules improves performance; perhaps the prompts can be refined further to focus on the target NEs or target concepts, reducing potential distractions for the model.

Another insightful approach is to employ multiple models equally in an ensemble. The top three approaches: GLiNER-LLM, Legal-BERT-mixed, and GLiNER-mixed, are likely the best candidates, as they all share similar overall performance (high F1-scores and being the best approach for five of ten labels and concepts), but each has its own specific strengths and weaknesses for certain labels. A majority-vote ensemble may therefore be informative for reliable annotations, though the voting scheme and the optimal combination of models remain to be determined empirically.

Finally, the performance gains seen when fine-tuning models using LLM-annotated data raises the question of whether the target domain data, LLM annotations, or an interaction of the two are attributable. As such, these effects can be isolated by asking human annotators and an LLM to independently label a short dataset of tax case law documents, and then comparing encoder performance when fine-tuned on each. This would help explain the effect of annotation source within a fixed domain, while comparable results would suggest that domain matching is the reason the performances increased in this study.

5.5 Conclusion

This research aimed to identify the best-performing approaches for identifying and classifying NEs and concepts in the legal and tax domains, to inform the IBFD on how to improve its current NERC and CE pipeline. In doing so, the variables of domain specificity and data scarcity are addressed, as the target domain is specialized and suffers from scarce labeled training data and limited generalizability from related domains.

From the experiments conducted, it was found that the company baseline was com-

petitive on generic NEs, but was unable to handle the domain-specific NEs and concepts as well. Using out-of-the-box models for zero-shot applications revealed that GLiNER performs better on concepts, while LLMs perform better on NEs. The Heuristics-based Gazetteer achieved higher performance (F1-scores of 0.14 in isolation, 0.40 with the automated components) than the Company Gazetteer (F1-scores of 0.10 in isolation, 0.37 with the automated components) despite being half its size, indicating that concepts with more stable surface forms benefit the most from the string matching facilitated by gazetteer lookup. LLM inference revealed that few-shot prompting (F1-score range between 0.45-0.50) outperforms zero-shot prompting (F1-score range between 0.38-0.42), and that the structure of the prompt output has a smaller impact than the details of the annotation guidelines. Fine-tuning encoder models on different datasets revealed that LLM-annotated target domain data yields the largest performance improvement (F1-scores of 0.53 for GLiNER, 0.51 for Legal-BERT, 0.33 for RoBERTa), followed by mixed data (F1-scores of 0.50 for GLiNER, 0.49 for Legal-BERT, 0.34 for RoBERTa), and lastly by approximate legal domain data (F1-scores of 0.44 for GLiNER, 0.26 for Legal-BERT, 0.10 for RoBERTa), shedding light on the importance of domain-matching. The effect of domain-matching is also presented by Legal-BERT outperforming RoBERTa in all conditions, and performing better than or competitively with GLiNER on certain target labels and concepts. No single model performed best across all labels; however, GLiNER on the LLM-annotated variant achieved the highest overall F1 score of 0.53, with its span-to-label embedding-matching mechanism giving it an advantage over traditional architectures.

Although the IBFD aimed to develop an automated NERC and CE labeling system, the approaches proposed in this thesis require further improvement before they can be applied directly to perform the task. The observed results, therefore, point to an automated system that instead requires human validation, thereby reducing the workload of manual annotation. As such, a recall-oriented approach is most appropriate. This can be achieved by using GLiNER-LLM alone, given that it outperforms all other approaches, both overall and in terms of recall, and by applying the well-performing JURISDICTION gazetteer (0.74 F1 score) as a post-processing correction filter.

This study intends to contribute to tax-case-law-specific NERC and CE research by presenting a holistic comparison across approach groups (rule-based, lookup, prompting, and fine-tuning) using a shared evaluation set, which, to the best of our knowledge, has not been done in this domain before. Furthermore, our findings also shed light on domain adaptation, indicating that related in-domain data can still be useful in providing models with target domain label coverage, but data from the target domain is ultimately what models benefit from the most, as using LLM-annotated target domain data for fine-tuning outperformed the use of approximate-domain human data. This effect cannot be clearly attributed to either domain differences or annotation sources (in the premises of this thesis but can be isolated as proposed in Section 5.4) although a major factor is that the target labels were not accounted for in the approximate legal domain datasets. Despite this, the results may still be informative for other domains also suffering from data scarcity.

When interpreting the results, it is important to consider the limitations faced. The main insights are, however, still considered valuable, and the next step would be to investigate the utility of the recommended solution for recognizing and classifying NEs and concepts in tax case law documents.

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Appendix A

Data-sets

A.1 Test-set composition

Table A.1: Case law collections used to create the IBFD test set.

Acronym	**Name of collection**
BIT	Bulletin for International Taxation
ECJD	European Court of Justice Documents
ECJI	European Court of Justice International
ET	European Taxation
GTTC2	Global Tax Treaty Commentaries – Country Policy & Practice
HRTE	Human Rights and Taxation in Europe and the World
IRS	US Internal Revenue Service
IVM	International VAT Monitor
TNS	Tax News Service
TTCLS	Tax Treaty Case Law Summaries
WHT	Withholding Rates Table Archive

A.2 Test-set composition statistics

Table A.2: Token counts per label in the IBFD test set, split by BIO position.

Label	**B-**	**I-**	**Total**
Named entities			
COURT	106	68	174
DATE	132	198	330
GPE	299	147	446
LAW	65	98	163
ORG	221	285	506
PERSON	186	174	360
PROVISION	87	283	370
Concepts			
JURISDICTION	153	159	312
TAX_CONCEPT	353	458	811
TAX_TYPE	86	98	184
Total (entity tokens)	**1688**	**1968**	**3656**

A.3 InLegalNER composition statistics

Table A.3: Token counts per label in the Indian Legal NER training set after label collapsing, split by BIO position.

Label	B-	I-	Total
COURT	2,367	10,264	12,631
DATE	1,885	3,292	5,177
GPE	1,398	296	1,694
LAW	1,804	4,121	5,925
ORG	1,441	3,002	4,443
PERSON	9,364	8,264	17,628
PROVISION	2,384	5,761	8,145
Total (entity tokens)	**20,643**	**35,000**	**55,643**

Table A.4: Token counts per label in the Indian Legal NER validation set after label collapsing, split by BIO position.

Label	B-	I-	Total
COURT	282	1,021	1,303
DATE	216	199	415
GPE	178	47	225
LAW	221	474	695
ORG	155	337	492
PERSON	981	981	1,962
PROVISION	257	614	871
Total (entity tokens)	**2,290**	**3,673**	**5,963**

A.4 E-NER composition statistics

Table A.5: Token counts per label in the E-NER training set, split by BIO position. Outside

Label	B-	I-	Total
COURT	15	86	101
GPE	951	652	1,603
LAW	1,033	2,006	3,039
ORG	3,514	5,504	9,018
PERSON	569	625	1,194
Total (entity tokens)	**6,082**	**8,873**	**14,955**

Table A.6: Token counts per label in the E-NER validation set, split by BIO position. Outside

Label	B-	I-	Total
COURT	15	53	68
GPE	76	61	137
LAW	138	246	384
ORG	845	2,266	3,111
PERSON	66	69	135
Total (entity tokens)	**1,140**	**2,695**	**3,835**

A.5 Combined Approximate legal domain data composition statistics

Table A.7: Token counts per label in the mixed (Indian_Legal_NER + E-NER) training set, split by BIO position.

Label	B-	I-	Total
COURT	2,382	10,350	12,732
DATE	1,885	3,292	5,177
GPE	2,349	948	3,297
LAW	2,837	6,127	8,964
ORG	4,955	8,506	13,461
PERSON	9,933	8,889	18,822
PROVISION	2,384	5,761	8,145
Total (entity tokens)	**26,725**	**43,873**	**70,598**

Table A.8: Token counts per label in the mixed (Indian + E-NER) validation set, split by BIO position.

Label	B-	I-	Total
COURT	297	1,074	1,371
DATE	216	199	415
GPE	254	108	362
LAW	359	720	1,079
ORG	1,000	2,603	3,603
PERSON	1,047	1,050	2,097
PROVISION	257	614	871
Total (entity tokens)	**3,430**	**6,368**	**9,798**

A.6 LLM-annotated in-domain data composition

Table A.9: Case law collections used to construct the LLM-labeled training data. Splits are performed at the document level; sentence counts are reported per collection.

Acr.	Name of collection	Train	Dev	Docs	Sents
BIT	Bulletin for International Taxation	4	1	5	1,595
ECJD	European Court of Justice Documents	27	3	30	6,698
ECJI	European Court of Justice International	4	1	5	1,106
ET	European Taxation	5	1	6	3,559
GTTC2	Global Tax Treaty Commentaries	13	2	15	903
HRTE	Human Rights and Taxation in Europe and the World	7	1	8	2,109
IRS	US Internal Revenue Service	9	1	10	1,737
IVM	International VAT Monitor	5	1	6	2,306
TNS	Tax News Service	72	8	80	997
TTCLS	Tax Treaty Case Law Summaries	54	6	60	1,704
WHT	Withholding Rates Table Archive	1	1	2	1,144
Total		**201**	**26**	**227**	**23,858**

A.7 LLM-annotated in-domain data composition statistics

Table A.10: Token counts per label in the LLM-labeled training set, split by BIO position.

Label	B-	I-	Total
Named entities			
COURT	3,890	5,698	9,588
DATE	3,649	3,964	7,613
GPE	4,628	1,477	6,197
LAW	4,720	13,195	17,915
ORG	4,677	5,138	9,815
PERSON	3,564	2,872	6,436
PROVISION	3,940	15,624	19,564
Concepts			
JURISDICTION	2,559	2,725	5,284
TAX_CONCEPT	6,052	7,262	13,314
TAX_TYPE	2,084	1,953	4,037
Total (entity tokens)	**39,803**	**59,908**	**99,711**

Table A.11: Token counts per label in the LLM-labeled validation set, split by BIO position.

Label	B-	I-	Total
Named entities			
COURT	228	447	675
DATE	481	532	1,013
GPE	437	117	554
LAW	532	1,469	2,001
ORG	646	947	1,593
PERSON	311	249	560
PROVISION	428	1,835	2,263
Concepts			
JURISDICTION	166	153	319
TAX_CONCEPT	692	791	1,483
TAX_TYPE	259	212	471
Total (entity tokens)	**4,180**	**6,752**	**10,932**

A.8 Mixed Approximate Domain Legal data and LLM-annotated data composition statistics

Table A.12: Token counts per label in the mixed (LLM-labeled + approximate legal domain data) training set, split by BIO position.

Label	B-	I-	Total
Named entities			
COURT	6,257	15,962	22,219
DATE	5,534	7,256	12,790
GPE	6,026	1,773	7,799
LAW	6,524	17,316	23,840
ORG	6,118	8,140	14,258
PERSON	12,928	11,136	24,064
PROVISION	6,324	21,385	27,709
Concepts			
JURISDICTION	2,559	2,725	5,284
TAX_CONCEPT	6,052	7,262	13,314
TAX_TYPE	2,084	1,953	4,037
Total (entity tokens)	**60,406**	**94,908**	**155,314**

Table A.13: Token counts per label in the mixed (LLM-labeled + approximate legal domain data) validation set, split by BIO position. Outside (O) tokens: 95,952.

Label	**B-**	**I-**	**Total**
Named entities			
COURT	510	1,468	1,978
DATE	697	731	1,428
GPE	615	164	779
LAW	753	1,943	2,696
ORG	801	1,284	2,085
PERSON	1,292	1,230	2,522
PROVISION	685	2,449	3,134
Concepts			
JURISDICTION	166	153	319
TAX_CONCEPT	692	791	1,483
TAX_TYPE	259	212	471
Total (entity tokens)	**6,470**	**10,425**	**16,895**

Appendix B

Gazetteer

B.1 Prompt for IBFD Glossary Parsing

The following prompt was provided to Claude (Anthropic, 2024) for parsing, alongside the glossary XML file as input. The prompt below represents a consolidated version of the instructions developed through an iterative dialogue with the AI assistant, during which intermediate outputs were inspected and design decisions were refined.

Prompt

Parse the IBFD International Tax Glossary XML file to extract all glossary term texts. The parser should:

1. Extract the term text from each `<glossentry>` → `<glossterm>` element, stripping all inline XML tags to produce plain text.
2. Handle country-specific entries by removing country parentheticals (e.g., "(France)", "(e.g. Ireland; UK)") from the term text, using the `<country>` XML tag to reliably identify which portion of the text is the country reference.
3. For redirect entries that contain only a `<seelist>` ("See: X") with no definition, store the reference only—do not resolve the chain.
4. Extract taxtopics from the v3.0 version only.
5. For definitions containing inline `<glossref>` cross-references, produce both a plain text version (all tags stripped) and a tagged version where glossrefs are marked with square brackets (e.g., `[tax base]`). Also capture the target IDs of glossref elements for potential graph-based lookups.

Return the output as a JSON file that can be iterated through.

Appendix C

LLM Zero-Shot Prompting

Claude was used to visualize these prompts.

C.1 GPT-NER - SIMPLE

Zero-shot prompt with simple label descriptions (GPT-NER format)

```
You are a Legal and Tax NER tagger. Given a sentence, return it exactly as
provided but with every entity marked using the GPT-NER inline format:

    ##entity text@@ <LABEL>

where ## opens the entity span and @@ closes it, followed by a space, the
label in angle brackets, and a space before the next token. Tokens that are
not part of any entity are left unmarked. Never leave an entity span unclosed.

--- LABELS ---

PERSON
  Covers: people, including judges, advocates, parties, and named individuals.

ORG
  Covers: companies, agencies, institutions, and international bodies.
  Excludes: courts and tribunals (use COURT).

GPE
  Covers: specific named countries, cities, states, and regions.
  Excludes: abstract jurisdictional references (use JURISDICTION).

LAW
  Covers: named laws, treaties, directives, regulations, and conventions.
  Excludes: specific article or section references (use PROVISION).

DATE
  Covers: absolute dates, relative dates, and date ranges or periods.

TAX_TYPE
  Covers: named tax instruments as they appear in context.
  Excludes: jurisdictional modifiers preceding the instrument; generic nouns
  alone; verbal or metaphorical uses.

TAX_CONCEPT
  Covers: domain-specific terms that are not named tax instruments -- legal
```

```
  doctrines, planning mechanisms, economic concepts, and behavioural
  descriptions.
  Excludes: accounting-standard vocabulary.

PROVISION
  Covers: specific articles, sections, paragraphs, or clauses within a law,
  including the law name when it appears as part of the reference.

JURISDICTION
  Covers: abstract or role-based jurisdictional references.
  Excludes: specific named countries or cities (use GPE).

COURT
  Covers: courts, tribunals, and judicial bodies, including generic references.

--- BOUNDARY RULES ---

1. Tag the minimal named span. Jurisdictional or descriptive modifiers before
   a TAX_TYPE are not part of the entity.
2. When a provision reference includes the law name, tag the entire reference
   as a single PROVISION.
3. Do not double-tag overlapping spans. Each token belongs to at most one
   entity.
4. Reproduce the original sentence exactly -- do not alter, reorder, or omit
   any word. Only insert ## and @@ markers with their labels.
```

C.2 IDX:TOK - SIMPLE

Zero-shot prompt with simple label descriptions (IDX:TOK format)

```
You are a Legal and Tax NER classifier. Given a JSON object containing a
sentence_number and a token_map (mapping token indices to tokens), identify
all named entities and return a JSON array of entity tokens.

--- LABELS ---

PERSON
  Covers: people, including judges, advocates, parties, and named individuals.

ORG
  Covers: companies, agencies, institutions, and international bodies.
  Excludes: courts and tribunals (use COURT).

GPE
  Covers: specific named countries, cities, states, and regions.
  Excludes: abstract jurisdictional references (use JURISDICTION).

LAW
  Covers: named laws, treaties, directives, regulations, and conventions.
  Excludes: specific article or section references (use PROVISION).

DATE
  Covers: absolute dates, relative dates, and date ranges or periods.

TAX_TYPE
  Covers: named tax instruments as they appear in context.
  Excludes: jurisdictional modifiers preceding the instrument; generic nouns
```

```
  alone; verbal or metaphorical uses.

TAX_CONCEPT
  Covers: domain-specific terms that are not named tax instruments -- legal
  doctrines, planning mechanisms, economic concepts, and behavioural
  descriptions.
  Excludes: accounting-standard vocabulary.

PROVISION
  Covers: specific articles, sections, paragraphs, or clauses within a law,
  including the law name when it appears as part of the reference.

JURISDICTION
  Covers: abstract or role-based jurisdictional references.
  Excludes: specific named countries or cities (use GPE).

COURT
  Covers: courts, tribunals, and judicial bodies, including generic references.

--- BOUNDARY RULES ---

1. Tag the minimal named span. Jurisdictional or descriptive modifiers before
   a TAX_TYPE are not part of the entity.
2. When a provision reference includes the law name, tag the entire reference
   as a single PROVISION.
3. Do not double-tag overlapping spans. Each token belongs to at most one
   entity.
4. Reproduce the original sentence exactly -- do not alter, reorder, or omit
   any word.

--- OUTPUT FORMAT ---

Return ONLY a JSON array. Each element represents one entity token:
{
  "sent_index": <sentence_number from input>,
  "token_index": <token position from token_map>,
  "token_text": <exact token string>,
  "entity_index": <integer grouping multi-token entities, starting at 0>,
  "label": "<one of the following integers representing the respective label:
    PERSON = 1, ORG = 2, GPE = 3, LAW = 4, DATE = 5, TAX_TYPE = 6,
    TAX_CONCEPT = 7, PROVISION = 8, JURISDICTION = 9, COURT = 10>"
}

Consecutive tokens sharing the same entity_index form one entity span.
Increment entity_index for each new entity.
```

C.3 GPT-NER - DETAILED

Zero-shot prompt with detailed label descriptions (GPT-NER format)

```
You are a Legal and Tax NER tagger. Given a sentence, return it exactly as
provided but with every entity marked using the GPT-NER inline format:

    ##entity text@@ <LABEL>

where ## opens the entity span and @@ closes it, followed by a space, the
label in angle brackets, and a space before the next token. Tokens that are
```

```
not part of any entity are left unmarked. Never leave an entity span unclosed.

--- LABELS ---

PERSON
  Covers: named or specifically referenced individuals in their human
  capacity, including judges, advocates general, named parties, and definite
  singular references resolvable to one specific individual in context. Spans
  include personal names with attached titles or roles.
  Excludes: generic plurals, corporate or legal persons even when named like
  individuals, and unresolved generic role references.

ORG
  Covers: named non-judicial organisations -- companies, agencies,
  supranational bodies, tax authorities, professional bodies). Spans include
  the full proper name with name-constitutive modifiers and standard
  legal-form suffixes.
  Excludes: courts and tribunals (use COURT), jurisdictional modifiers that
  merely localise a generic institution, generic descriptions without naming,
  and entity-type concepts (use TAX_CONCEPT).

GPE
  Covers: specific, named geopolitical entities -- countries, cities, regions,
  supranational unions with concrete identity. Spans cover the bare name;
  articles are included only when part of the official name.
  Excludes: demonyms and adjectival forms entirely are never tagged on their
  own and are stripped when they modify another entity, abstract role-based
  references (use JURISDICTION), and geographic regions without political
  identity.

LAW
  Covers: named statutory or treaty-level instruments referred to as a whole,
  without article or section pinpointing. Spans include the full official
  title, standard acronyms, definite back-references that unambiguously
  resolve to a previously named instrument, and years embedded in a statutory
  title.
  Excludes: pinpoint references (use PROVISION), case law and judgments, and
  generic legal-concept terms.

DATE
  Covers: absolute or relative temporal expressions denoting a point or
  period. Spans cover only the date expression itself; contextualizing nouns
  that label what kind of period it is are stripped.
  Excludes: durations not anchored to a calendar point, vague temporal
  references, and dates embedded inside another entity's name.

TAX_TYPE:
  Covers: named tax instruments as they appear in context.
  Exclude: jurisdictional modifiers, generic nouns alone, and verbal or
  metaphorical uses.

TAX_CONCEPT:
  Covers: domain terms that aren't named instruments: legal doctrines,
  planning mechanisms, behaviours, and economic concepts.
  Excludes: accounting-standard vocabulary.

PROVISION
  Covers: specific articles, sections, paragraphs, or sub-paragraphs within a
  named instrument, including the host instrument when cited together. Spans
```

```
  cover the full citation as a single entity -- pinpoint plus host instrument
  plus subdivisions -- and coordinated pinpoints sharing one host as one span.
  Excludes: the host instrument cited alone without a pinpoint (use LAW),
  case-paragraph references, and recital references unless project-scoped in.

JURISDICTION
  Covers: abstract or role-based references to a state in its legal or fiscal
  capacity, where the specific country is unnamed or generalized. Spans
  include the role descriptor with role-defining qualifiers.
  Excludes: named countries even when filling such a role (use GPE) and purely
  geographic terms without legal-role meaning.

COURT
  Covers: judicial bodies, tribunals, and adjudicative offices. Spans include
  the full proper name with name-constitutive modifiers, standard acronyms,
  and case-specific unnamed references.
  Excludes: jurisdictional modifiers that merely localize a generic court,
  individual judges or advocates general by name (use PERSON), and
  metaphorical uses.

--- BOUNDARY RULES ---

1. Tag the minimal named span. Jurisdictional or descriptive modifiers before
   a TAX_TYPE are not part of the entity.
2. When a provision reference includes the law name, tag the entire reference
   as a single PROVISION.
3. Do not double-tag overlapping spans. Each token belongs to at most one
   entity.
4. Reproduce the original sentence exactly -- do not alter, reorder, or omit
   any word. Only insert ## and @@ markers with their labels.
```

C.4 IDX:TOK - DETAILED

Zero-shot prompt with detailed label descriptions (IDX:TOK format)

```
You are a Legal and Tax NER classifier. Given a JSON object containing a
sentence_number and a token_map (mapping token indices to tokens), identify
all named entities and return a JSON array of entity tokens.

--- LABELS ---

PERSON
  Covers: named or specifically referenced individuals in their human
  capacity, including judges, advocates general, named parties, and definite
  singular references resolvable to one specific individual in context. Spans
  include personal names with attached titles or roles.
  Excludes: generic plurals, corporate or legal persons even when named like
  individuals, and unresolved generic role references.

ORG
  Covers: named non-judicial organisations -- companies, agencies,
  supranational bodies, tax authorities, professional bodies). Spans include
  the full proper name with name-constitutive modifiers and standard
  legal-form suffixes.
  Excludes: courts and tribunals (use COURT), jurisdictional modifiers that
  merely localise a generic institution, generic descriptions without naming,
  and entity-type concepts (use TAX_CONCEPT).
```

```
GPE
  Covers: specific, named geopolitical entities -- countries, cities, regions,
  supranational unions with concrete identity. Spans cover the bare name;
  articles are included only when part of the official name.
  Excludes: demonyms and adjectival forms entirely are never tagged on their
  own and are stripped when they modify another entity, abstract role-based
  references (use JURISDICTION), and geographic regions without political
  identity.

LAW
  Covers: named statutory or treaty-level instruments referred to as a whole,
  without article or section pinpointing. Spans include the full official
  title, standard acronyms, definite back-references that unambiguously
  resolve to a previously named instrument, and years embedded in a statutory
  title.
  Excludes: pinpoint references (use PROVISION), case law and judgments, and
  generic legal-concept terms.

DATE
  Covers: absolute or relative temporal expressions denoting a point or
  period. Spans cover only the date expression itself; contextualizing nouns
  that label what kind of period it is are stripped.
  Excludes: durations not anchored to a calendar point, vague temporal
  references, and dates embedded inside another entity's name.

TAX_TYPE:
  Covers: named tax instruments as they appear in context.
  Exclude: jurisdictional modifiers, generic nouns alone, and verbal or
  metaphorical uses.

TAX_CONCEPT:
  Covers: domain terms that aren't named instruments: legal doctrines,
  planning mechanisms, behaviours, and economic concepts.
  Excludes: accounting-standard vocabulary.

PROVISION
  Covers: specific articles, sections, paragraphs, or sub-paragraphs within a
  named instrument, including the host instrument when cited together. Spans
  cover the full citation as a single entity -- pinpoint plus host instrument
  plus subdivisions -- and coordinated pinpoints sharing one host as one span.
  Excludes: the host instrument cited alone without a pinpoint (use LAW),
  case-paragraph references, and recital references unless project-scoped in.

JURISDICTION
  Covers: abstract or role-based references to a state in its legal or fiscal
  capacity, where the specific country is unnamed or generalized. Spans
  include the role descriptor with role-defining qualifiers.
  Excludes: named countries even when filling such a role (use GPE) and purely
  geographic terms without legal-role meaning.

COURT
  Covers: judicial bodies, tribunals, and adjudicative offices. Spans include
  the full proper name with name-constitutive modifiers, standard acronyms,
  and case-specific unnamed references.
  Excludes: jurisdictional modifiers that merely localize a generic court,
  individual judges or advocates general by name (use PERSON), and
  metaphorical uses.
```

```
--- BOUNDARY RULES ---

1. Tag the minimal named span. Jurisdictional or descriptive modifiers before
   a TAX_TYPE are not part of the entity.
2. When a provision reference includes the law name, tag the entire reference
   as a single PROVISION.
3. Do not double-tag overlapping spans. Each token belongs to at most one
   entity.
4. Reproduce the original sentence exactly -- do not alter, reorder, or omit
   any word.

--- OUTPUT FORMAT ---

Return ONLY a JSON array. Each element represents one entity token:
{
  "sent_index": <sentence_number from input>,
  "token_index": <token position from token_map>,
  "token_text": <exact token string>,
  "entity_index": <integer grouping multi-token entities, starting at 0>,
  "label": "<one of the following integers representing the respective label:
    PERSON = 1, ORG = 2, GPE = 3, LAW = 4, DATE = 5, TAX_TYPE = 6,
    TAX_CONCEPT = 7, PROVISION = 8, JURISDICTION = 9, COURT = 10>"
}

Consecutive tokens sharing the same entity_index form one entity span.
Increment entity_index for each new entity.
```

Appendix D

LLM Few-Shot Prompting

Claude was used to visualize these prompts.

D.1 GPT-NER - SIMPLE

Few-shot prompt with simple label descriptions and examples (GPT-NER format)

```
You are a Legal and Tax NER tagger. Given a sentence, return it exactly as
provided but with every entity marked using the GPT-NER inline format:

    ##entity text@@ <LABEL>

where ## opens the entity span and @@ closes it, followed by a space, the
label in angle brackets, and a space before the next token. Tokens that are
not part of any entity are left unmarked. Never leave an entity span unclosed.

--- LABELS ---

PERSON
  Covers: people, including judges, advocates, parties, and named individuals.

ORG
  Covers: companies, agencies, institutions, and international bodies.
  Excludes: courts and tribunals (use COURT).

GPE
  Covers: specific named countries, cities, states, and regions.
  Excludes: abstract jurisdictional references (use JURISDICTION).

LAW
  Covers: named laws, treaties, directives, regulations, and conventions.
  Excludes: specific article or section references (use PROVISION).

DATE
  Covers: absolute dates, relative dates, and date ranges or periods.

TAX_TYPE
  Covers: named tax instruments as they appear in context.
  Excludes: jurisdictional modifiers preceding the instrument; generic nouns
  alone; verbal or metaphorical uses.

TAX_CONCEPT
```

```
  Covers: domain-specific terms that are not named tax instruments -- legal
  doctrines, planning mechanisms, economic concepts, and behavioural
  descriptions.
  Excludes: accounting-standard vocabulary.

PROVISION
  Covers: specific articles, sections, paragraphs, or clauses within a law,
  including the law name when it appears as part of the reference.

JURISDICTION
  Covers: abstract or role-based jurisdictional references.
  Excludes: specific named countries or cities (use GPE).

COURT
  Covers: courts, tribunals, and judicial bodies, including generic references.

--- EXAMPLES ---

Example 1:
Text: "The outcome of this case was quite predictable and must be considered
correct"
Expected output: "The outcome of this case was quite predictable and must be
considered correct"

Example 2:
Text: "Residents are subject to world-wide taxation by virtue of
Article 2 of the LIR.
Expected output: "Residents are subject to world-wide taxation by virtue of
##Article 2 of the LIR@@ <PROVISION>."

Example 3:
Text: "On 15 March 2017, the European Commission issued a decision
requiring Luxembourg to recover the unlawful state aid."
Expected output: "On ##15 March 2017@@<DATE>, the ##European Commission@@<ORG>
issued a decision requiring ##Luxembourg@@<GPE> to recover the
unlawful state aid."

Example 4:

Text: "The Dutch corporate income tax does not apply to profits
attributed to a permanent establishment in cases involving transfer pricing
adjustments."
Expected output: "The Dutch ##corporate income tax@@<TAX_TYPE> does not apply to
profits attributed to a ##permanent establishment@@<TAX_CONCEPT>
in cases involving ##transfer pricing@@<TAX_CONCEPT> adjustments."

Example 5:

Text: "Advocate General Kokott argued before the Court of Justice that the
residence state must grant relief from double taxation."
Expected output: "##Advocate General Kokott@@<PERSON> argued before the ##Court
of Justice@@<COURT> that the ##residence state@@<JURISDICTION> must grant relief
from ##double taxation@@<TAX_CONCEPT>."


Example 6:
Text: "Under Article 49 TFEU, Belgium may not impose withholding tax on dividends
paid to a parent company in another Member State"
Expected output: "Under ##Article 49 TFEU@@<PROVISION>, ##Belgium@@<GPE> may not
```

```
impose ##withholding tax@@<TAX_TYPE> on dividends paid to a parent company in
another ##Member State@@<JURISDICTION>"


--- BOUNDARY RULES ---

1. Tag the minimal named span. Jurisdictional or descriptive modifiers before
   a TAX_TYPE are not part of the entity: `Dutch ## corporate income tax @@
   <TAX_TYPE>'.
2. When a provision reference includes the law name, tag the entire reference
   as a single PROVISION: `## Article 49 TFEU @@ <PROVISION>'.
3. Do not double-tag overlapping spans. Each token belongs to at most one
   entity.
4. Reproduce the original sentence exactly -- do not alter, reorder, or omit
   any word. Only insert ## and @@ markers with their labels.
```

D.2 IDX:TOK - SIMPLE

Few-shot prompt with simple label descriptions and examples (IDX:TOK format)

```
You are a Legal and Tax NER classifier. Given a JSON object containing a
sentence_number and a token_map (mapping token indices to tokens), identify
all named entities and return a JSON array of entity tokens.

--- LABELS ---

PERSON
  Covers: people, including judges, advocates, parties, and named individuals.

ORG
  Covers: companies, agencies, institutions, and international bodies.
  Excludes: courts and tribunals (use COURT).

GPE
  Covers: specific named countries, cities, states, and regions.
  Excludes: abstract jurisdictional references (use JURISDICTION).

LAW
  Covers: named laws, treaties, directives, regulations, and conventions.
  Excludes: specific article or section references (use PROVISION).

DATE
  Covers: absolute dates, relative dates, and date ranges or periods.

TAX_TYPE
  Covers: named tax instruments as they appear in context.
  Excludes: jurisdictional modifiers preceding the instrument; generic nouns
  alone; verbal or metaphorical uses.

TAX_CONCEPT
  Covers: domain-specific terms that are not named tax instruments -- legal
  doctrines, planning mechanisms, economic concepts, and behavioural
  descriptions.
  Excludes: accounting-standard vocabulary.

PROVISION
```

```
  Covers: specific articles, sections, paragraphs, or clauses within a law,
  including the law name when it appears as part of the reference.

JURISDICTION
  Covers: abstract or role-based jurisdictional references.
  Excludes: specific named countries or cities (use GPE).

COURT
  Covers: courts, tribunals, and judicial bodies, including generic references.

--- EXAMPLES ---

Example 1:
Input: {"sentence_number": 0, "token_map": {"0": "The", "1": "outcome", "2": "of",
↪  "3": "this", "4": "case", "5": "was", "6": "quite", "7": "predictable", "8":
↪  "."}}
Output: []

Example 2:
Input: {"sentence_number": 1, "token_map": {"0": "Residents", "1": "are", "2":
↪  "subject", "3": "to", "4": "world-wide", "5": "taxation", "6": "by", "7":
↪  "virtue", "8": "of", "9": "Article", "10": "2", "11": "of", "12": "the", "13":
↪  "LIR", "14": "."}}
Output: [
  {"sent_index": 1, "token_index": 9, "token_text": "Article", "entity_index": 0,
  ↪  "label": 8},
  {"sent_index": 1, "token_index": 10, "token_text": "2", "entity_index": 0,
  ↪  "label": 8},
  {"sent_index": 1, "token_index": 11, "token_text": "of", "entity_index": 0,
  ↪  "label": 8},
  {"sent_index": 1, "token_index": 12, "token_text": "the", "entity_index": 0,
  ↪  "label": 8},
  {"sent_index": 1, "token_index": 13, "token_text": "LIR", "entity_index": 0,
  ↪  "label": 8}
]

Example 3:
Input: {"sentence_number": 2, "token_map": {"0": "On", "1": "15", "2": "March",
↪  "3": "2017", "4": ",", "5": "the", "6": "European", "7": "Commission", "8":
↪  "issued", "9": "a", "10": "decision", "11": "requiring", "12": "Luxembourg",
↪  "13": "to", "14": "recover", "15": "the", "16": "unlawful", "17": "state",
↪  "18": "aid", "19": "."}}
Output: [
  {"sent_index": 2, "token_index": 1, "token_text": "15", "entity_index": 0,
  ↪  "label": 5},
  {"sent_index": 2, "token_index": 2, "token_text": "March", "entity_index": 0,
  ↪  "label": 5},
  {"sent_index": 2, "token_index": 3, "token_text": "2017", "entity_index": 0,
  ↪  "label": 5},
  {"sent_index": 2, "token_index": 6, "token_text": "European", "entity_index": 1,
  ↪  "label": 2},
  {"sent_index": 2, "token_index": 7, "token_text": "Commission", "entity_index":
  ↪  1, "label": 2},
  {"sent_index": 2, "token_index": 12, "token_text": "Luxembourg", "entity_index":
  ↪  2, "label": 3}
]

Example 4:
```

```
Input: {"sentence_number": 3, "token_map": {"0": "The", "1": "Dutch", "2":
↪  "corporate", "3": "income", "4": "tax", "5": "does", "6": "not", "7": "apply",
↪  "8": "to", "9": "profits", "10": "attributed", "11": "to", "12": "a", "13":
↪  "permanent", "14": "establishment", "15": "in", "16": "cases", "17":
↪  "involving", "18": "transfer", "19": "pricing", "20": "adjustments", "21":
↪  "."}}
Output: [
  {"sent_index": 3, "token_index": 2, "token_text": "corporate", "entity_index": 0,
  ↪  "label": 6},
  {"sent_index": 3, "token_index": 3, "token_text": "income", "entity_index": 0,
  ↪  "label": 6},
  {"sent_index": 3, "token_index": 4, "token_text": "tax", "entity_index": 0,
  ↪  "label": 6},
  {"sent_index": 3, "token_index": 13, "token_text": "permanent", "entity_index":
  ↪  1, "label": 7},
  {"sent_index": 3, "token_index": 14, "token_text": "establishment",
  ↪  "entity_index": 1, "label": 7},
  {"sent_index": 3, "token_index": 18, "token_text": "transfer", "entity_index": 2,
  ↪  "label": 7},
  {"sent_index": 3, "token_index": 19, "token_text": "pricing", "entity_index": 2,
  ↪  "label": 7}
]

Example 5:
Input: {"sentence_number": 4, "token_map": {"0": "Advocate", "1": "General", "2":
↪  "Kokott", "3": "argued", "4": "before", "5": "the", "6": "Court", "7": "of",
↪  "8": "Justice", "9": "that", "10": "the", "11": "residence", "12": "state",
↪  "13": "must", "14": "grant", "15": "relief", "16": "from", "17": "double",
↪  "18": "taxation", "19": "."}}
Output: [
  {"sent_index": 4, "token_index": 0, "token_text": "Advocate", "entity_index": 0,
  ↪  "label": 1},
  {"sent_index": 4, "token_index": 1, "token_text": "General", "entity_index": 0,
  ↪  "label": 1},
  {"sent_index": 4, "token_index": 2, "token_text": "Kokott", "entity_index": 0,
  ↪  "label": 1},
  {"sent_index": 4, "token_index": 6, "token_text": "Court", "entity_index": 1,
  ↪  "label": 10},
  {"sent_index": 4, "token_index": 7, "token_text": "of", "entity_index": 1,
  ↪  "label": 10},
  {"sent_index": 4, "token_index": 8, "token_text": "Justice", "entity_index": 1,
  ↪  "label": 10},
  {"sent_index": 4, "token_index": 11, "token_text": "residence", "entity_index":
  ↪  2, "label": 9},
  {"sent_index": 4, "token_index": 12, "token_text": "state", "entity_index": 2,
  ↪  "label": 9},
  {"sent_index": 4, "token_index": 17, "token_text": "double", "entity_index": 3,
  ↪  "label": 7},
  {"sent_index": 4, "token_index": 18, "token_text": "taxation", "entity_index": 3,
  ↪  "label": 7}
]

Example 6:
Input: {"sentence_number": 5, "token_map": {"0": "Under", "1": "Article", "2":
↪  "49", "3": "TFEU", "4": ",", "5": "Belgium", "6": "may", "7": "not", "8":
↪  "impose", "9": "withholding", "10": "tax", "11": "on", "12": "dividends", "13":
↪  "paid", "14": "to", "15": "a", "16": "parent", "17": "company", "18": "in",
↪  "19": "another", "20": "Member", "21": "State", "22": "."}}
```

```
Output: [
  {"sent_index": 5, "token_index": 1, "token_text": "Article", "entity_index": 0,
  ↪  "label": 8},
  {"sent_index": 5, "token_index": 2, "token_text": "49", "entity_index": 0,
  ↪  "label": 8},
  {"sent_index": 5, "token_index": 3, "token_text": "TFEU", "entity_index": 0,
  ↪  "label": 8},
  {"sent_index": 5, "token_index": 5, "token_text": "Belgium", "entity_index": 1,
  ↪  "label": 3},
  {"sent_index": 5, "token_index": 9, "token_text": "withholding", "entity_index":
  ↪  2, "label": 6},
  {"sent_index": 5, "token_index": 10, "token_text": "tax", "entity_index": 2,
  ↪  "label": 6},
  {"sent_index": 5, "token_index": 20, "token_text": "Member", "entity_index": 3,
  ↪  "label": 9},
  {"sent_index": 5, "token_index": 21, "token_text": "State", "entity_index": 3,
  ↪  "label": 9}
]

--- BOUNDARY RULES ---

1. Tag the minimal named span. Jurisdictional modifiers before a TAX_TYPE are
   not part of the entity: in `Dutch corporate income tax', only `corporate
   income tax' is TAX_TYPE.
2. When a provision reference includes the law name, tag the entire reference
   as a single PROVISION.
3. Each token belongs to at most one entity.
4. Only return tokens that are part of entities. Do not return O labels.

--- OUTPUT FORMAT ---

Return ONLY a JSON array. Each element represents one entity token:
{
  "sent_index": <sentence_number from input>,
  "token_index": <token position from token_map>,
  "token_text": <exact token string>,
  "entity_index": <integer grouping multi-token entities, starting at 0>,
  "label": "<one of the following integers representing the respective label:
    PERSON = 1, ORG = 2, GPE = 3, LAW = 4, DATE = 5, TAX_TYPE = 6,
    TAX_CONCEPT = 7, PROVISION = 8, JURISDICTION = 9, COURT = 10>"
}

Consecutive tokens sharing the same entity_index form one entity span.
Increment entity_index for each new entity.
```

D.3 GPT-NER - DETAILED

Few-shot prompt with detailed label descriptions and examples (GPT-NER format)

```
You are a Legal and Tax NER tagger. Given a sentence, return it exactly as
provided but with every entity marked using the GPT-NER inline format:

    ##entity text@@ <LABEL>

where ## opens the entity span and @@ closes it, followed by a space, the
label in angle brackets, and a space before the next token. Tokens that are
```

```
not part of any entity are left unmarked. Never leave an entity span unclosed.

--- LABELS ---

PERSON
  Covers: named or specifically referenced individuals in their human
  capacity, including judges, advocates general, named parties, and definite
  singular references resolvable to one specific individual in context (e.g.,
  Mr. Bosal, Advocate General Konll, the applicant). Spans include personal
  names with attached titles or roles.
  Excludes: generic plurals (taxpayers, judges), corporate or legal persons
  even when named like individuals (use ORG), and unresolved generic role
  references (a taxpayer in this situation).

ORG
  Covers: named non-judicial organizations -- companies, agencies,
  supranational bodies, tax authorities, professional bodies (e.g., Bosal
  Holding BV, European Commission, OECD, HMRC). Spans include the full proper
  name with name-constitutive modifiers (European in European Commission,
  Internal in Internal Revenue Service) and standard legal-form suffixes (BV,
  plc, GmbH).
  Excludes: courts and tribunals (use COURT), jurisdictional modifiers that
  merely localize a generic institution, generic descriptions without naming
  (the company, the holding), and entity-type concepts (holding company,
  subsidiary) (use TAX_CONCEPT).

GPE
  Covers: specific, named geopolitical entities -- countries, cities, regions,
  supranational unions with concrete identity (e.g., Belgium, Brussels,
  European Union, EU, California). Spans cover the bare name; articles are
  included only when part of the official name (The Hague).
  Excludes: demonyms and adjectival forms (Belgian, Dutch, European) entirely
  are never tagged on their own and are stripped when they modify another
  entity, abstract role-based references (use JURISDICTION), and geographic
  regions without political identity (Europe as a continent).

LAW
  Covers: named statutory or treaty-level instruments referred to as a whole,
  without article or section pinpointing (e.g., TFEU, EC Treaty,
  Parent-Subsidiary Directive, Income Tax Act 2007). Spans include the full
  official title, standard acronyms, definite back-references that
  unambiguously resolve to a previously named instrument (the Treaty, the
  Directive, the Code), and years embedded in a statutory title.
  Excludes: pinpoint references (use PROVISION), case law and judgments, and
  generic legal-concept terms (the law, legislation).

DATE
  Covers: absolute or relative temporal expressions denoting a point or
  period (e.g., 18 September 2003, 1990, 2019). Spans cover only the date
  expression itself; contextualizing nouns that label what kind of period it
  is are stripped.
  Excludes: durations not anchored to a calendar point (for five years), vague
  temporal references (recently, previously, last year), and dates embedded
  inside another entity's name (the 2007 in Income Tax Act 2007 stays inside
  the LAW span).

TAX_TYPE
  Covers: named tax instruments as they appear in context (e.g., corporate
  income tax, withholding tax, VAT).
```

```
  Exclude: jurisdictional modifiers, generic nouns alone (tax, rate), and
  verbal or metaphorical uses (withholding documents, a tax on patience).

TAX_CONCEPT
  Covers: domain terms that aren't named instruments: legal doctrines,
  planning mechanisms, behaviors, and economic concepts (e.g., transfer
  pricing, arm's length, permanent establishment).
  Excludes: accounting-standard vocabulary such as GAAP, IFRS, or depreciation.

PROVISION
  Covers: specific articles, sections, paragraphs, or sub-paragraphs within a
  named instrument, including the host instrument when cited together (e.g.,
  Article 49 TFEU, Article 4(1) of the Parent-Subsidiary Directive, Section
  3(2), paragraph 3). Spans cover the full citation as a single entity --
  pinpoint plus host instrument plus subdivisions -- and coordinated pinpoints
  sharing one host as one span (Articles 43 and 49 EC).
  Excludes: the host instrument cited alone without a pinpoint (use LAW),
  case-paragraph references (paragraph 23 of the judgment), and recital
  references unless project-scoped in.

JURISDICTION
  Covers: abstract or role-based references to a state in its legal or fiscal
  capacity, where the specific country is unnamed or generalized (e.g., Member
  State, residence state, source state, host state, contracting state, third
  country). Spans include the role descriptor with role-defining qualifiers
  (the Member State concerned, the State of residence).
  Excludes: named countries even when filling such a role (use GPE) and purely
  geographic terms without legal-role meaning.

COURT
  Covers: judicial bodies, tribunals, and adjudicative offices (e.g., Court of
  Justice, European Court of Justice, CJEU, Hoge Raad, Bundesfinanzhof). Spans
  include the full proper name with name-constitutive modifiers (European in
  European Court of Justice), standard acronyms, and case-specific unnamed
  references (the referring court, the national court).
  Excludes: jurisdictional modifiers that merely localize a generic court,
  individual judges or advocates general by name (use PERSON), and
  metaphorical uses (the court of public opinion).

--- EXAMPLES ---

Example 1:
Text: "The outcome of this case was quite predictable and must be considered
correct"
Expected output: "The outcome of this case was quite predictable and must be
considered correct"

Example 2:
Text: "Residents are subject to world-wide taxation by virtue of
Article 2 of the LIR.
Expected output: "Residents are subject to world-wide taxation by virtue of
##Article 2 of the LIR@@ <PROVISION>."

Example 3:
Text: "On 15 March 2017, the European Commission issued a decision
requiring Luxembourg to recover the unlawful state aid."
Expected output: "On ##15 March 2017@@<DATE>, the ##European Commission@@<ORG>
issued a decision requiring ##Luxembourg@@<GPE> to recover the
unlawful state aid."
```

```
Example 4:

Text: "The Dutch corporate income tax does not apply to profits
attributed to a permanent establishment in cases involving transfer pricing
adjustments."
Expected output: "The Dutch ##corporate income tax@@<TAX_TYPE> does not apply to
profits attributed to a ##permanent establishment@@<TAX_CONCEPT>
in cases involving ##transfer pricing@@<TAX_CONCEPT> adjustments."

Example 5:

Text: "Advocate General Kokott argued before the Court of Justice that the
residence state must grant relief from double taxation."
Expected output: "##Advocate General Kokott@@<PERSON> argued before the ##Court
of Justice@@<COURT> that the ##residence state@@<JURISDICTION> must grant relief
from ##double taxation@@<TAX_CONCEPT>."


Example 6:
Text: "Under Article 49 TFEU, Belgium may not impose withholding tax on dividends
paid to a parent company in another Member State"
Expected output: "Under ##Article 49 TFEU@@<PROVISION>, ##Belgium@@<GPE> may not
impose ##withholding tax@@<TAX_TYPE> on dividends paid to a parent company in
another ##Member State@@<JURISDICTION>"

--- BOUNDARY RULES ---

1. Tag the minimal named span. Jurisdictional or descriptive modifiers before
   a TAX_TYPE are not part of the entity: `Dutch ## corporate income tax @@
   <TAX_TYPE>'.
2. When a provision reference includes the law name, tag the entire reference
   as a single PROVISION: `## Article 49 TFEU @@ <PROVISION>'.
3. Do not double-tag overlapping spans. Each token belongs to at most one
   entity.
4. Reproduce the original sentence exactly -- do not alter, reorder, or omit
   any word. Only insert ## and @@ markers with their labels.
```

D.4 IDX:TOK - DETAILED

Few-shot prompt with detailed label descriptions and examples (IDX:TOK format)

```
You are a Legal and Tax NER classifier. Given a JSON object containing a
sentence_number and a token_map (mapping token indices to tokens), identify
all named entities and return a JSON array of entity tokens.

--- LABELS ---

PERSON
  Covers: named or specifically referenced individuals in their human
  capacity, including judges, advocates general, named parties, and definite
  singular references resolvable to one specific individual in context (e.g.,
  Mr. Bosal, Advocate General Konll, the applicant). Spans include personal
  names with attached titles or roles.
  Excludes: generic plurals (taxpayers, judges), corporate or legal persons
  even when named like individuals (use ORG), and unresolved generic role
```

```
  references (a taxpayer in this situation).

ORG
  Covers: named non-judicial organizations -- companies, agencies,
  supranational bodies, tax authorities, professional bodies (e.g., Bosal
  Holding BV, European Commission, OECD, HMRC). Spans include the full proper
  name with name-constitutive modifiers (European in European Commission,
  Internal in Internal Revenue Service) and standard legal-form suffixes (BV,
  plc, GmbH).
  Excludes: courts and tribunals (use COURT), jurisdictional modifiers that
  merely localize a generic institution, generic descriptions without naming
  (the company, the holding), and entity-type concepts (holding company,
  subsidiary) (use TAX_CONCEPT).

GPE
  Covers: specific, named geopolitical entities -- countries, cities, regions,
  supranational unions with concrete identity (e.g., Belgium, Brussels,
  European Union, EU, California). Spans cover the bare name; articles are
  included only when part of the official name (The Hague).
  Excludes: demonyms and adjectival forms (Belgian, Dutch, European) entirely
  are never tagged on their own and are stripped when they modify another
  entity, abstract role-based references (use JURISDICTION), and geographic
  regions without political identity (Europe as a continent).

LAW
  Covers: named statutory or treaty-level instruments referred to as a whole,
  without article or section pinpointing (e.g., TFEU, EC Treaty,
  Parent-Subsidiary Directive, Income Tax Act 2007). Spans include the full
  official title, standard acronyms, definite back-references that
  unambiguously resolve to a previously named instrument (the Treaty, the
  Directive, the Code), and years embedded in a statutory title.
  Excludes: pinpoint references (use PROVISION), case law and judgments, and
  generic legal-concept terms (the law, legislation).

DATE
  Covers: absolute or relative temporal expressions denoting a point or
  period (e.g., 18 September 2003, 1990, 2019). Spans cover only the date
  expression itself; contextualizing nouns that label what kind of period it
  is are stripped.
  Excludes: durations not anchored to a calendar point (for five years), vague
  temporal references (recently, previously, last year), and dates embedded
  inside another entity's name (the 2007 in Income Tax Act 2007 stays inside
  the LAW span).

TAX_TYPE
  Covers: named tax instruments as they appear in context (e.g., corporate
  income tax, withholding tax, VAT).
  Exclude: jurisdictional modifiers, generic nouns alone (tax, rate), and
  verbal or metaphorical uses (withholding documents, a tax on patience).

TAX_CONCEPT
  Covers: domain terms that aren't named instruments: legal doctrines,
  planning mechanisms, behaviors, and economic concepts (e.g., transfer
  pricing, arm's length, permanent establishment).
  Excludes: accounting-standard vocabulary such as GAAP, IFRS, or depreciation.

PROVISION
  Covers: specific articles, sections, paragraphs, or sub-paragraphs within a
  named instrument, including the host instrument when cited together (e.g.,
```

```
  Article 49 TFEU, Article 4(1) of the Parent-Subsidiary Directive, Section
  3(2), paragraph 3). Spans cover the full citation as a single entity --
  pinpoint plus host instrument plus subdivisions -- and coordinated pinpoints
  sharing one host as one span (Articles 43 and 49 EC).
  Excludes: the host instrument cited alone without a pinpoint (use LAW),
  case-paragraph references (paragraph 23 of the judgment), and recital
  references unless project-scoped in.

JURISDICTION
  Covers: abstract or role-based references to a state in its legal or fiscal
  capacity, where the specific country is unnamed or generalized (e.g., Member
  State, residence state, source state, host state, contracting state, third
  country). Spans include the role descriptor with role-defining qualifiers
  (the Member State concerned, the State of residence).
  Excludes: named countries even when filling such a role (use GPE) and purely
  geographic terms without legal-role meaning.

COURT
  Covers: judicial bodies, tribunals, and adjudicative offices (e.g., Court of
  Justice, European Court of Justice, CJEU, Hoge Raad, Bundesfinanzhof). Spans
  include the full proper name with name-constitutive modifiers (European in
  European Court of Justice), standard acronyms, and case-specific unnamed
  references (the referring court, the national court).
  Excludes: jurisdictional modifiers that merely localize a generic court,
  individual judges or advocates general by name (use PERSON), and
  metaphorical uses (the court of public opinion).

--- EXAMPLES ---

Example 1:
Input: {"sentence_number": 0, "token_map": {"0": "The", "1": "outcome", "2": "of",
↪  "3": "this", "4": "case", "5": "was", "6": "quite", "7": "predictable", "8":
↪  "."}}
Output: []

Example 2:
Input: {"sentence_number": 1, "token_map": {"0": "Residents", "1": "are", "2":
↪  "subject", "3": "to", "4": "world-wide", "5": "taxation", "6": "by", "7":
↪  "virtue", "8": "of", "9": "Article", "10": "2", "11": "of", "12": "the", "13":
↪  "LIR", "14": "."}}
Output: [
  {"sent_index": 1, "token_index": 9, "token_text": "Article", "entity_index": 0,
  ↪  "label": 8},
  {"sent_index": 1, "token_index": 10, "token_text": "2", "entity_index": 0,
  ↪  "label": 8},
  {"sent_index": 1, "token_index": 11, "token_text": "of", "entity_index": 0,
  ↪  "label": 8},
  {"sent_index": 1, "token_index": 12, "token_text": "the", "entity_index": 0,
  ↪  "label": 8},
  {"sent_index": 1, "token_index": 13, "token_text": "LIR", "entity_index": 0,
  ↪  "label": 8}
]

Example 3:
Input: {"sentence_number": 2, "token_map": {"0": "On", "1": "15", "2": "March",
↪  "3": "2017", "4": ",", "5": "the", "6": "European", "7": "Commission", "8":
↪  "issued", "9": "a", "10": "decision", "11": "requiring", "12": "Luxembourg",
↪  "13": "to", "14": "recover", "15": "the", "16": "unlawful", "17": "state",
↪  "18": "aid", "19": "."}}
```

```
Output: [
  {"sent_index": 2, "token_index": 1, "token_text": "15", "entity_index": 0,
  ↪  "label": 5},
  {"sent_index": 2, "token_index": 2, "token_text": "March", "entity_index": 0,
  ↪  "label": 5},
  {"sent_index": 2, "token_index": 3, "token_text": "2017", "entity_index": 0,
  ↪  "label": 5},
  {"sent_index": 2, "token_index": 6, "token_text": "European", "entity_index": 1,
  ↪  "label": 2},
  {"sent_index": 2, "token_index": 7, "token_text": "Commission", "entity_index":
  ↪  1, "label": 2},
  {"sent_index": 2, "token_index": 12, "token_text": "Luxembourg", "entity_index":
  ↪  2, "label": 3}
]

Example 4:
Input: {"sentence_number": 3, "token_map": {"0": "The", "1": "Dutch", "2":
↪  "corporate", "3": "income", "4": "tax", "5": "does", "6": "not", "7": "apply",
↪  "8": "to", "9": "profits", "10": "attributed", "11": "to", "12": "a", "13":
↪  "permanent", "14": "establishment", "15": "in", "16": "cases", "17":
↪  "involving", "18": "transfer", "19": "pricing", "20": "adjustments", "21":
↪  "."}}
Output: [
  {"sent_index": 3, "token_index": 2, "token_text": "corporate", "entity_index": 0,
  ↪  "label": 6},
  {"sent_index": 3, "token_index": 3, "token_text": "income", "entity_index": 0,
  ↪  "label": 6},
  {"sent_index": 3, "token_index": 4, "token_text": "tax", "entity_index": 0,
  ↪  "label": 6},
  {"sent_index": 3, "token_index": 13, "token_text": "permanent", "entity_index":
  ↪  1, "label": 7},
  {"sent_index": 3, "token_index": 14, "token_text": "establishment",
  ↪  "entity_index": 1, "label": 7},
  {"sent_index": 3, "token_index": 18, "token_text": "transfer", "entity_index": 2,
  ↪  "label": 7},
  {"sent_index": 3, "token_index": 19, "token_text": "pricing", "entity_index": 2,
  ↪  "label": 7}
]

Example 5:
Input: {"sentence_number": 4, "token_map": {"0": "Advocate", "1": "General", "2":
↪  "Kokott", "3": "argued", "4": "before", "5": "the", "6": "Court", "7": "of",
↪  "8": "Justice", "9": "that", "10": "the", "11": "residence", "12": "state",
↪  "13": "must", "14": "grant", "15": "relief", "16": "from", "17": "double",
↪  "18": "taxation", "19": "."}}
Output: [
  {"sent_index": 4, "token_index": 0, "token_text": "Advocate", "entity_index": 0,
  ↪  "label": 1},
  {"sent_index": 4, "token_index": 1, "token_text": "General", "entity_index": 0,
  ↪  "label": 1},
  {"sent_index": 4, "token_index": 2, "token_text": "Kokott", "entity_index": 0,
  ↪  "label": 1},
  {"sent_index": 4, "token_index": 6, "token_text": "Court", "entity_index": 1,
  ↪  "label": 10},
  {"sent_index": 4, "token_index": 7, "token_text": "of", "entity_index": 1,
  ↪  "label": 10},
  {"sent_index": 4, "token_index": 8, "token_text": "Justice", "entity_index": 1,
  ↪  "label": 10},
```

```
  {"sent_index": 4, "token_index": 11, "token_text": "residence", "entity_index":
  ↪  2, "label": 9},
  {"sent_index": 4, "token_index": 12, "token_text": "state", "entity_index": 2,
  ↪  "label": 9},
  {"sent_index": 4, "token_index": 17, "token_text": "double", "entity_index": 3,
  ↪  "label": 7},
  {"sent_index": 4, "token_index": 18, "token_text": "taxation", "entity_index": 3,
  ↪  "label": 7}
]

Example 6:
Input: {"sentence_number": 5, "token_map": {"0": "Under", "1": "Article", "2":
↪  "49", "3": "TFEU", "4": ",", "5": "Belgium", "6": "may", "7": "not", "8":
↪  "impose", "9": "withholding", "10": "tax", "11": "on", "12": "dividends", "13":
↪  "paid", "14": "to", "15": "a", "16": "parent", "17": "company", "18": "in",
↪  "19": "another", "20": "Member", "21": "State", "22": "."}}
Output: [
  {"sent_index": 5, "token_index": 1, "token_text": "Article", "entity_index": 0,
  ↪  "label": 8},
  {"sent_index": 5, "token_index": 2, "token_text": "49", "entity_index": 0,
  ↪  "label": 8},
  {"sent_index": 5, "token_index": 3, "token_text": "TFEU", "entity_index": 0,
  ↪  "label": 8},
  {"sent_index": 5, "token_index": 5, "token_text": "Belgium", "entity_index": 1,
  ↪  "label": 3},
  {"sent_index": 5, "token_index": 9, "token_text": "withholding", "entity_index":
  ↪  2, "label": 6},
  {"sent_index": 5, "token_index": 10, "token_text": "tax", "entity_index": 2,
  ↪  "label": 6},
  {"sent_index": 5, "token_index": 20, "token_text": "Member", "entity_index": 3,
  ↪  "label": 9},
  {"sent_index": 5, "token_index": 21, "token_text": "State", "entity_index": 3,
  ↪  "label": 9}
]

--- BOUNDARY RULES ---

1. Tag the minimal named span. Jurisdictional modifiers before a TAX_TYPE are
   not part of the entity: in `Dutch corporate income tax', only `corporate
   income tax' is TAX_TYPE.
2. When a provision reference includes the law name, tag the entire reference
   as a single PROVISION.
3. Each token belongs to at most one entity.
4. Only return tokens that are part of entities. Do not return O labels.

--- OUTPUT FORMAT ---

Return ONLY a JSON array. Each element represents one entity token:
{
  "sent_index": <sentence_number from input>,
  "token_index": <token position from token_map>,
  "token_text": <exact token string>,
  "entity_index": <integer grouping multi-token entities, starting at 0>,
  "label": "<one of the following integers representing the respective label:
    PERSON = 1, ORG = 2, GPE = 3, LAW = 4, DATE = 5, TAX_TYPE = 6,
    TAX_CONCEPT = 7, PROVISION = 8, JURISDICTION = 9, COURT = 10>"
}

Consecutive tokens sharing the same entity_index form one entity span.
```

```
Increment entity_index for each new entity.
```

Appendix E

Results

E.1 Baselines and Gazetteers

Table E.1: Per-label precision, recall, and F1 for the gazetteers, individual automated components of the company baseline and gazetteers in combination with the automated component on the IBFD test set. The highest F1 score is in bold. Systems: Heur = heuristic gazetteer only; Comp = company gazetteer only; spaCy = spaCy only; spaCy+Re = spaCy + regex; +Comp = spaCy + regex + company gazetteer; +Heur = spaCy + regex + heuristic gazetteer. †+Comp is the reported company baseline.

	Heur			Comp			spaCy			spaCy+Re			+Comp†			+Heur		
Label	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Named entities																		
COURT	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.35	0.37	**0.36**	0.35	0.37	**0.36**	0.35	0.37	**0.36**
DATE	0.00	0.00	0.00	0.00	0.00	0.00	0.51	0.64	**0.57**	0.51	0.64	**0.57**	0.51	0.64	**0.57**	0.51	0.64	**0.57**
GPE	0.00	0.00	0.00	0.00	0.00	0.00	0.68	0.65	**0.66**	0.68	0.65	**0.66**	0.68	0.65	**0.66**	0.68	0.65	**0.66**
LAW	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.02	0.01	0.03	0.02	**0.02**	0.03	0.02	**0.02**	0.03	0.02	**0.02**
ORG	0.00	0.00	0.00	0.00	0.00	0.00	0.14	0.36	0.20	0.16	0.34	**0.22**	0.16	0.34	**0.22**	0.16	0.34	**0.22**
PERSON	0.00	0.00	0.00	0.00	0.00	0.00	0.53	0.52	**0.53**	0.53	0.52	**0.53**	0.53	0.52	**0.53**	0.53	0.52	**0.53**
PROVISION	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.40	0.34	**0.37**	0.40	0.34	**0.37**	0.40	0.34	**0.37**
Concepts																		
JURISDICTION	0.65	0.86	**0.74**	0.65	0.86	**0.74**	0.00	0.00	0.00	0.00	0.00	0.00	0.68	0.73	0.71	0.68	0.73	0.71
TAX_CONCEPT	0.42	0.10	**0.16**	0.07	0.26	0.11	0.00	0.00	0.00	0.00	0.00	0.00	0.08	0.24	0.12	0.39	0.08	0.13
TAX_TYPE	0.49	0.47	**0.48**	0.06	0.47	0.11	0.00	0.00	0.00	0.00	0.00	0.00	0.07	0.47	0.12	0.49	0.47	**0.48**
micro avg	0.56	0.12	0.20	0.12	0.16	0.14	0.34	0.27	0.30	0.39	0.31	0.35	0.25	0.45	0.32	0.43	0.42	0.42
macro avg	0.16	0.14	0.14	0.08	0.16	0.10	0.19	0.22	0.20	0.27	0.29	0.27	0.35	0.43	0.37	0.42	0.42	**0.40**

E.2 Zero-shot

Table E.2: Zero-shot LLM prompting and GLiNER: per-label precision, recall, and F1 on the IBFD test set. The highest F1 score is in bold. GPT-NER uses inline `##`/`@@` span marking; IDX:TOK uses the index–token dictionary format. Both prompt formats are shown in simple and detailed variants. GLiNER is evaluated zero-shot with label-name mapping.

	GPT-NER simple			GPT-NER detailed			IDX:TOK simple			IDX:TOK detailed			GLiNER		
Label	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Named entities															
COURT	0.33	0.68	0.44	0.39	0.60	**0.47**	0.26	0.76	0.39	0.32	0.73	0.45	0.29	0.46	0.36
DATE	0.66	0.46	0.54	0.68	0.45	0.55	0.76	0.73	**0.75**	0.71	0.64	0.67	0.72	0.79	**0.75**
GPE	0.65	0.60	0.62	0.68	0.60	0.64	0.56	0.69	0.62	0.61	0.78	**0.68**	0.55	0.77	0.64
LAW	0.22	0.51	**0.31**	0.23	0.42	0.30	0.19	0.46	0.27	0.19	0.43	0.26	0.13	0.62	0.21
ORG	0.38	0.33	0.36	0.47	0.35	**0.40**	0.38	0.32	0.35	0.41	0.38	0.39	0.20	0.43	0.27
PERSON	0.71	0.79	**0.75**	0.77	0.66	0.71	0.61	0.74	0.67	0.68	0.67	0.68	0.46	0.81	0.59
PROVISION	0.32	0.39	0.35	0.36	0.46	**0.41**	0.24	0.54	0.33	0.31	0.57	0.40	0.09	0.18	0.12
Concepts															
JURISDICTION	0.46	0.19	0.27	0.38	0.16	0.23	0.26	0.37	**0.30**	0.17	0.14	0.15	0.29	0.07	0.11
TAX_CONCEPT	0.34	0.07	0.12	0.29	0.03	0.05	0.24	0.22	0.23	0.21	0.10	0.13	0.32	0.33	**0.32**
TAX_TYPE	0.17	0.08	0.11	0.17	0.02	0.04	0.26	0.36	0.31	0.19	0.21	0.20	0.35	0.62	**0.45**
micro avg	0.47	0.39	0.43	0.51	0.36	0.42	0.37	0.49	0.43	0.41	0.45	0.43	0.34	0.51	0.41
macro avg	0.43	0.41	0.39	0.44	0.38	0.38	0.38	0.52	**0.42**	0.38	0.46	0.40	0.34	0.51	0.38

E.3 Few-shot

Table E.3: Few-shot LLM prompting: per-label precision, recall, and F1 on the IBFD test set. The highest F1 score is in bold. GPT-NER uses inline `##`/`@@` span marking; IDX:TOK uses the index–token dictionary format. Both formats are shown in simple and detailed variants.

	GPT-NER simple			GPT-NER detailed			IDX:TOK simple			IDX:TOK detailed		
Label	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Named entities												
COURT	0.34	0.73	0.46	0.39	0.64	0.48	0.27	0.67	0.39	0.37	0.72	**0.49**
DATE	0.70	0.58	0.63	0.59	0.55	0.57	0.73	0.70	0.72	0.75	0.73	**0.74**
GPE	0.68	0.65	0.67	0.70	0.66	0.68	0.60	0.71	0.65	0.67	0.75	**0.71**
LAW	0.21	0.46	0.29	0.23	0.45	**0.30**	0.15	0.34	0.21	0.13	0.42	0.20
ORG	0.48	0.36	0.41	0.46	0.43	**0.44**	0.35	0.30	0.32	0.42	0.45	0.43
PERSON	0.79	0.82	**0.80**	0.67	0.68	0.67	0.51	0.68	0.58	0.46	0.53	0.49
PROVISION	0.37	0.64	0.47	0.39	0.57	0.46	0.25	0.59	0.35	0.37	0.68	**0.48**
Concepts												
JURISDICTION	0.45	0.49	0.47	0.56	0.71	0.62	0.45	0.61	0.52	0.57	0.71	**0.63**
TAX_CONCEPT	0.38	0.11	0.18	0.37	0.08	0.13	0.25	0.38	**0.30**	0.32	0.27	0.29
TAX_TYPE	0.54	0.36	0.43	0.52	0.36	0.42	0.46	0.48	0.47	0.44	0.53	**0.48**
micro avg	0.51	0.48	0.49	0.52	0.48	0.50	0.39	0.54	0.45	0.45	0.55	0.49
macro avg	0.50	0.52	0.48	0.49	0.51	0.48	0.40	0.54	0.45	0.45	0.58	**0.50**

E.4 Finetuned Encoders: Approximate Legal Domain Data (ALDD)

Table E.4: Fine-tuned encoders trained on approximate legal domain data (ALDD): per-label precision, recall, and F1 on the IBFD test set. The highest F1 score is in bold. Concept labels are not present in this dataset; hence the zero scores for RoBERTa and LegalBERT. GLiNER predicts them zero-shot via its embedding-matching architecture.

	RoBERTa			LegalBERT			GLiNER		
Label	P	R	F1	P	R	F1	P	R	F1
Named entities									
COURT	0.22	0.08	0.11	0.57	0.41	0.47	0.64	0.69	**0.66**
DATE	0.36	0.08	0.12	0.54	0.20	0.29	0.79	0.39	**0.53**
GPE	0.41	0.13	0.20	0.66	0.54	0.59	0.65	0.69	**0.67**
LAW	0.00	0.00	0.00	0.07	0.15	0.09	0.18	0.66	**0.28**
ORG	0.11	0.10	0.11	0.54	0.17	0.26	0.44	0.35	**0.39**
PERSON	0.38	0.25	0.30	0.78	0.64	0.70	0.80	0.69	0.74
PROVISION	0.11	0.11	0.11	0.14	0.16	**0.15**	0.08	0.15	0.11
Concepts									
JURISDICTION	0.00	0.00	0.00	0.00	0.00	0.00	0.40	0.16	**0.23**
TAX_CONCEPT	0.00	0.00	0.00	0.00	0.00	0.00	0.42	0.22	**0.29**
TAX_TYPE	0.00	0.00	0.00	0.00	0.00	0.00	0.55	0.49	**0.52**
micro avg	0.21	0.08	0.12	0.49	0.24	0.32	0.47	0.44	0.45
macro avg	0.16	0.08	0.10	0.33	0.23	0.26	0.50	0.45	**0.44**

E.5 Finetuned Encoders: LLM-labeled Data

Table E.5: Fine-tuned encoders trained on LLM-labeled data: per-label precision, recall, and F1 on the IBFD test set. The highest F1 score is in bold.

	RoBERTa			LegalBERT			GLiNER		
Label	P	R	F1	P	R	F1	P	R	F1
Named entities									
COURT	0.34	0.75	0.47	0.41	0.79	0.54	0.43	0.74	**0.55**
DATE	0.70	0.74	0.72	0.76	0.80	**0.78**	0.69	0.82	0.75
GPE	0.57	0.62	0.59	0.58	0.84	0.68	0.71	0.84	**0.77**
LAW	0.12	0.43	0.19	0.19	0.51	**0.28**	0.20	0.49	**0.28**
ORG	0.21	0.29	0.25	0.42	0.57	**0.48**	0.39	0.49	0.43
PERSON	0.26	0.13	0.18	0.43	0.54	0.48	0.49	0.81	**0.61**
PROVISION	0.08	0.11	0.09	0.36	0.67	**0.47**	0.35	0.51	0.42
Concepts									
JURISDICTION	0.39	0.39	0.39	0.57	0.82	**0.67**	0.54	0.85	0.66
TAX_CONCEPT	0.13	0.06	0.08	0.31	0.21	0.25	0.22	0.32	**0.26**
TAX_TYPE	0.33	0.38	0.35	0.36	0.55	0.44	0.50	0.63	**0.55**
micro avg	0.32	0.36	0.34	0.45	0.59	0.51	0.44	0.63	0.52
macro avg	0.31	0.39	0.33	0.44	0.63	0.51	0.45	0.65	**0.53**

E.6 Finetuned Encoders: Mix of approximate domain legal data and LLM-labeled data

Table E.6: Fine-tuned encoders trained on mixed data (ELD + LLM): per-label precision, recall, and F1 on the IBFD test set. The highest F1 score is in bold.

	RoBERTa			LegalBERT			GLiNER		
Label	P	R	F1	P	R	F1	P	R	F1
Named entities									
COURT	0.32	0.68	0.43	0.40	0.77	**0.53**	0.40	0.79	**0.53**
DATE	0.66	0.72	0.69	0.75	0.75	**0.75**	0.62	0.86	0.72
GPE	0.62	0.56	0.59	0.66	0.85	0.74	0.68	0.83	**0.75**
LAW	0.13	0.40	0.20	0.14	0.45	0.21	0.17	0.54	**0.26**
ORG	0.23	0.30	0.26	0.45	0.50	**0.47**	0.43	0.49	0.46
PERSON	0.30	0.30	0.30	0.33	0.44	0.38	0.40	0.51	**0.45**
PROVISION	0.10	0.11	0.11	0.39	0.60	**0.47**	0.31	0.49	0.38
Concepts									
JURISDICTION	0.45	0.48	0.46	0.57	0.81	**0.67**	0.51	0.82	0.62
TAX_CONCEPT	0.13	0.06	0.08	0.28	0.18	0.22	0.29	0.36	**0.32**
TAX_TYPE	0.21	0.33	0.26	0.45	0.52	0.48	0.43	0.62	**0.51**
micro avg	0.33	0.36	0.35	0.45	0.56	0.50	0.43	0.61	0.51
macro avg	0.31	0.39	0.34	0.44	0.59	0.49	0.42	0.63	**0.50**

Appendix F

Error Analysis

F.1 Token IDs

Table F.1: SentencePiece subword tokenization of specific examples seen in the error analysis. The underscore prefix marks a word-initial subword. Note that "taxpayer" is a single vocabulary item, whereas "tax payer" splits into two subword tokens, yielding distinct token IDs.

Term	**Token IDs**	**Subword pieces**
taxpayer	13250	▁taxpayer
tax payer	1189, 37061	▁tax, ▁payer
taxpayers	11353	▁taxpayers
tax	1189	▁tax
withholding tax	25994, 1189	▁withholding, ▁tax
taxable income	19246, 1766	▁taxable, ▁income
tax credit	1189, 1236	▁tax, ▁credit
Grand	3029	▁Grand
grand	4176	▁grand